

Business Requirements Document

Online Marketplace Storefront, Seller Portal and Administration

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Conforms to	ISO/IEC/IEEE 29148:2018 (Business Requirements Specification), IIBA BABOK® Guide v3

This document defines the business needs, scope, business rules, functional and non-functional requirements for the HexCommerce marketplace, derived from a complete review of the current user interface. It is the authoritative reference for design, build, test and acceptance.

Document Control

Revision History

Version	Date	Author	Description of change
1.0	2026-09-28	Business Analysis	Initial issue, derived from a review of the UI source code and the running application (27 routes). No stakeholder input yet.
1.1	2026-09-28	Business Analysis	Added business case and delivery plan, RACI, cutover, migration, training and hypercare; system context, process, data-model and state diagrams; 12 production (R2) modules with wireframes and a notification catalogue; sizing assumptions; UI state gallery; requirements-to-test traceability; validation-status markers. Removed revision entries that did not reflect actual drafts.

Approvals

By signing, approvers confirm that the requirements herein accurately reflect the business need and are accepted as the baseline for solution delivery. Changes after approval follow the change-control process in Section 15.2.

Role	Name	Signature	Date
Executive Sponsor			
Product Owner			
Head of Marketplace / Seller Operations			
Head of Customer Service			
Engineering Lead			
QA / Test Lead			
Data Protection Officer			

Distribution

Recipient group	Purpose
Product, UX and Engineering	Design and build reference
Quality Assurance	Test basis; acceptance criteria for system and user acceptance testing
Marketplace, Customer Service and Finance Operations	Validation of processes, rules and reporting
Legal, Security and Data Protection	Compliance review of Sections 11 and 12

Referenced Standards

Standard	Application in this document
ISO/IEC/IEEE 29148:2018	Structure and content of the business requirements specification; requirement quality attributes (necessary, unambiguous, verifiable, traceable).
IIBA BABOK® Guide v3	Elicitation, stakeholder analysis, requirements classification (business, stakeholder, solution, transition).
ISO/IEC 25010:2023	Product quality model used to classify non-functional requirements.
W3C WCAG 2.2 / EN 301 549	Accessibility conformance target (Level AA).
IETF RFC 2119 / RFC 8174	Normative keywords SHALL, SHOULD, MAY.
ISO 8601, ISO 4217, BCP 47	Date/time, currency and locale representation.
PCI DSS v4.0; UK/EU GDPR; PECR/ePrivacy	Payment, privacy and consent obligations for production release.
OWASP ASVS v4.0; NIST SP 800-63B	Application security and authentication requirements.

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1. Introduction

1.1 Purpose

This Business Requirements Document (BRD) specifies what the HexCommerce online marketplace must do to meet its business objectives. It was produced by systematically reviewing every screen, component, state, business rule and data structure of the current HexCommerce user interface, and translating the observed behaviour into verifiable requirements. Where the current interface only simulates a capability (for example payment or authentication), the BRD states the production requirement explicitly so that the gap is visible and plannable.

1.2 Scope of this Document

The document covers three user-facing applications delivered in one web front end:

- **Customer storefront** – discovery, search, product evaluation, reviews, wish list, basket, checkout, orders, account and customer service.
- **Seller portal** – seller recruitment and application, dashboard, listing management with image upload, deal management, order fulfilment and business details.
- **Marketplace administration** – review and decision of seller applications.

It also covers cross-cutting configuration (branding, theming, currency, locale, delivery and tax parameters) and the non-functional qualities expected of the solution.

1.3 Intended Audience

Audience	How to use this document
Executive Sponsor, Product Owner	Sections 2–4 and 15–17: objectives, scope, business case, gaps, risks, decisions and transition.
UX / UI Design	Sections 5–8 and Appendices A–B: journeys, screens, states, wireframes and functional behaviour.
Engineering	Sections 7–11 and 14: functional requirements, business rules, data, NFRs and interfaces.
Quality Assurance	Acceptance criteria in Section 8, traceability in Section 18 and test traceability in Appendix D.
Operations (Marketplace, CS, Finance)	Sections 6, 9, 13 and 17: processes, rules, reporting and transition.
Legal, Security, Data Protection	Sections 10, 11 and 12.

1.4 Document Conventions

1.4.1 Normative language

Requirements use the keywords of RFC 2119: **SHALL** denotes a mandatory requirement; **SHOULD** a recommended requirement that may be deviated from with justification; **MAY** an optional capability.

1.4.2 Requirement identifiers

Each requirement has a unique, stable identifier of the form <Type>-<Area>-<nn>, where Type is BO (business objective), FR (functional), BR (business rule), NFR (non-functional), GAP, RSK or OI; and Area is a module code (e.g. CHK = Checkout). Identifiers are never reused.

1.4.3 Prioritisation (MoSCoW)

Code	Meaning
M – Must	Essential for the release; the solution is not acceptable without it.
S – Should	Important and expected, but a workaround exists for the short term.
C – Could	Desirable; included if time and budget allow.
W – Won't (this time)	Explicitly out of scope for the release; recorded for future planning.

1.4.4 Release designation

Code	Meaning
R1	Behaviour present in the current UI baseline (high-fidelity front end operating on sample data with browser-local persistence). Acceptance can be verified against the current build.
R2	Required for production go-live (real identity, payments, persistence, integrations and compliance). Not present in the current baseline.
R1/R2	Partially present in R1; the acceptance criteria state what R2 adds.

R1 acceptance criteria describe what the current UI does. They are not, by themselves, evidence that the behaviour is what the business wants: prototype values (for example the hard-coded 5.99 delivery fee, or tax added on top of prices) may have been captured as requirements. Where such values were spotted they are logged as gaps (Section 15); the remainder must be confirmed in the validation workshops.

1.4.5 Validation status

This document was produced by analysing the UI source code and the running application. No stakeholder interviews or workshops have yet taken place. The following content is therefore **Proposed** and must be validated before the document is baselined:

Content	Status	Validation owner
Business objectives and KPI targets (Section 2.3)	Proposed by Business Analysis – targets are illustrative	Executive Sponsor
MoSCoW priorities on all requirements	Proposed by Business Analysis	Product Owner
R1 acceptance criteria	Derived from current UI behaviour – confirm intent	Product Owner, UX Lead
R2 requirements, wireframes and delivery plan	Proposed – based on common marketplace practice	Product Owner, Engineering Lead
Sizing assumptions (Section 11.2)	Planning assumptions – replace with forecast	Finance, Engineering Lead
Retention periods and compliance obligations (Sections 10.6, 12)	Indicative – not legal advice	Legal / DPO
Durations, costs and benefits (Section 4)	Indicative / to be estimated	Executive Sponsor, Finance

1.5 Definitions and Acronyms

Term	Definition
AOV	Average order value.
Basket / Cart	The customer's list of items intended for purchase (used interchangeably).
Buy box	The purchase panel on the product page (price, delivery, stock, quantity, Add to cart, Buy now).
Department / Sub-department	Top-level catalogue category and its narrower grouping (e.g. Electronics › Audio).
Facet	A filter dimension on a listing page (brand, price, rating, availability).
KPI	Key performance indicator.
KYC / KYB	Know Your Customer / Know Your Business verification.
Lightning deal	Time-limited, quantity-limited marketplace discount on a catalogue product.
Listing	A product offered for sale by a third-party seller.
PDP	Product detail page.
PSP	Payment service provider.
Referral fee	Commission the marketplace deducts from each seller sale.
Saved for later	Checkout-time parking list of items removed from the basket, distinct from the wish list.
Seller deal	Time-limited percentage discount a seller runs on one of their own listings.
Wish list	Durable personal list of saved products.

2. Business Context

2.1 Background

HexCommerce is a general-merchandise online marketplace. Customers buy from a catalogue spanning eight departments — Electronics; Home & Kitchen; Fashion; Books; Sports & Outdoors; Beauty & Personal Care; Toys & Games; and Grocery — with 29 sub-departments. Independent businesses can apply to sell on the marketplace and, once approved, manage their own listings, promotions and fulfilment. The current front end is a complete, themeable, responsive implementation of these journeys operating on representative sample data (41 products, 6 orders, 3 saved addresses, 3 payment cards, 4 seller listings, 5 seller orders, 2 seller deals, 3 seller applications).

2.2 Business Problem and Opportunity

Customers expect a fast, familiar and trustworthy marketplace experience on any device; sellers expect a low-friction route to market with clear fees and simple tools. The business needs a single, brand-configurable platform that:

- converts browsing into purchases through strong discovery, transparent pricing and delivery promises;
- grows supply by letting sellers self-onboard, list products and run promotions without assistance;
- reduces customer-service cost through structured self-service issue reporting;
- can be re-branded and localised for new markets or partners purely by configuration.

2.3 Business Objectives

Proposed targets

The KPI targets below are illustrative proposals made during analysis; they are not derived from business data and must be confirmed or replaced by the Executive Sponsor (Section 1.4.5).

Table 1: Business objectives and success measures (proposed)

ID	Objective	Success measure (proposed KPI and target)
BO-01	Deliver a complete, conventional discover-to-purchase journey that maximises conversion.	Visit-to-order conversion $\geq 3.0\%$ within 6 months of launch; checkout completion $\geq 65\%$ of checkout starts.
BO-02	Increase average order value through free-delivery thresholds, deals, related products and saved lists.	AOV $\geq$ free-delivery threshold + 20%; $\geq 25\%$ of orders include a deal item.
BO-03	Grow marketplace supply by enabling third-party sellers to self-onboard and self-manage.	Application decision ≤ 2 business days; time from approval to first live listing ≤ 1 day (median).
BO-04	Reduce cost-to-serve with self-service customer support.	$\geq 70\%$ of post-purchase contacts raised via the help desk; first response ≤ 24 h for 95% of issues.
BO-05	Build customer trust through reviews, verified purchases and transparent pricing and delivery information.	$\geq 60\%$ of product pages with ≥ 5 reviews; return-reason “not as described” $\leq 2\%$ of orders.
BO-06	Enable white-labelling and market expansion by configuration only.	New brand/theme/currency deployment in ≤ 1 day with zero code change.
BO-07	Be inclusive and usable on every device.	WCAG 2.2 AA conformance; Core Web Vitals “good” for $\geq 75\%$ of mobile page loads.

2.4 Stakeholders

Table 2: Stakeholder register

Stakeholder	Interest / responsibility	Influence
Executive Sponsor	Funding, strategic alignment, final approval.	High
Product Owner	Backlog ownership, prioritisation, acceptance.	High
Customers (shoppers)	Finding, evaluating and buying products; post-purchase support.	High
Sellers	Listing products, promotions, fulfilment, payouts.	High
Marketplace Administrators	Seller onboarding decisions, marketplace integrity.	Medium
Customer Service Operations	Handling issues, SLAs, returns and refunds.	Medium
Merchandising / Marketing	Hero content, promotional tiles, deals, rails.	Medium
Finance	Tax, delivery fees, referral fees, payouts, invoicing.	Medium
Engineering, UX and QA	Design, build, test and operate the solution.	Medium
Legal, Security, Data Protection Officer	Regulatory compliance, security posture, privacy.	High

2.5 User Classes and Personas

Table 3: User classes

User class	Description	Primary goals	Key screens
Guest visitor	Unauthenticated visitor browsing the store.	Discover products, compare prices, decide to buy.	Home, Department, Search, Product, Deals
Customer (signed in)	Registered shopper. Persona: “Alex Morgan”, member since 2019, Manchester.	Buy quickly, track orders, get help, save items.	Cart, Checkout, Orders, Account, Wish list, Customer service
Prospective seller	Business applying to sell.	Understand fees, apply, learn the decision.	Sell, Register as a seller
Seller	Approved business. Persona: “Northgate Goods”, verified limited company, Home & Kitchen.	List products, run deals, fulfil orders, monitor sales.	Seller dashboard, Listings, Deals, Seller orders, Business details
Marketplace administrator	Internal staff member reviewing sellers.	Approve legitimate sellers quickly; reject with reasons.	Admin › Seller applications
Store operator	Technical/brand owner deploying the storefront.	Brand, theme and localise without code change.	Configuration (environment)

3. Scope

3.1 In Scope

Table 4: In-scope capabilities

Area	Capabilities
Storefront	Global navigation; home merchandising; department browsing; faceted filtering; sorting; pagination; search with type-ahead; product detail; ratings and reviews; wish list; basket and saved-for-later; checkout and confirmation; order history and tracking; account hub; customer-service help desk; deals; sign-in and registration; informational pages (About); 404 handling.
Seller portal	Recruitment page; seller application with status tracking and re-application; dashboard and KPIs; listing management with four-step wizard, live preview and image upload; deal management; seller order fulfilment; business details.
Administration	Seller application review queue with approve, reject-with-reason and reconsider.
Configuration	Site name and tagline; six colour themes and per-token overrides; currency; locale; free-delivery threshold; tax rate.
Production capabilities (R2)	Multi-seller orders and shipping rules; vouchers; returns and refunds; seller payouts and settlement; fraud and risk; buyer–seller messaging; notifications; CS agent console; moderation; user, role and order administration; merchandising CMS; SEO.
Quality	Responsive design from 320 px; keyboard and screen-reader accessibility; performance; resilience to storage failure.

3.2 Out of Scope

- Native mobile applications (iOS/Android); the responsive web experience serves mobile users.
- Warehouse management, carrier label generation and last-mile logistics systems (integration points only).
- Advertising platform, affiliate programme and careers content (footer links are placeholders).
- Loyalty/subscription programmes and gift cards (explicitly removed from the current UI).
- Multi-language content translation (R2 enables externalised strings only; translation is a later phase).
- Back-office ERP and general-ledger accounting (integration points only; settlement rules are in scope – FR-PAY).

3.3 Current State versus Target State

The current UI is a production-grade front end whose behaviour is real for browsing and list management but simulated for identity, payment and fulfilment. The table below summarises which capabilities are functional in R1 and which depend on R2 back-end services.

Table 5: Current (R1) and target (R2) state

Capability	R1 – current UI baseline	R2 – production target
Catalogue, search, filters, sort, pagination	Functional against in-memory sample catalogue.	Catalogue service and search engine; seller listings included.
Basket, wish list, saved for later	Functional; persisted in the browser.	Persisted to customer account; cross-device.
Reviews	Write/delete own reviews; persisted in the browser.	Central store, moderation, helpful votes.

Capability	R1 – current UI baseline	R2 – production target
Checkout and payment	Simulated: generates reference, clears basket; no payment.	PSP authorisation, order creation, e-mail confirmation.
Orders and tracking	Sample history with tracker.	Live order and carrier events; invoices; returns.
Identity	Persona always signed in; forms presentational.	Authentication, sessions, RBAC, MFA.
Customer service	Functional threads, persisted in the browser; auto-acknowledgement.	Agent console, notifications, SLAs, attachments.
Seller portal and admin	Functional workflows persisted in the browser.	Server-side persistence, KYC/KYB, payouts, audit.

3.4 Assumptions

1. The visual design, information architecture and interaction patterns of the current UI are approved as the basis for production.
2. Sample data is representative of production data shapes; field names and constraints in Section 10 are indicative of the target data model.
3. A single currency and tax regime applies per deployment (per-market deployments are acceptable for launch).
4. Sellers ship their own orders at launch unless decided otherwise (OI-02).
5. Third-party services (identity, PSP, search, e-mail, media) will be procured rather than built.

3.5 Constraints

1. The front end is built with Next.js (App Router), React 19, TypeScript and Tailwind CSS; requirements must be implementable within this stack.
2. Configuration values prefixed NEXT_PUBLIC_ are inlined at build time; changing them requires a rebuild/redeploy.
3. Card data must never touch HexCommerce servers (PCI DSS SAQ A scope).
4. Accessibility conformance (WCAG 2.2 AA) is a release gate, not an enhancement.

3.6 Dependencies

- Identity provider (customer, seller, admin roles).
- Payment service provider with hosted card fields and 3-D Secure.
- Order management and inventory services.
- Search service supporting relevance, facets and type-ahead.
- Transactional e-mail/notification service.
- Media storage and CDN for product imagery.

4. Business Case and Delivery Plan

This section frames the investment decision. Monetary values have deliberately **not** been invented: baselines, costs and budget are marked “To be estimated” and must be supplied by Finance and Engineering during Phase 0 (validation). Durations are indicative.

4.1 Expected Benefits

Table 6: Benefits register

Benefit	Measure	Baseline	Proposed target	Owner
Revenue from marketplace sales	Gross merchandise value (GMV); referral-fee revenue	To be established	Set in Phase 0	Executive Sponsor
Higher conversion	Visit-to-order conversion; checkout completion	To be established	BO-01 (proposed)	Product Owner
Larger baskets	Average order value; deal and voucher attach rate	To be established	BO-02 (proposed)	Merchandising
Assortment growth without inventory risk	Active sellers; live listings; time to first listing	Zero (new capability)	BO-03 (proposed)	Seller Operations
Lower cost-to-serve	Contacts per 100 orders; % self-service; cost per contact	To be established	BO-04 (proposed)	CS Operations
Faster launches for new brands/markets	Days to launch a re-branded storefront	To be established	BO-06 (proposed)	Engineering Lead
Reduced regulatory and fraud exposure	Chargeback rate; compliance findings	To be established	Within PSP and audit thresholds	Legal / Security

4.2 Cost Categories

Table 7: Cost model (to be estimated)

Category	Basis of estimate	One-off	Recurring	Owner
Delivery team	Team size × duration of Phases 0–5 (Section 4.4)	To be estimated	—	Engineering Lead
Third-party services	Identity, search, notifications, media/CDN, fraud, KYC/KYB – vendor quotes at forecast volumes (Section 11.2)	Set-up fees	Subscription / usage	Procurement
Payment processing	PSP fee per transaction × forecast orders; chargeback fees	—	Usage-based	Finance
Infrastructure	Hosting, databases, environments, monitoring at forecast load	To be estimated	Monthly	Engineering Lead
Assurance	Accessibility audit, penetration test, PCI SAQ A, legal review	To be estimated	Annual re-tests	Security / Legal
Operations	CS agents, Trust & Safety moderators, seller onboarding staff, training	Training	Headcount	CS Ops / Seller Ops
Contingency	Typically 10–20% of one-off costs, set by the Sponsor	To be set	—	Executive Sponsor

4.3 Budget and Funding Approval

Budget is **to be confirmed** by the Executive Sponsor at the end of Phase 0, once estimates and vendor quotes are available. Funding is released per phase against the gate criteria in Section 4.5; the business case is re-validated at each gate against the benefits register.

4.4 Release Plan

The plan sequences the R2 capabilities so that each phase delivers a usable increment. Durations are **indicative** and assume one cross-functional delivery team per phase; they must be re-estimated in Phase 0.

Table 8: Indicative release plan

Phase	Scope	Indicative duration	Outcome / exit criterion	Modules
Phase 0 – Validation	Stakeholder workshops; validate objectives, KPIs, priorities, sizing and open issues; select vendors (identity, PSP, search, notifications).	4 weeks	BRD v2.0 approved; vendor shortlist	All
Phase 1 – Foundation	Identity and roles, server-side persistence for basket/lists/reviews/support, catalogue and search services, seller listings in catalogue.	8–10 weeks	Customers sign in; data survives devices	AUT, CRT, REV, CS, SRC, SLS, UAD (part)
Phase 2 – Commerce	Payments, order management, tax, multi-seller orders and shipping, notifications, invoices, fraud screening.	10–12 weeks	Real orders placed end to end (internal pilot)	CHK, ORD, MSO, NTF, FRD, CFG
Phase 3 – Marketplace	KYC/KYB, payouts and settlement, moderation, seller order fulfilment with tracking, seller deals purchasable.	8–10 weeks	First external sellers paid	SEL, ADM, PAY, MOD, SOR, SDL, DEA
Phase 4 – Service & growth	Returns and refunds, CS agent console, messaging, vouchers, CMS, SEO.	8–10 weeks	Self-service returns live; marketing self-sufficient	RET, CSA, MSG, PRM, CMS, SEO
Phase 5 – Launch readiness	Accessibility audit, penetration test, PCI attestation, performance test, UAT, training, cutover rehearsal.	4–6 weeks	Go/no-go approval	All
Hypercare	Heightened support after go-live; daily defect triage; KPI monitoring.	4 weeks	Exit criteria met; handover to BAU	All

4.5 Stage Gates

Gate	When	Decision criteria	Decision maker
G0 – Mandate	Start of Phase 0	Problem statement and sponsor confirmed	Executive Sponsor
G1 – Requirements baseline	End of Phase 0	BRD validated and approved; estimates and budget; vendors selected	Executive Sponsor
G2 – Commerce ready	End of Phase 2	Internal pilot orders placed, paid and refunded end to end	Product Owner
G3 – Marketplace ready	End of Phase 3	Pilot sellers onboarded, verified and paid	Product Owner
G4 – Go / no-go	End of Phase 5	All Must requirements accepted; audits passed; UAT signed off; cutover rehearsed	Executive Sponsor

Gate	When	Decision criteria	Decision maker
G5 – Hypercare exit	Go-live + 4 weeks	Hypercare exit criteria met (Section 17.5)	Product Owner

4.6 Roles and Responsibilities (RACI)

R = Responsible, A = Accountable, C = Consulted, I = Informed. Roles: SP = Executive Sponsor; PO = Product Owner; BA = Business Analyst; UX = UX Lead; ENG = Engineering Lead; QA = QA Lead; SO = Seller Operations; CS = Customer Service Operations; FIN = Finance; LGL = Legal / Data Protection Officer; SEC = Security.

Table 9: RACI matrix

Activity	SP	PO	BA	UX	ENG	QA	SO	CS	FIN	LGL	SEC
Approve BRD and scope	A	R	C	C	C	C	C	C	C	C	C
Prioritise backlog	I	A/R	C	C	C	I	C	C	I	I	I
Maintain requirements & traceability	I	A	R	C	C	C	I	I	I	I	I
UX design & accessibility	I	A	C	R	C	C	I	I	I	I	I
Solution architecture & build	I	C	C	C	A/R	C	I	I	I	I	C
Test strategy & execution	I	C	C	I	C	A/R	I	I	I	I	C
User acceptance testing	I	A	R	C	C	R	R	R	R	I	I
Fees, tax and payout rules	A	C	C	I	C	I	C	I	R	C	I
Compliance sign-off (GDPR, PCI, consumer law)	A	C	C	I	C	I	I	I	C	R	R
Seller onboarding operations	I	C	I	I	I	I	A/R	I	C	C	C
Customer service operations & training	I	C	I	I	I	I	I	A/R	I	C	I
Go / no-go decision	A	R	C	C	C	C	C	C	C	C	C

5. Solution Overview

5.1 System Context

The context diagram shows the people who use the platform (solid lines) and the external systems it depends on in production (dashed lines, see Section 14). In the R1 baseline none of the external systems is connected.

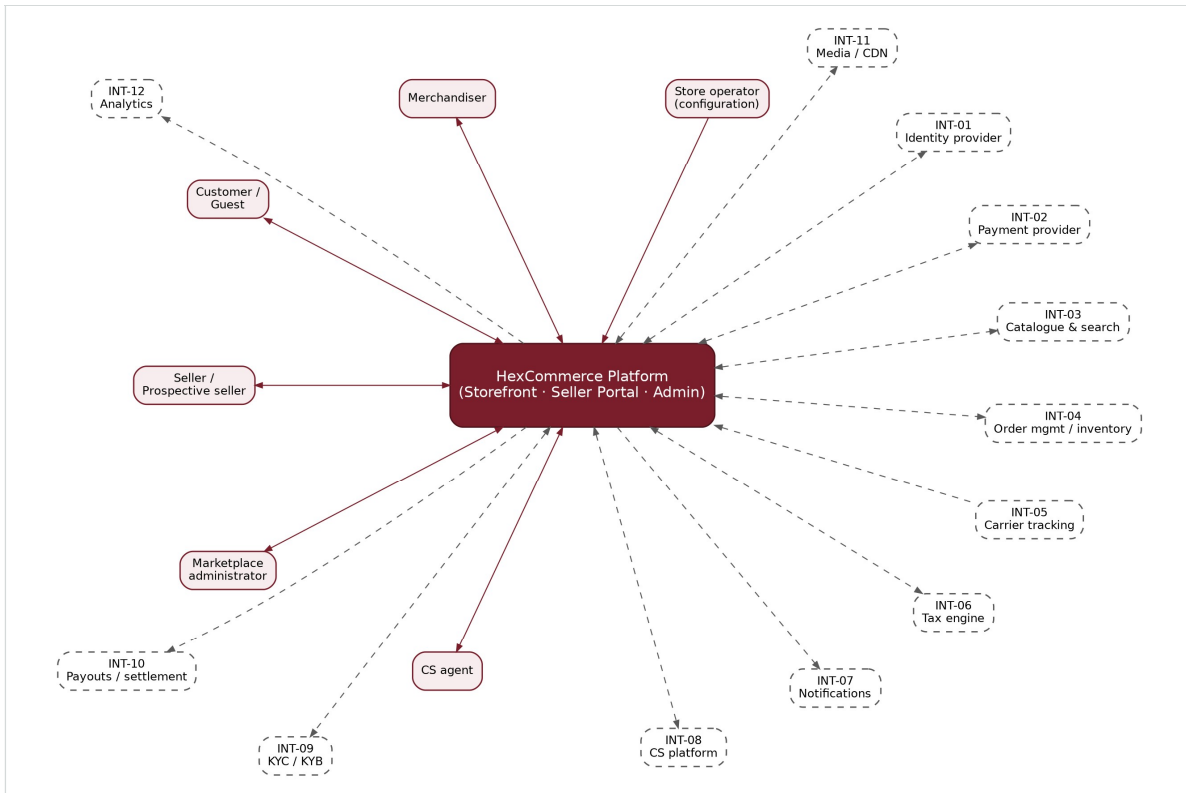


Figure 1: System context diagram (R2 target)

5.2 Information Architecture

The solution comprises 27 routes plus a not-found page. The site map below lists every screen, its route and its audience.

Table 10: Site map and screen inventory

#	Screen	Route	Audience	Module
1	Home	/	All	HOM
2	Department listing	/category/{slug}	All	CAT
3	Search results / All products	/search?q=...	All	SRC
4	Product detail	/product/{slug}	All	PDP, REV
5	Today's Deals	/deals	All	DEA
6	Shopping cart	/cart	Customer	CRT
7	Checkout	/checkout	Customer	CHK
8	Wish list	/wishlist	Customer	WSH

#	Screen	Route	Audience	Module
9	Your orders	/orders	Customer	ORD
10	Order details	/orders/{id}	Customer	ORD
11	Your account	/account	Customer	ACC
12	Customer service	/customer-service	Customer	CS
13	Sign in	/signin	Guest	AUT
14	Create account	/register	Guest	AUT
15	About	/about	All	GLB
16	Sell on HexCommerce	/sell	Prospective seller	SEL
17	Register as a seller	/sell/apply	Prospective seller	SEL
18	Seller dashboard	/sell/dashboard	Seller	SDB
19	Your listings	/sell/listings	Seller	SLS
20	Add a product	/sell/listings/new	Seller	SLS, SIM
21	Edit listing	/sell/listings/{id}/edit	Seller	SLS, SIM
22	Your deals	/sell/deals	Seller	SDL
23	Create a deal	/sell/deals/new	Seller	SDL
24	Edit deal	/sell/deals/{id}/edit	Seller	SDL
25	Seller orders	/sell/orders	Seller	SOR
26	Business details	/sell/register	Seller	SBD
27	Seller applications (admin)	/admin/sellers	Administrator	ADM
—	Page not found	(any unknown route)	All	GLB

5.3 Navigation Model

Navigation is provided through five complementary mechanisms, ensuring every destination is reachable within two interactions from any page:

- **Header** – logo, deliver-to, search, Account & Lists flyout, Returns & Orders, Cart.
- **Department strip** – the eight departments, Today's Deals and Customer Service.
- **"All" slide-over menu** – full department list, account links and seller links.
- **Footer** – corporate, selling, delivery/returns and help links; department index; admin entry point.
- **Mobile tab bar** – Home, Deals, Saved, Orders, Cart on phones.
- **Breadcrumbs** – on all interior pages for hierarchical way-finding.

5.4 Responsive Behaviour

The layout adapts across breakpoints. On phones the search field moves to its own row, the department strip scrolls horizontally, filters move to a drawer, the order summary moves above the basket lines, and the bottom tab bar appears.

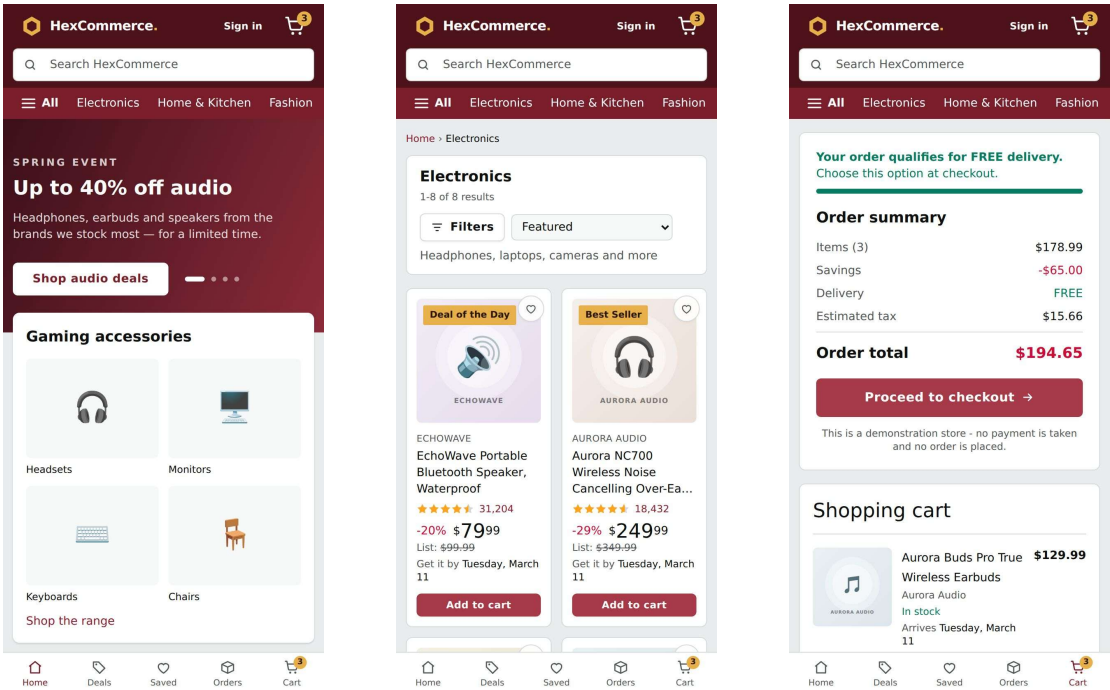


Figure 2: Phone layouts: home, department listing with Filters button, and basket with bottom tab bar

6. Business Processes

The following end-to-end processes describe how actors achieve business outcomes using the solution. Each step references the functional requirements that support it; flowcharts are provided for the principal processes.

6.1 Discover and evaluate a product (Customer)

Actors	Customer
Outcome	A product is added to the basket or wish list.

1. Customer lands on Home and engages with the hero, deals strip or a product rail (FR-HOM-01...09); or selects a department (FR-GLB-05).
2. Customer narrows results with facets and sort; filtered view is reflected in the URL (FR-CAT-02...08).
3. Alternatively the customer searches, optionally choosing a type-ahead suggestion (FR-SRC-01...05).
4. Customer opens a product, reviews gallery, price, delivery promise, stock, features, specifications and reviews (FR-PDP-01...07, FR-REV-01).
5. Customer selects a quantity and chooses Add to cart, Buy now or Save for later (FR-PDP-08, FR-WSH-01).

6.2 Purchase (Customer)

Actors	Customer
Outcome	Order placed and confirmed; basket emptied.

1. Customer opens the basket, adjusts quantities, saves items for later and reviews totals and free-delivery progress (FR-CRT-01...05).
2. Customer proceeds to checkout (FR-CRT-04).
3. Customer confirms delivery address, payment method and delivery speed (FR-CHK-03...05).
4. Customer reviews items and totals and places the order (FR-CHK-06, FR-CHK-07).
5. R2: payment is authorised, the order is created, stock reserved and confirmation e-mailed (FR-CHK-08).

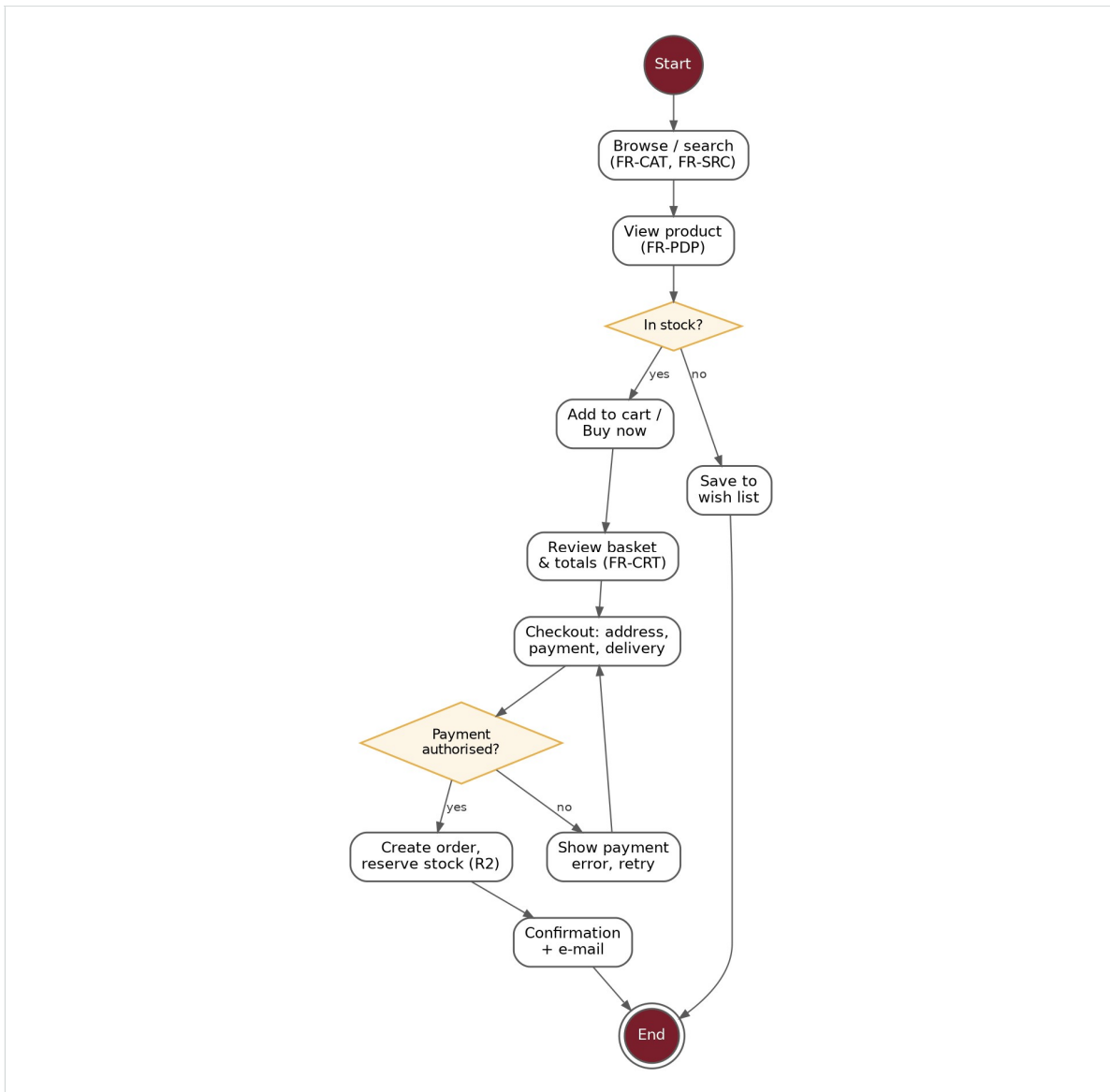


Figure 3: Process flow: purchase

6.3 Track an order and obtain help (Customer)

Actors	Customer
Outcome	Issue resolved or return initiated.

1. Customer opens Your orders and selects an order to see the tracker (FR-ORD-01, FR-ORD-03).
2. If there is a problem, customer opens Customer service, selects the item and issue type and submits (FR-CS-01...03).
3. Customer receives an acknowledgement and follows the thread, replying when asked (FR-CS-04...06).
4. Support resolves the issue; the customer may reopen it by replying (BR-CS-01).

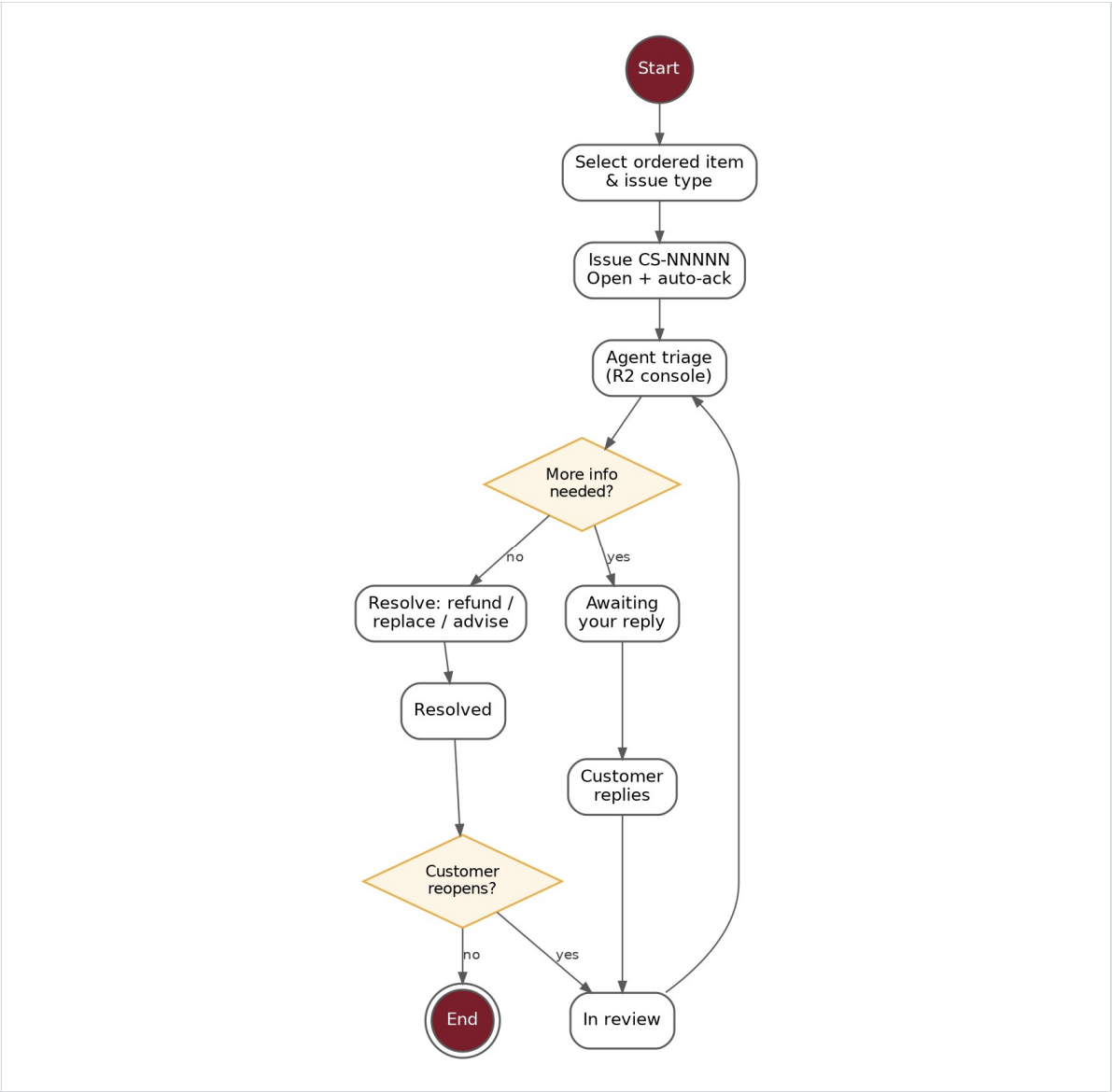


Figure 4: Process flow: support issue handling

6.4 Write a review (Customer)

Actors	Customer
Outcome	Review visible on the product page and aggregate updated.

1. Customer opens the product and selects “Write a customer review” (FR-REV-02).
2. Customer rates, adds a headline and body; the form validates (FR-REV-03, BR-REV-01).
3. Review appears at the top as “Your review” with verified status where applicable (FR-REV-04, FR-REV-05).

6.5 Seller onboarding (Prospective seller, Administrator)

Actors	Prospective seller; Administrator
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Outcome	Seller approved and able to list; or rejected with reason.
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1. Prospect reads the Sell page and selects “Create a seller account” (FR-SEL-01).
2. Prospect completes and submits the application (FR-SEL-02...04).
3. Administrator filters the queue to Pending and reviews the application (FR-ADM-01, FR-ADM-02).
4. Administrator approves, or rejects with a reason (FR-ADM-03).
5. Prospect sees the decision on /sell and /sell/apply; if rejected, may apply again with the form pre-filled (FR-SEL-05...07).

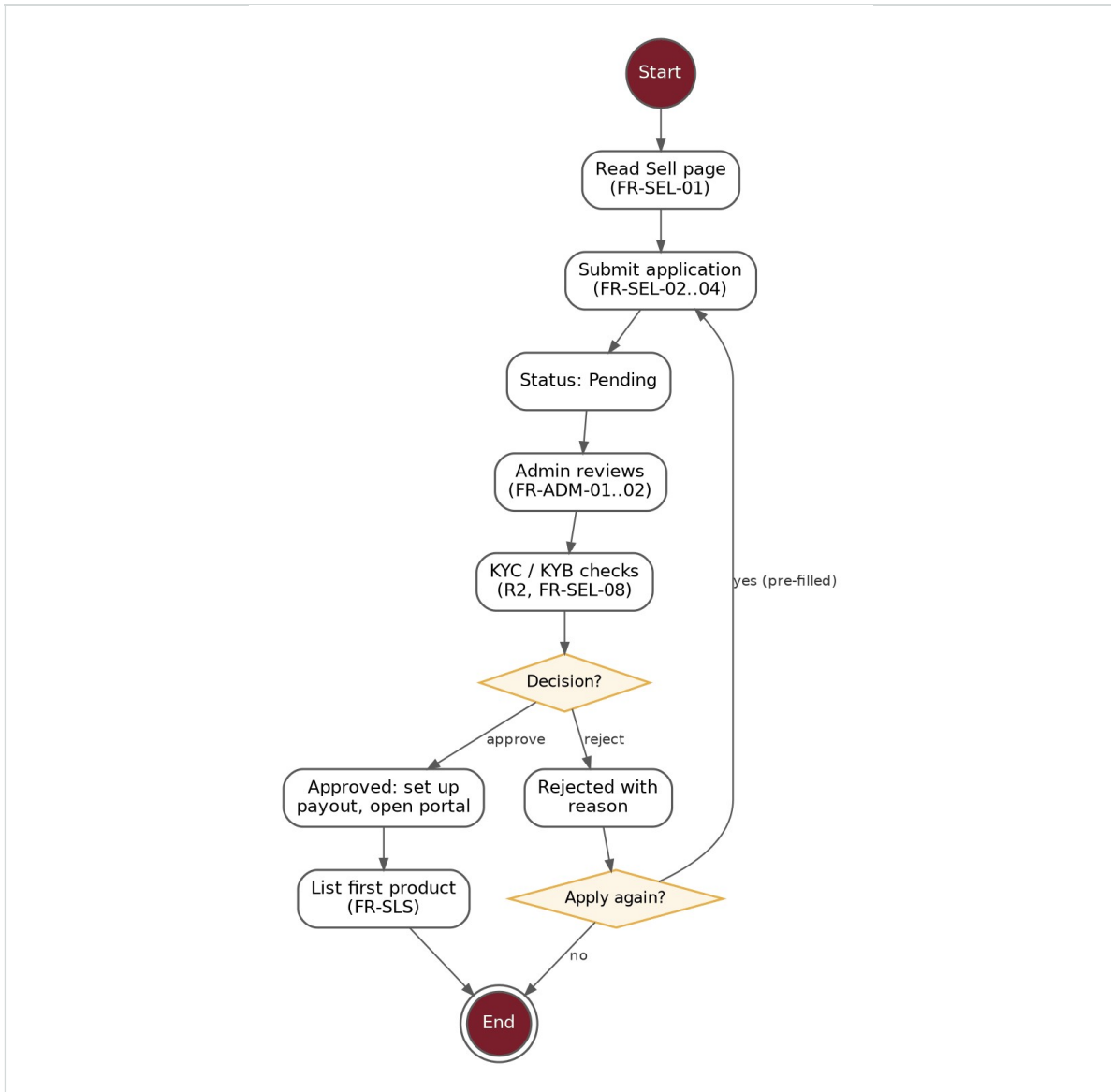


Figure 5: Process flow: seller onboarding

6.6 List a product (Seller)

Actors	Seller
Outcome	Listing Active (or Draft / Out of stock).

1. Seller selects “Add a product” from the dashboard, flyout or menu (FR-SDB-01).
2. Seller completes Product details including images (FR-SLS-04, FR-SIM-01...03).
3. Seller sets pricing and stock, then description and features (FR-SLS-05, FR-SLS-06).
4. Seller reviews and publishes or saves as draft; status set per BR-SLR-02 (FR-SLS-07).
5. Seller manages the listing from Your listings: edit, unpublish, publish, remove (FR-SLS-01, FR-SLS-02).

6.7 Run a deal (Seller)

Actors	Seller
Outcome	Deal Live (visible to customers) or Scheduled.

1. Seller opens Your deals and selects “Create a deal” (FR-SDL-01).
2. Seller chooses an Active listing, discount, units and dates, watching the live preview (FR-SDL-03, FR-SDL-04).
3. Seller starts the deal; it is Live or Scheduled per start date (BR-SLR-03).
4. Live deal appears in “Deals from our sellers” on /deals (FR-DEA-03).
5. Seller may start a scheduled deal early or end any deal (FR-SDL-02).

6.8 Fulfil an order (Seller)

Actors	Seller
Outcome	Order Delivered.

1. Seller reviews New orders in Seller orders (FR-SOR-01).
2. Seller packs and ships, then selects “Confirm dispatch” (FR-SOR-02).
3. On delivery, seller selects “Mark delivered” (FR-SOR-02). R2 captures tracking and notifies the buyer (FR-SOR-03).

6.9 Return an item (Customer, Seller) – R2

Actors	Customer; Seller; CS agent
Outcome	Refund issued or replacement dispatched; or return rejected with reason.

1. Customer opens a delivered order and selects “Return or replace items” within the return window (FR-RET-01).
2. Customer selects items, reason, photos and preferred resolution (FR-RET-02).
3. System issues a QR code or label for the chosen method (FR-RET-03).
4. Seller (or marketplace) receives and inspects the item within SLA (FR-RET-06).
5. Refund is issued to the original payment method or a replacement dispatched; cost allocated in settlement (FR-RET-05, FR-RET-07).

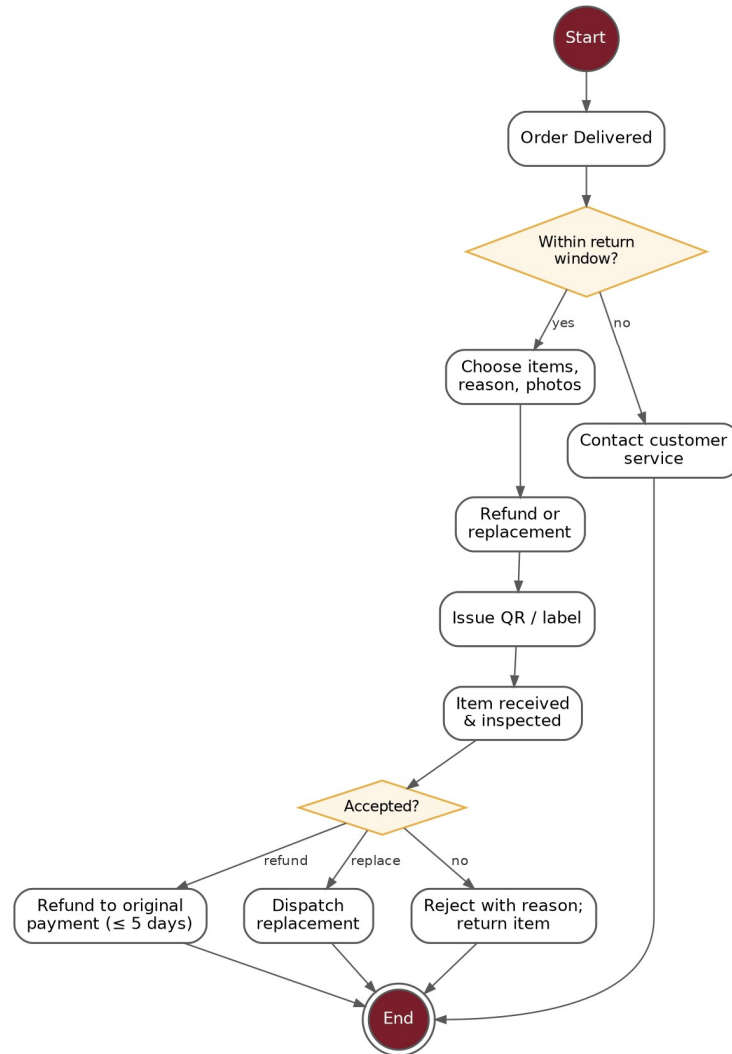


Figure 6: Process flow: returns and refunds (R2)

7. Functional Requirements Overview

Functional requirements are grouped into 35 modules: 23 modules describing the current UI (with R2 extensions) and 12 production-only (R2) modules that have no screen yet. The table summarises requirement counts and priority mix; detailed requirements follow in Section 8.

Table 11: Functional requirements by module

Code	Module	Total	Must	Should	Could	In R1
GLB	Global Navigation and Site Chrome	12	6	5	1	12
HOM	Home Page and Merchandising	11	3	7	1	10
CAT	Catalogue Browsing, Filtering, Sorting and Pagination	12	10	2	0	12
SRC	Search and Type-ahead	8	4	4	0	7
PDP	Product Detail Page	10	7	2	1	10
REV	Ratings and Customer Reviews	8	5	3	0	7
WSH	Wish List	4	4	0	0	4
CRT	Shopping Cart and Saved for Later	8	7	1	0	8
CHK	Checkout and Order Placement	10	9	1	0	7
ORD	Order History and Tracking	6	3	2	1	3
ACC	Customer Account Hub	5	4	1	0	3
CS	Customer Service Help Desk	9	7	2	0	7
DEA	Deals and Promotions (Customer-facing)	5	2	3	0	3
AUT	Customer Identity: Sign-in and Registration	5	5	0	0	2
SEL	Seller Recruitment and Application	8	6	2	0	7
ADM	Marketplace Administration: Seller Application Review	5	4	1	0	4
SDB	Seller Dashboard	6	2	4	0	5
SLS	Seller Listing Management	12	9	3	0	10
SIM	Seller Product Image Upload	5	4	1	0	4
SDL	Seller Deal Management	7	4	3	0	6
SOR	Seller Order Fulfilment	4	3	1	0	2
SBD	Seller Business Details	2	1	1	0	1
CFG	Configuration and White-labelling	5	4	1	0	4
MSO	Multi-seller Orders, Shipping Rules and Zones	7	6	1	0	0
PRM	Vouchers and Promotion Codes	6	4	2	0	0
RET	Returns and Refunds	7	6	1	0	0
PAY	Seller Payouts and Settlement	6	5	1	0	0
FRD	Fraud and Risk Management	5	3	2	0	0

Code	Module	Total	Must	Should	Could	In R1
MSG	Buyer–Seller Messaging	5	3	2	0	0
NTF	Notifications (E-mail and In-app)	5	3	2	0	0
CSA	Customer Service Agent Console	6	5	1	0	0
MOD	Catalogue, Review and Content Moderation	5	4	1	0	0
UAD	User, Role and Order Administration	5	5	0	0	0
CMS	Merchandising Content Management	5	3	2	0	0
SEO	Search Engine Optimisation and Discoverability	5	3	2	0	1
Total		234	163	67	4	139

8. Functional Requirements

Each module lists its purpose, a reference screenshot of the current UI (or, for R2 modules, an annotated low-fidelity wireframe) where applicable, and its requirements with priority, acceptance criteria and release designation. Screens for additional states (errors, empty states, confirmations) are in Appendix A.

8.1 Global Navigation and Site Chrome (GLB)

The persistent header, department strip, slide-over menu, footer and mobile tab bar that frame every page. These elements provide consistent way-finding and give constant access to search, account, orders and basket.

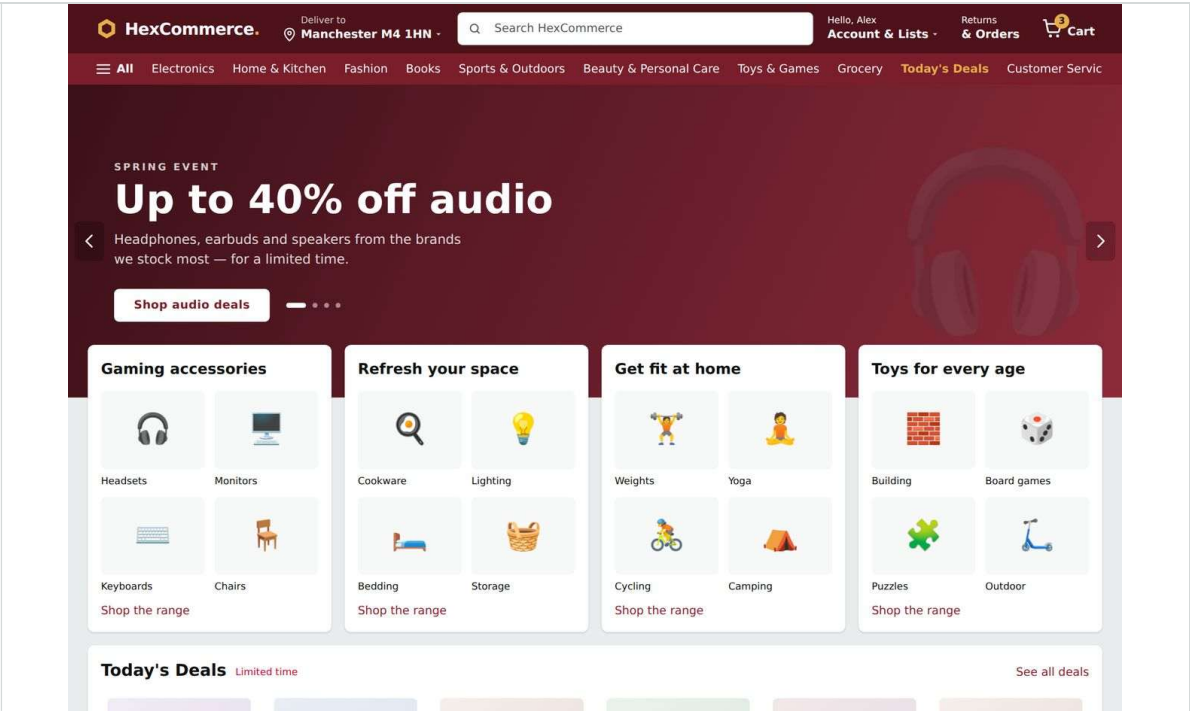


Figure 7: Home page with persistent header, deliver-to control, search, account flyout trigger, cart counter and department strip

Table 12: Global Navigation and Site Chrome requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-GLB-01	The system shall display a persistent header on every page containing: the brand logo (links to Home), a “Deliver to” address control, the search field, the “Account & Lists” control, a “Returns & Orders” link and the Cart link.	M	Header is present on all 27 routes, including the 404 page; logo navigates to “/”.	R1
FR-GLB-02	The “Deliver to” control shall display the city and postcode of the selected delivery address and open a picker listing all saved addresses (label, name, line 1, city, postcode) as single-select options, plus an “Add or edit an address” link.	S	Selecting an address updates the header label immediately and the choice survives a page refresh in the same browser. Picker closes on selection, Escape, or an outside click.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-GLB-03	The “Account & Lists” flyout shall greet the customer by first name and present: Sign in, Register; Your Lists (Wish list, Saved for later); Your Account (Account, Orders, Customer service); Selling (Seller dashboard, Add a product, Deals).	S	Flyout opens on pointer hover and on click/keyboard activation; closes on Escape or outside click; exposes role="menu" with menu items.	R1
FR-GLB-04	The Cart link shall show a badge with the total number of units in the basket (sum of line quantities) and an accessible name of the form “Shopping cart, N items”.	M	Adding 2 units of one item and 1 of another shows “3”. Badge shows 0 until the persisted basket has loaded (no flash of incorrect count).	R1
FR-GLB-05	A department strip shall list all eight departments followed by “Today’s Deals” and “Customer Service”. On narrow viewports the strip shall scroll horizontally without widening the page.	M	No horizontal page scroll at 320 px width; each item links to its route.	R1
FR-GLB-06	An “All” button shall open a modal slide-over menu containing: Shop by department (with glyphs), Programmes & features, Your account (Account, Orders, Wish list, Customer service, Sign out) and Selling (Dashboard, Listings, Add a product, Deals, Seller orders).	S	Menu is a dialog with aria-modal; closes on Escape, scrim click, close button, or any link; background scroll is locked while open.	R1
FR-GLB-07	The footer shall contain four link columns (Get to know us; Make money with us; Payment & delivery; Let us help you), a brand bar with the configured tagline, a department index, a legal line with the current year, and the active theme name.	M	All footer links resolve to an existing route; site name and tagline reflect configuration.	R1
FR-GLB-08	On phone-width viewports (below the “md” breakpoint) a fixed bottom tab bar shall provide Home, Deals, Saved, Orders and Cart, with a unit-count badge on Cart and the active tab indicated.	S	Active tab carries aria-current="page"; tab bar is hidden at ≥ 768 px; page content is not obscured (bottom padding applied).	R1
FR-GLB-09	A “Skip to main content” link shall be the first focusable element on every page.	M	First Tab press reveals the link; activating it moves focus to the main landmark.	R1
FR-GLB-10	Unknown routes, and unknown product, department or order identifiers, shall render a branded “We could not find that page” page with links to Home, Today’s Deals and every department.	M	HTTP 404 status returned; page offers at least 10 onward links.	R1
FR-GLB-11	Every page shall have a unique document title in the form “<Page> <Site name>” (home: “<Site name> - <Tagline>”) and a meta description.	S	Titles verified for all routes; product and department titles are derived from their records.	R1
FR-GLB-12	The browser UI theme colour shall follow the header colour of the active theme.	C	Mobile browser chrome matches the header colour for each preset.	R1

8.2 Home Page and Merchandising (HOM)

The home page is the primary merchandising surface. It combines promotional content with algorithmically curated product rails to drive discovery and deal uptake.

Table 13: Home Page and Merchandising requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-HOM-01	The home page shall display a hero carousel of promotional slides, each with an eyebrow, headline, body copy, call-to-action and gradient artwork.	S	Four slides are shown; each CTA navigates to its configured destination (e.g. “Shop audio deals” → Electronics › Audio).	R1
FR-HOM-02	The carousel shall auto-advance every 7 seconds, pause while hovered or while a control has focus, and shall not auto-advance when the user has requested reduced motion.	M	With prefers-reduced-motion: reduce the slide never changes on its own. Previous/next arrows (≥ 32px) and per-slide indicators allow manual navigation.	R1
FR-HOM-03	Four promotional tiles, each showing four themed entry points and a “Shop the range” link, shall overlap the lower edge of the hero.	S	Tiles render above the hero on all breakpoints and remain clickable.	R1
FR-HOM-04	A “Today’s Deals” strip shall show six featured deals with percentage-off badge, remaining-time label, product title and price, plus a “See all deals” link.	M	Six cards displayed; each links to its product detail page.	R1
FR-HOM-05	A “Best sellers across the store” rail shall show the 12 highest-ranked products using the popularity score defined in BR-CAT-01.	M	Order matches the score; “Browse all” opens search sorted by Most Reviewed.	R1
FR-HOM-06	A “Biggest savings this week” rail shall show up to 12 discounted products ordered by percentage saving, descending.	S	Only products with list price > selling price appear.	R1
FR-HOM-07	Two feature blocks (“Top rated in Electronics”, “Popular in Home & Kitchen”) shall each list four products with rating, review count, price and “Save N%” where applicable.	S	Each block has a “See more” link to its department.	R1
FR-HOM-08	Additional rails shall show “More in Electronics” (10 items) and “Recently viewed”, followed by a “Shop by department” directory.	S	All eight departments are listed in the directory.	R1
FR-HOM-09	Product rails shall scroll horizontally with snap alignment. On desktop, left/right arrow buttons shall scroll by ~85% of the visible width and hide when no further scrolling is possible; on touch devices the rail is swiped.	S	Left arrow hidden at scroll start; right arrow hidden at scroll end.	R1
FR-HOM-10	An editorial banner shall promote free next-day delivery with a call-to-action to Today’s Deals.	C	Banner uses theme brand colour and inverted button.	R1
FR-HOM-11	“Recently viewed” shall reflect the customer’s actual browsing history rather than a fixed list.	S	Viewing a product places it first in the rail on the next home page visit.	R2

8.3 Catalogue Browsing, Filtering, Sorting and Pagination (CAT)

Department listing pages and the search results page share one results layout. All facet, sort and page state is carried in the URL so every view is shareable, bookmarkable and correctly handled by browser back/forward navigation.

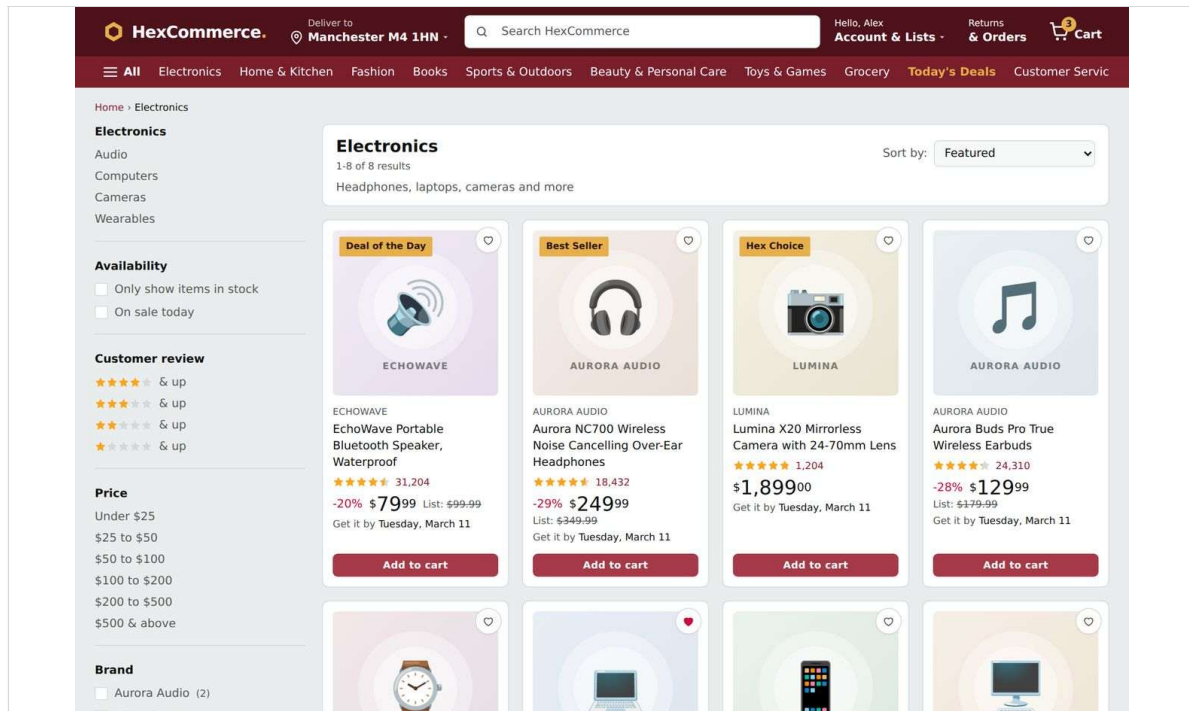


Figure 8: Department listing (Electronics) with filter sidebar, sort control, result count and product grid

Table 14: Catalogue Browsing, Filtering, Sorting and Pagination requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CAT-01	The system shall provide a listing page for each department at /category/{slug}, showing the department name, descriptive blurb and breadcrumb (Home › Department).	M	All eight departments render; an unknown slug returns the 404 page.	R1
FR-CAT-02	A filter sidebar shall offer the facets: Sub-department (single-select toggle), Availability (“Only show items in stock”, “On sale today”), Customer review (4, 3, 2, 1 stars & up), Price (six bands) and Brand (multi-select with counts, shown only when more than one brand is present), plus “Clear all filters”.	M	Each facet narrows results correctly per BR-CAT-03; selecting an active facet again removes it.	R1
FR-CAT-03	Brand facet counts shall be computed before the brand filter is applied so that ticking one brand does not hide the others.	S	With “Brand A” ticked, “Brand B (n)” remains visible and selectable.	R1
FR-CAT-04	All filter, sort and pagination state shall be represented in URL query parameters (q, category, subcategory, brand, min, max, rating, stock, deals, sort, page).	M	Copying the URL into a new browser reproduces the same result set and order.	R1
FR-CAT-05	Changing any filter or the sort order shall reset pagination to page 1.	M	Applying a filter from page 3 lands on page 1.	R1
FR-CAT-06	Active filters shall be displayed as removable chips above the results, each removing only its own filter, with a “Clear all” action.	M	Chip labels: sub-department name, brand, price range, “N stars & up”, “In stock”, “On sale”.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-CAT-07	A “Sort by” control shall offer: Featured (default), Price: Low to High, Price: High to Low, Avg. Customer Review, Most Reviewed, Newest Arrivals.	M	Sorting follows BR-CAT-02; Featured removes the sort parameter from the URL.	R1
FR-CAT-08	Results shall be paginated at 12 per page with a result-range label (“13-24 of 41 results”), numbered page links, and Previous/Next controls that are disabled at the boundaries.	M	For more than 7 pages, the first, last, current and adjacent pages are shown with ellipses; out-of-range page numbers are clamped.	R1
FR-CAT-09	Below the “lg” breakpoint the sidebar shall be replaced by a “Filters” button (with a badge showing the number of active filters) that opens a modal drawer; selecting a facet closes the drawer.	M	Drawer closes on Escape, scrim tap, close button or “Show results”; background scroll is locked.	R1
FR-CAT-10	When no products match, an empty state shall suggest removing filters or checking spelling and offer “Clear filters” and “Back to home”.	M	Empty state shown for an impossible combination (e.g. price $\geq$ \$500 in Grocery).	R1
FR-CAT-11	A skeleton placeholder shall be displayed while a listing or search segment is loading.	S	No blank screen during navigation between listing pages.	R1
FR-CAT-12	Each product card shall show: wish-list heart, image, merchandising badge, “Out of stock” overlay when stock = 0, brand, title (clamped), star rating and count, price with discount badge and struck list price, “Get it by <date>” (non-compact cards, in-stock only) and an “Add to cart” button.	M	Out-of-stock cards show a disabled “Unavailable” button. After adding, the button shows “Added ✓” for ~1.6 s.	R1

8.4 Search and Type-ahead (SRC)

Free-text search is available from every page. It combines an accessible type-ahead with relevance-ranked results that reuse the catalogue results layout and facets.

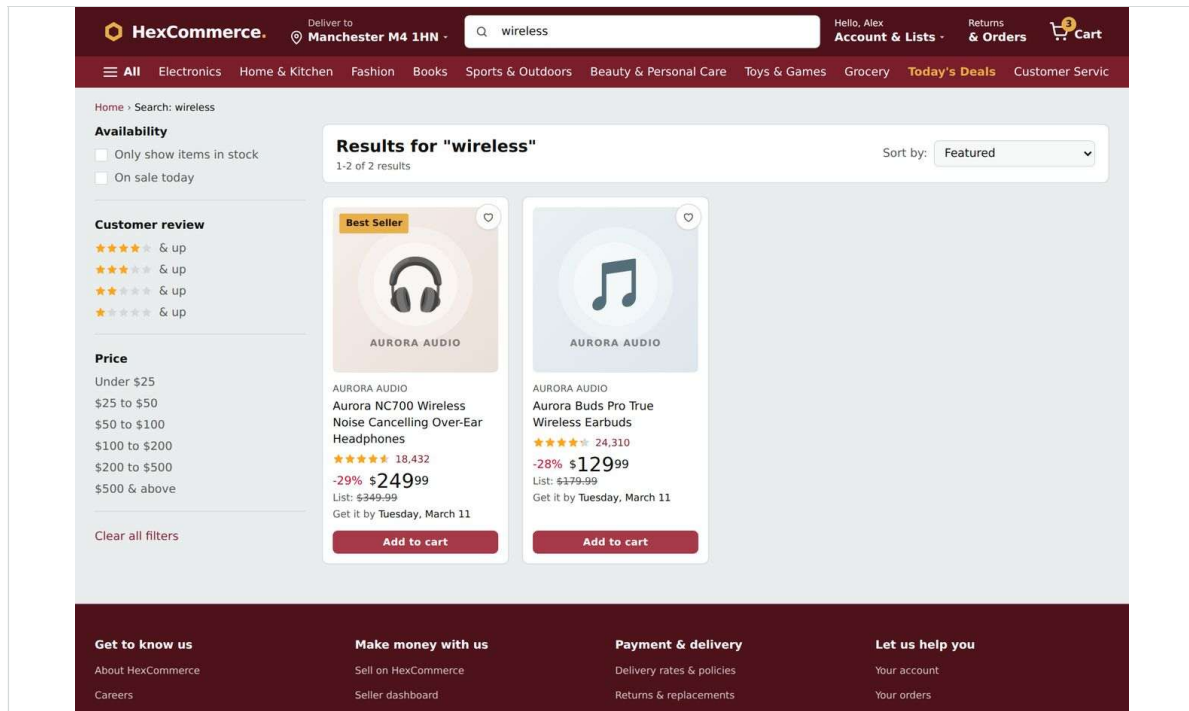


Figure 9: Search results for “wireless” reusing the shared results layout

Table 15: Search and Type-ahead requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SRC-01	A search field shall be available in the header on all pages (on its own row on phones) with the placeholder “Search <Site name>”. Submitting with Enter or the search icon shall open /search?q=<term>; an empty submission shall open the all-products listing.	M	Leading/trailing whitespace is trimmed before navigation.	R1
FR-SRC-02	After 2 or more characters the field shall display up to 8 type-ahead suggestions drawn from product titles, brands and tags.	M	Suggestions update on every keystroke with no perceptible delay.	R1
FR-SRC-03	Suggestions shall be keyboard operable: ArrowDown/ArrowUp move the highlight (wrapping), Enter selects the highlighted suggestion, Escape closes the list; an outside click closes the list.	M	Field exposes role="combobox" with aria-expanded and aria-controls; list is role="listbox" with role="option" items.	R1
FR-SRC-04	The search field shall reflect the current q parameter, including after back/forward navigation.	S	Navigating back from a result page restores the previous term in the field.	R1
FR-SRC-05	Search shall return only products in which every query term appears in the title, brand, summary, tags or department, ranked by relevance per BR-SRC-01 with average rating as tie-breaker.	M	“wireless headphones” returns only products containing both terms.	R1
FR-SRC-06	Relevance order shall be preserved unless the customer explicitly selects a non-Featured sort.	S	Choosing “Price: Low to High” re-orders; returning to Featured restores relevance order.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-SRC-07	Search results shall support an optional department scope; when present the breadcrumb shall include the department and a subtitle “Searching within <Department>.” shall be shown.	S	/search?q=x&category=books shows Books in the breadcrumb.	R1
FR-SRC-08	Search should tolerate common misspellings and support synonyms and zero-result recovery suggestions.	S	“hedphones” returns headphone results.	R2

8.5 Product Detail Page (PDP)

The product detail page (PDP) presents everything a customer needs to make a purchase decision, following the familiar marketplace three-column layout: gallery, details, buy box.

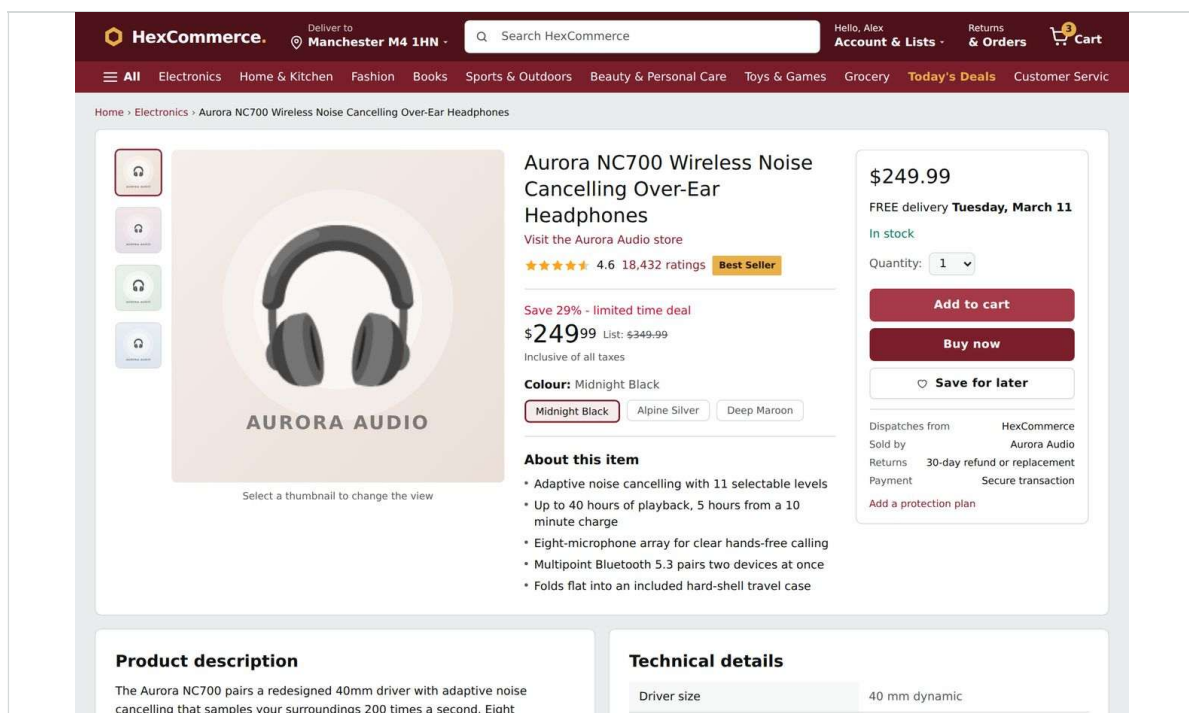


Figure 10: Product detail page: gallery, details column and buy box

Table 16: Product Detail Page requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-PDP-01	Each product shall have a detail page at /product/{slug} with breadcrumb Home › Department › Product title.	M	Unknown slug returns 404; page title is the product title; meta description is the summary.	R1
FR-PDP-02	A gallery shall show a main image and a thumbnail strip (vertical on desktop, horizontal on mobile). Thumbnails shall change the main image on hover, focus or tap.	M	Four generated views are shown when a product has no photographs; when real images exist, one thumbnail per image.	R1
FR-PDP-03	The details column shall show the title, a “Visit the <Brand> store” link, star rating with numeric value, a “N ratings” link that jumps to reviews, and any merchandising badge.	M	Brand link opens a search for the brand name.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-PDP-04	The price block shall show “Save N% - limited time deal” when discounted, the price in large superscript format, the struck-through list price, and a tax-inclusion statement.	M	Discount % = round((list – price) / list × 100).	R1
FR-PDP-05	Where a product has colour or size variants, selectable variant chips shall be displayed with the current selection named.	S	R1: the first option is shown as selected (presentational). R2: selecting a variant updates price, stock, images and the item added to the basket.	R1/R2
FR-PDP-06	An “About this item” bullet list, a “Product description” section and a “Technical details” table (all specification rows plus Brand and Item model number) shall be displayed.	M	Table uses row headers (th scope="row").	R1
FR-PDP-07	The buy box shall show: price; delivery message (FREE delivery with next-day date when price ≥ free-delivery threshold, otherwise the fee and threshold); stock status per BR-INV-01; quantity selector (1 to min(10, stock)); “Add to cart”; “Buy now”; “Save for later”; and reassurance rows (Dispatches from, Sold by, Returns, Payment).	M	All purchase controls are disabled when stock = 0 and the primary button reads “Currently unavailable”.	R1
FR-PDP-08	“Add to cart” shall add the selected quantity and confirm inline (“Added to cart ✓” for ~1.8 s). “Buy now” shall add the selected quantity and navigate directly to checkout.	M	Basket badge increments by the chosen quantity; Buy now lands on /checkout with the item present.	R1
FR-PDP-09	A “More from <Department>” rail shall show up to 10 related products: same sub-department first, then the rest of the department, excluding the current product.	S	Current product never appears in its own rail.	R1
FR-PDP-10	An “Add a protection plan” link shall be offered in the buy box.	C	R1 routes to Customer Service; R2 offers purchasable protection plans.	R1/R2

8.6 Ratings and Customer Reviews (REV)

Reviews build trust and support purchase decisions. Customers can read the rating distribution and individual reviews, and write, view and delete their own reviews.

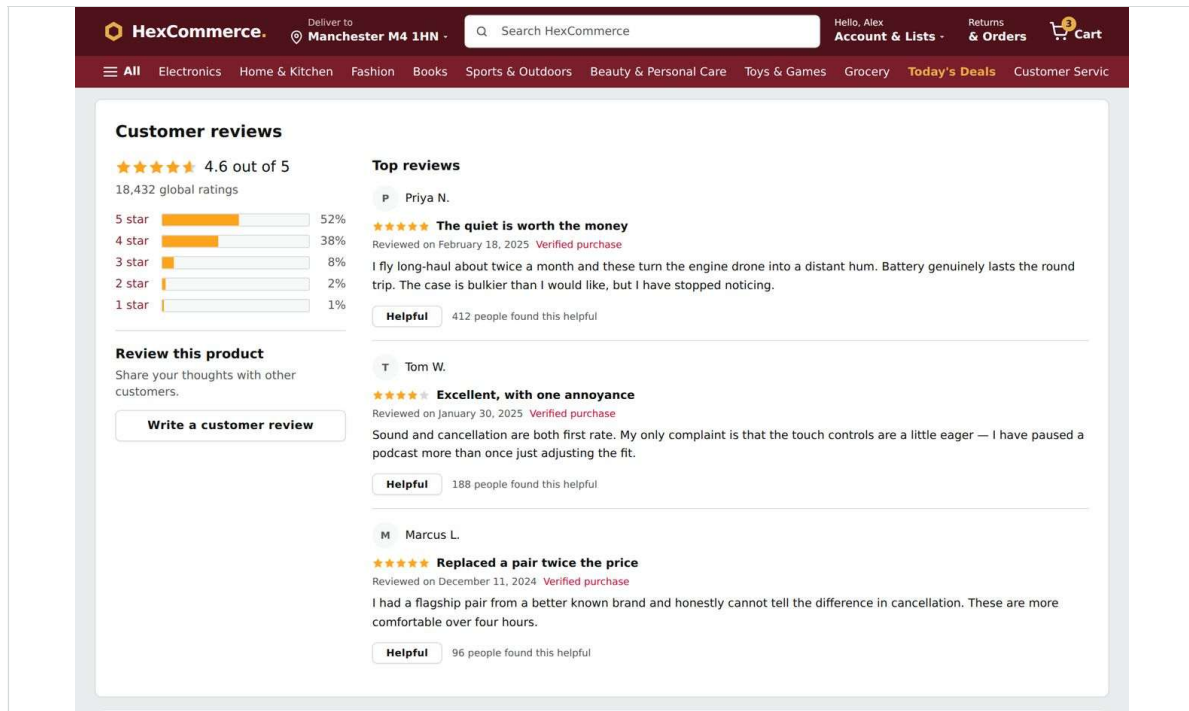


Figure 11: Customer reviews: rating summary, histogram, “Write a customer review” and top reviews

Table 17: Ratings and Customer Reviews requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-REV-01	The reviews section shall show the average rating (stars and “x.x out of 5”), the global rating count, and a five-bar histogram with percentages for 5 to 1 stars.	M	Histogram percentages are derived from the average (BR-REV-03) and sum to approximately 100%.	R1
FR-REV-02	A “Write a customer review” control shall open an inline review form; its label shall change to “Close” while open and to “Write another review” once the customer has written one.	M	Control exposes aria-expanded.	R1
FR-REV-03	The review form shall capture an overall star rating (1–5), a headline and a review body, and validate per BR-REV-01 with inline error messages.	M	Star picker is a radiogroup operable by arrow keys, with words Poor / Fair / Good / Very good / Excellent.	R1
FR-REV-04	The form shall indicate “✓ Verified purchase” when the customer has ordered the product, otherwise “Posting as <name>”.	S	Verified status is derived from order history (BR-REV-02).	R1
FR-REV-05	On submission the review shall appear first in the list tagged “Your review”, a confirmation message shall show for ~6 s, and the average and rating count shall be recalculated to include it.	M	Review persists after refresh in the same browser.	R1
FR-REV-06	Customers shall be able to delete their own reviews.	S	Delete removes the review and recalculates the aggregate.	R1
FR-REV-07	Each review shall show the author initial avatar, author, star rating, headline, review date, “Verified purchase” flag, body text, a “Helpful” button and the helpful count.	S	R1: Helpful is presentational. R2: one vote per customer per review, count increments.	R1/R2

ID	Requirement	Priority	Acceptance criteria	Release
FR-REV-08	Reviews shall be stored centrally, moderated before publication, and attributed to authenticated customers.	M	Moderation queue with approve/reject; profanity and PII screening.	R2

8.7 Wish List (WSH)

A durable personal collection of products the customer is interested in, distinct from the checkout-time “Saved for later” list.

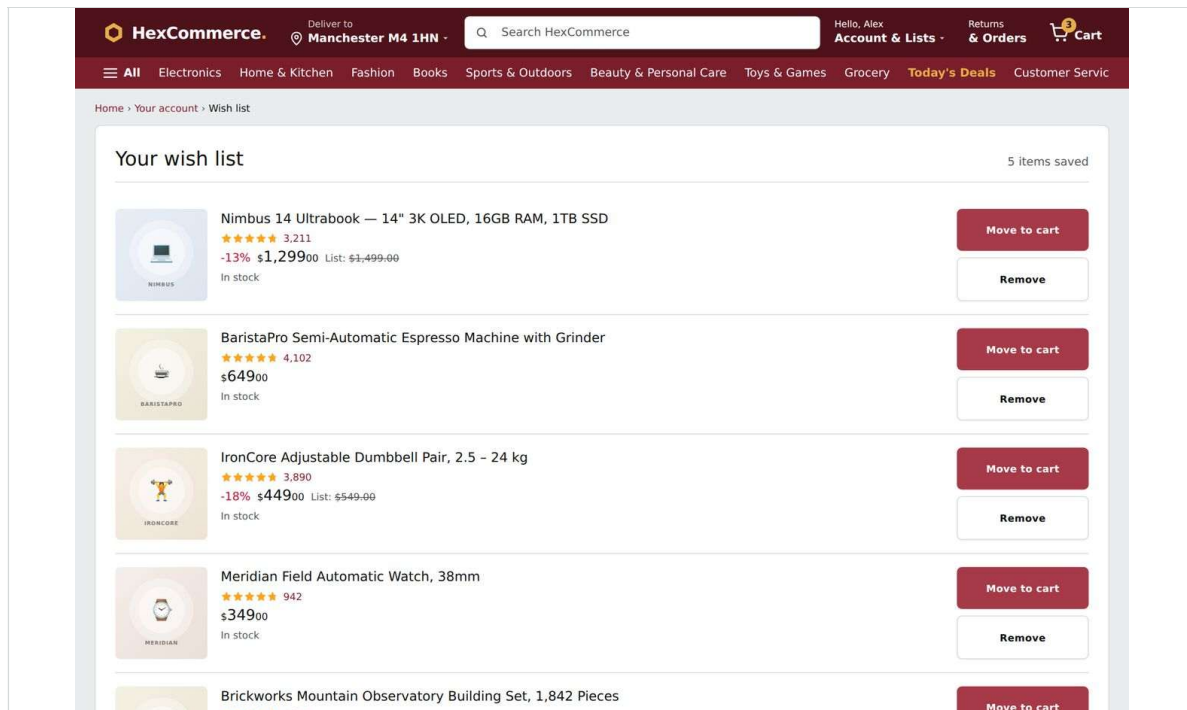


Figure 12: Wish list with move-to-cart and remove actions

Table 18: Wish List requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-WSH-01	A heart toggle on every product card and on the product page shall add or remove the product from the wish list, newest first.	M	Button exposes aria-pressed and a label of “Save <title> for later” / “Remove <title> from wish list”.	R1
FR-WSH-02	The /wishlist page shall list saved items with image, title, rating, price (with discount badge), stock status, “Move to cart” and “Remove”.	M	Header shows “N items saved”.	R1
FR-WSH-03	“Move to cart” shall remove the item from the wish list and add one unit to the basket (incrementing an existing line, subject to BR-CRT-01). It shall be disabled for out-of-stock items.	M	Basket badge increases by 1; item disappears from the wish list.	R1
FR-WSH-04	An empty state shall invite the customer to tap the heart on any product and offer “Start browsing”.	M	Shown when the list is empty.	R1

8.8 Shopping Cart and Saved for Later (CRT)

The basket lets the customer review and adjust intended purchases, see savings, delivery qualification and totals, and proceed to checkout.

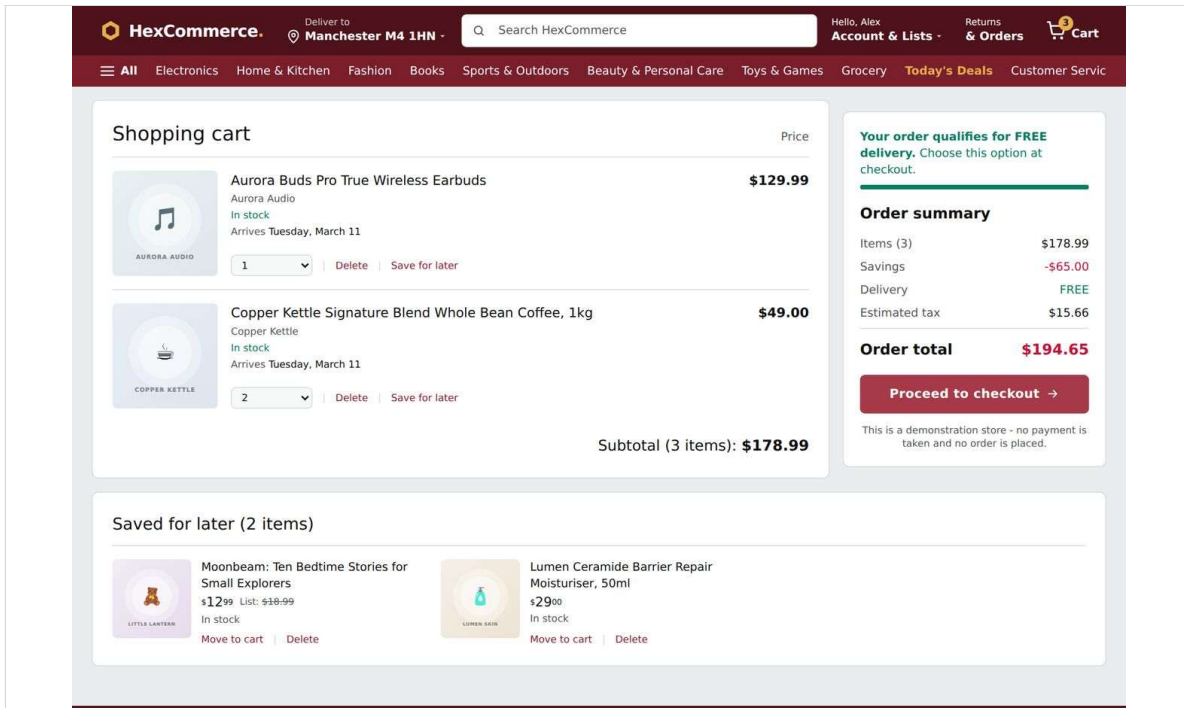


Figure 13: Shopping cart with order summary and free-delivery progress

Table 19: Shopping Cart and Saved for Later requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CRT-01	The /cart page shall list each basket line with image, title, brand, stock status, estimated arrival date, line total, a quantity selector “0 (Delete)” to 10, and “Delete” and “Save for later” actions.	M	Selecting 0 removes the line; changes update all totals immediately.	R1
FR-CRT-02	A subtotal line shall read “Subtotal (N items): <amount>”, using singular “item” when N = 1.	M	N equals the total units, not the number of lines.	R1
FR-CRT-03	An order-summary panel (shared with checkout) shall show a free-delivery progress bar and message, Items (N), Savings (when > 0), Delivery (FREE or fee), Estimated tax and Order total, calculated per BR-PRC-01 to BR-PRC-05.	M	Below threshold: “Add <amount> of eligible items to qualify for FREE delivery.” At/above: “Your order qualifies for FREE delivery.”	R1
FR-CRT-04	A “Proceed to checkout” button shall navigate to /checkout.	M	Button marked aria-disabled when the basket is empty.	R1
FR-CRT-05	“Save for later” shall move the whole line to a “Saved for later” section below the basket (anchor #saved-for-later) offering “Move to cart” (one unit; disabled if out of stock) and “Delete”.	M	Section heading shows the count; section hidden when empty.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-CRT-06	An empty-basket state shall show “Your cart is empty”, mention any saved-for-later items and offer “Shop today’s deals” and “View saved items”.	M	Saved-for-later section still renders below the empty state.	R1
FR-CRT-07	Basket, wish list and saved-for-later lists shall persist across page refreshes and be seeded with sample content on first visit.	M	R1: persisted in browser storage; failure to access storage degrades gracefully to in-session state. R2: persisted server-side against the customer account and merged on sign-in.	R1/R2
FR-CRT-08	A loading skeleton shall be displayed until the persisted basket has been read.	S	No flash of the default basket before persisted data loads.	R1

8.9 Checkout and Order Placement (CHK)

A single-page, four-section checkout collects delivery address, payment method and delivery speed, shows a final review, and ends in an order confirmation.

The screenshot displays the HexCommerce checkout interface. At the top, the navigation bar includes the HexCommerce logo, delivery location (Manchester M4 1HN), a search bar, and user account links. Below the navigation bar, a category menu is visible. The main checkout area is titled 'Checkout (3 items)' and contains three numbered sections:

- 1 Delivery address:** Shows three address options for Alex Morgan (Home, Work, and Mum's house) in Manchester and Chester.
- 2 Payment method:** Shows three payment options: Visa (ending in 4242), Mastercard (ending in 8317), and Amex (ending in 9004).
- 3 Delivery options:** Partially visible at the bottom.

On the right side, an order summary box displays the following details:

- Items (3): \$178.99
- Savings: -\$65.00
- Delivery: FREE
- Estimated tax: \$15.66
- Order total: \$194.65**

A 'Place your order' button is located below the summary, with a note stating: 'This is a demonstration store - no payment is taken and no order is placed.'

Figure 14: Checkout: numbered sections with order summary and “Place your order”

Table 20: Checkout and Order Placement requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CHK-01	The checkout page shall display “Checkout (N items)” and four numbered sections: 1 Delivery address, 2 Payment method, 3 Delivery options, 4 Review your items.	M	Sections render in order on all breakpoints.	R1
FR-CHK-02	If the basket is empty, checkout shall show “There is nothing to check out” with a “Continue shopping” link instead of the form.	M	Direct navigation to /checkout with an empty basket shows the guard.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-CHK-03	Section 1 shall list saved addresses as radio options (name, label, full address) with the default address pre-selected.	M	Exactly one address selected at all times.	R1
FR-CHK-04	Section 2 shall list saved payment methods (brand, “ending in” last four digits, expiry) with the default pre-selected, and state that the billing address equals the selected delivery address.	M	Only masked card data is displayed.	R1
FR-CHK-05	Section 3 shall offer “Tomorrow, <date> – FREE next-day delivery” (default) and “<date + 4 days> – FREE standard delivery”.	M	Arrival date in Section 4 updates when the speed changes.	R1
FR-CHK-06	Section 4 shall list each item with image, title, quantity and line total, and state “Delivering to <name>, arriving <date>”.	M	Matches basket contents.	R1
FR-CHK-07	“Place your order” shall generate an order reference in the format 204-NNNNNNN-NNNNNNN, empty the basket and show a confirmation with a personalised thank-you, confirmation e-mail address, order reference, estimated arrival and links to “View your orders” and “Continue shopping”.	M	Basket badge shows 0 after placement.	R1
FR-CHK-08	Order placement shall authorise payment through a PCI DSS-compliant payment service provider, create a persistent order, reserve stock, send a confirmation e-mail and add the order to the customer’s history.	M	Order appears in /orders within 5 seconds; failed authorisation returns the customer to Section 2 with an error.	R2
FR-CHK-09	Customers shall be able to add a new address and a new payment method during checkout.	S	New address is validated and optionally saved to the account.	R2
FR-CHK-10	Delivery charges and tax shown at checkout shall be consistent with the delivery speed selected and the destination jurisdiction.	M	Totals reconcile to the confirmation and invoice.	R2

8.10 Order History and Tracking (ORD)

Customers can review past orders, see their fulfilment status, reorder items and access returns and invoices.

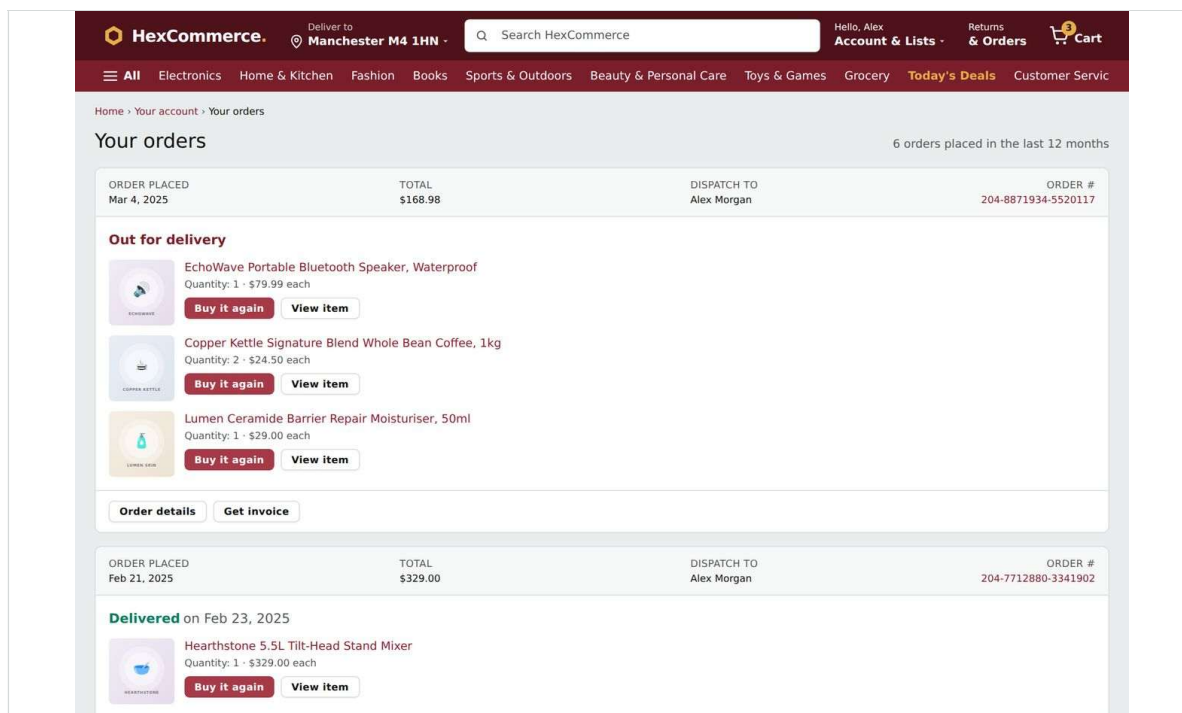


Figure 15: Order history with status, items and post-purchase actions

Table 21: Order History and Tracking requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-ORD-01	The /orders page shall list orders newest first, each with a header strip (Order placed, Total, Dispatch to, Order #), a colour-coded status (with delivery date when Delivered) and its items (image, title, quantity, unit price, “Buy it again”, “View item”).	M	Status colours: Delivered = success, Shipped/Out for delivery = brand, Processing = muted, Cancelled = sale red.	R1
FR-ORD-02	Each order shall offer “Order details”, “Get invoice” and, for Delivered orders only, “Return or replace items”.	M	Return action hidden for non-delivered orders.	R1
FR-ORD-03	The order detail page (/orders/{id}) shall show order date and number, a four-stage tracker (Processing → Shipped → Out for delivery → Delivered) highlighting reached stages, items with “Sold by” and line prices, delivery address, payment method and an order summary (Items, Delivery, Tax, Order total).	M	Cancelled orders replace the tracker with “Order cancelled – no payment was taken”. Unknown id returns 404.	R1
FR-ORD-04	“Get invoice” shall produce a downloadable tax-compliant invoice (PDF).	S	Invoice totals match the order record.	R2
FR-ORD-05	“Return or replace items” shall start a structured returns flow (item, reason, refund/replacement, drop-off method) within the statutory return window.	S	Return label or QR code generated; refund status visible on the order.	R2
FR-ORD-06	Customers should be able to filter order history by time period and search by product.	C	Filter “Last 30 days / 3 months / 12 months / year”.	R2

8.11 Customer Account Hub (ACC)

A single hub for account destinations, recent orders, saved addresses, payment methods and recently viewed items.

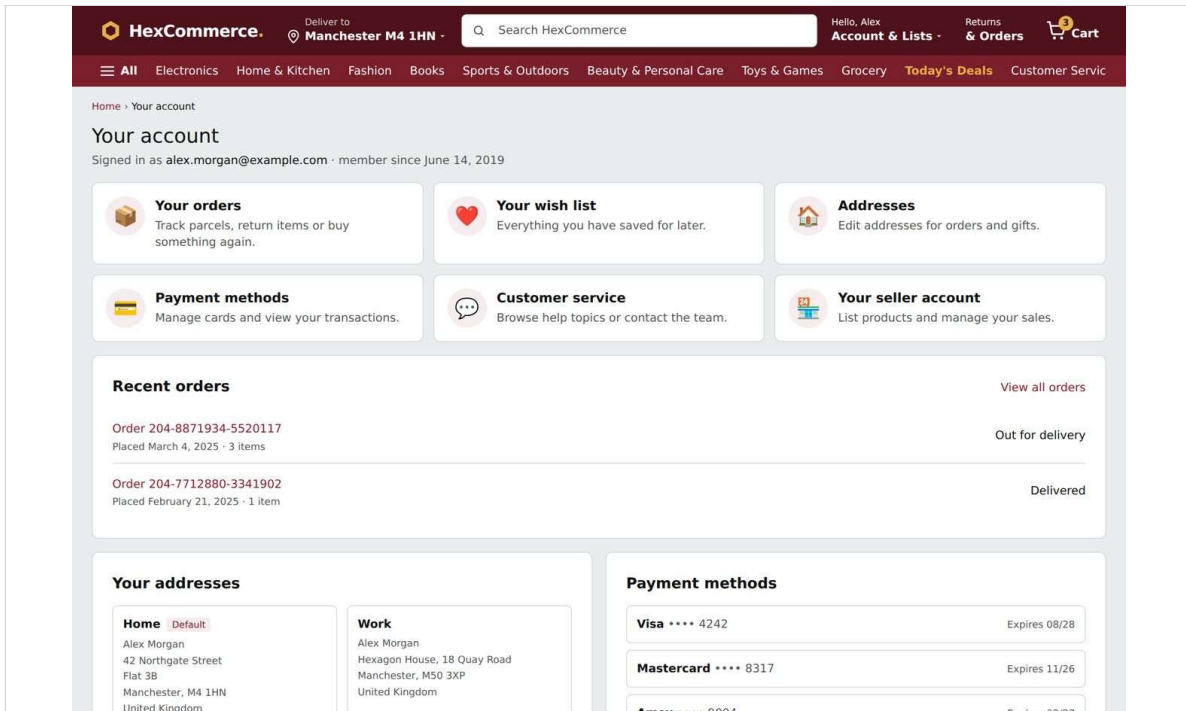


Figure 16: Account hub with destination tiles, recent orders, addresses and payment methods

Table 22: Customer Account Hub requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-ACC-01	The /account page shall show the signed-in e-mail and member-since date and six destination tiles: Your orders, Your wish list, Addresses, Payment methods, Customer service, Your seller account.	M	Each tile links to its destination.	R1
FR-ACC-02	A “Recent orders” card shall list the two most recent orders with number, date, item count and status, and a “View all orders” link.	S	Item count uses correct singular/plural.	R1
FR-ACC-03	Saved addresses (with label and “Default” badge) and payment methods (brand, masked number, expiry) shall be displayed.	M	No full card numbers are ever shown.	R1
FR-ACC-04	Customers shall be able to add, edit, delete and set default addresses and payment methods.	M	Address validation and card tokenisation via the payment provider.	R2
FR-ACC-05	Customers shall be able to manage profile details, password, communication preferences, and request export or deletion of personal data.	M	Data subject requests completed within statutory deadlines.	R2

8.12 Customer Service Help Desk (CS)

A self-service help desk where customers raise an issue against a purchased item and follow a threaded conversation with the support team.

The screenshot displays the HexCommerce Customer Service Help Desk interface. At the top, there's a navigation bar with the HexCommerce logo, delivery location (Manchester M4 1HN), a search bar, and user account links. Below this is a category menu. The main content area is titled 'Customer service' and includes a 'Report an issue' form. The form has two steps: 'Which item is this about?' and 'What went wrong?'. The first step lists three items: EchoWave Portable Bluetooth Speaker, Copper Kettle Signature Blend Whole Bean Coffee, and Lumen Ceramide Barrier Repair Moisturiser. The second step has a dropdown for 'Select an issue...' and a text area for 'Add details (optional)'. On the right, the 'Your issues' section shows two issues: 'Item arrived damaged or defective' (status: Your reply needed) and 'Refund not received' (status: Resolved).

Figure 17: Customer service: two-step issue form and issue list with status pills

Table 23: Customer Service Help Desk requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CS-01	The /customer-service page shall provide a “Report an issue” form in two steps: (1) choose the item from non-cancelled orders, shown with image, title, order number and delivery/status; (2) choose an issue type and optionally add details.	M	Validation: “Choose the item your issue is about.” / “Choose what went wrong.”	R1
FR-CS-02	Issue types shall be: Item arrived damaged or defective; Item never arrived; Wrong item delivered; Parts or accessories missing; Item not as described; Request a return; Refund not received; Billing or payment query; Something else.	M	Exactly nine options plus the placeholder.	R1
FR-CS-03	Submitting shall create an issue with reference CS-NNNNN and status Open, containing the customer’s message and an automated acknowledgement promising a reply within 24 hours; the new issue shall open expanded at the top of the list.	M	Confirmation shows for ~6 s; form resets.	R1
FR-CS-04	The “Your issues” list shall show “N total · M open” and, per issue, the product image, topic, unread indicator, reference, product, last-updated date and a status pill; rows expand to show the thread.	M	Opening an issue marks it read.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-CS-05	The thread shall show issue metadata (reference, raised date, order link), messages in chronological order with author initials, name and timestamp, and a reply box whose send button is disabled while empty.	M	Customer messages visually distinct from support messages.	R1
FR-CS-06	Status transitions shall follow BR-CS-01. For a Resolved issue the reply control shall read “Reopen & send” with an explanatory note.	M	Replying to a Resolved issue changes its status to In review.	R1
FR-CS-07	Where no orders exist the form shall show “Nothing to report yet”; where no issues exist the list shall show an explanatory empty state.	S	—	R1
FR-CS-08	Issues shall be routed to a support agent console with SLA tracking, and customers notified of replies by e-mail and in-app.	M	First response within 24 h for 95% of issues.	R2
FR-CS-09	Customers should be able to attach photos (e.g. evidence of damage) to an issue.	S	JPEG/PNG up to 10 MB, virus-scanned.	R2

8.13 Deals and Promotions (Customer-facing) (DEA)

A dedicated “Today’s Deals” destination combining marketplace lightning deals, general discounts and live seller-run deals.

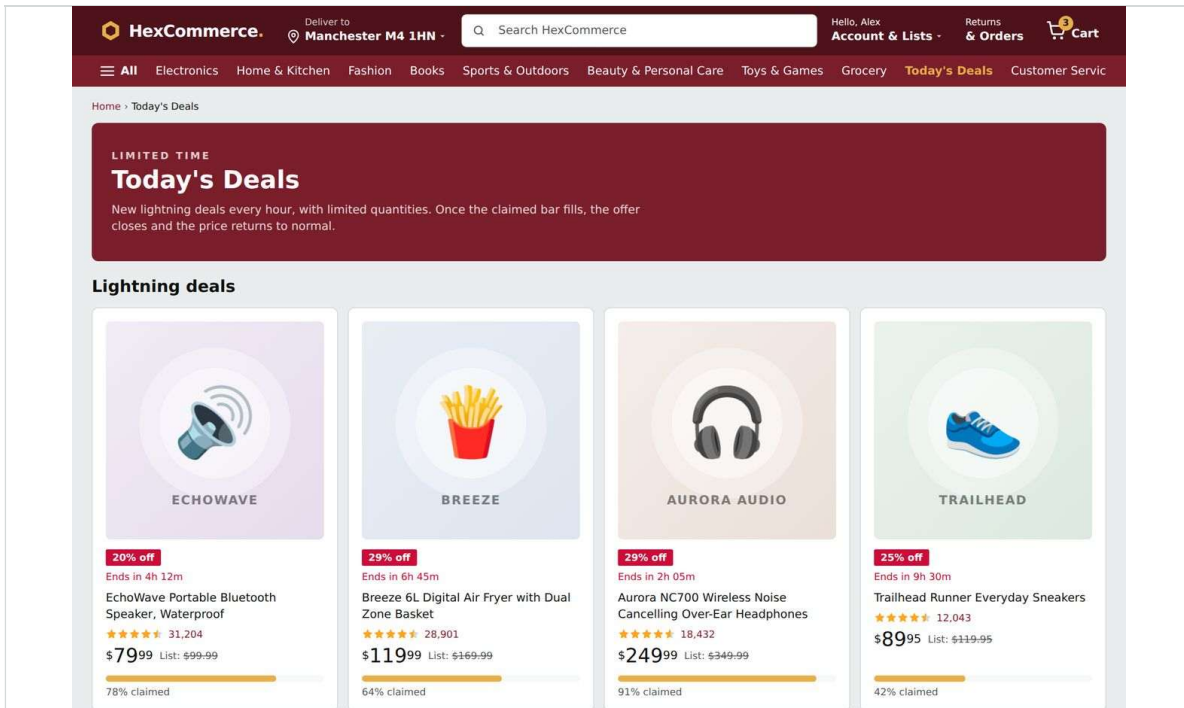


Figure 18: Today's Deals with lightning deals and claimed-progress bars

Table 24: Deals and Promotions (Customer-facing) requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-DEA-01	The /deals page shall show a banner explaining lightning deals and a “Lightning deals” section of four featured deals, each with image, “N% off” badge, remaining-time label, title, rating, price with list price and a claimed-percentage progress bar.	M	Progress bar exposes role="progressbar" with value and label.	R1
FR-DEA-02	An “All deals” grid shall show the remaining deals and an “Also reduced this week” grid the top 10 discounted products.	S	Cards use the standard product card (FR-CAT-12).	R1
FR-DEA-03	A “Deals from our sellers” section shall list every Live seller deal with image, “N% off”, end date, title, deal price vs normal price, claimed percentage and “Sold by <seller>”; the section is hidden when there are none.	S	A deal started by a seller appears immediately; an ended deal disappears.	R1
FR-DEA-04	Deal countdowns shall be live timers and deals shall close automatically when time expires or claimed quantity reaches 100%.	S	Price reverts to normal after close.	R2
FR-DEA-05	Seller deal cards shall be purchasable (add to basket at the deal price).	M	Claimed count increments per unit sold.	R2

8.14 Customer Identity: Sign-in and Registration (AUT)

Entry points for authentication and account creation. In the R1 baseline these are presentational; a persona (“Alex Morgan”) is treated as always signed in.

The screenshot shows the HexCommerce sign-in page. The header is dark blue with the HexCommerce logo, delivery location (Manchester M4 1HN), a search bar, and user account links (Hello, Alex, Account & Lists, Returns & Orders, Cart). The navigation menu includes categories like Electronics, Home & Kitchen, Fashion, Books, Sports & Outdoors, Beauty & Personal Care, Toys & Games, Grocery, Today's Deals, and Customer Service. The main content area has a light blue background with a 'Sign in' form. The form includes fields for 'Email or mobile phone number' and 'Password', a 'Continue' button, and a link for 'Forgotten password?'. Below the form is a link for 'New to HexCommerce?' and a 'Create your HexCommerce account' button. The footer contains four columns of links: 'Get to know us', 'Make money with us', 'Payment & delivery', and 'Let us help you'.

Figure 19: Sign-in page

Table 25: Customer Identity: Sign-in and Registration requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-AUT-01	The /signin page shall capture e-mail or mobile number and password, and offer “Forgotten password?”, “Continue” and “Create your <Site> account”.	M	Fields use correct autocomplete tokens (email, current-password).	R1
FR-AUT-02	The /register page shall capture name, e-mail, password (hint: at least 6 characters) and password confirmation, with a link back to sign-in.	M	Fields use autocomplete tokens name, email, new-password.	R1
FR-AUT-03	The system shall authenticate customers, maintain secure sessions, enforce the password policy (NFR-SEC-03), support password reset by e-mail and offer optional multi-factor authentication.	M	Sessions expire after inactivity; account lock-out after repeated failures.	R2
FR-AUT-04	“Sign out” shall terminate the session and clear personal data from the device.	M	Back button after sign-out does not reveal account pages.	R2
FR-AUT-05	Registration shall capture explicit consent to terms and privacy notice and optional marketing consent.	M	Consent timestamp and version stored.	R2

8.15 Seller Recruitment and Application (SEL)

Prospective sellers learn about selling on the marketplace and submit a business application that is reviewed by a marketplace administrator.

Figure 20: Register as a seller: business application form

Table 26: Seller Recruitment and Application requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SEL-01	The /sell page shall present the value proposition (Reach, Fulfilment, Insights, Simple fees), “Getting started” steps, fee information (referral fee 8%–15% by category, no listing fee or monthly subscription on the standard plan) and CTAs “Create a seller account” and “Go to your Seller Dashboard”.	M	CTAs link to /sell/apply and /sell/dashboard.	R1
FR-SEL-02	The application form shall capture: legal business name, business type (Sole trader, Partnership, Limited company, Non-profit), primary contact, business e-mail, business phone, years trading, website (optional), description of what will be sold, registered address (line 1, line 2 optional, city, state/county, postcode, country) and tax registration number.	M	Contact name and e-mail are pre-filled from the customer profile; country defaults to United Kingdom.	R1
FR-SEL-03	The form shall validate per BR-SEL-01 and display one error message at a time.	M	Invalid e-mail shows “Enter a valid business email.”	R1
FR-SEL-04	On submission the application shall be given reference SA-NNNN, status Pending and a submitted timestamp, and a confirmation page shall be shown.	M	Application appears at the top of the admin queue.	R1
FR-SEL-05	Revisiting /sell/apply shall show the applicant’s current state: Pending (awaiting decision), Approved (link to Seller Dashboard) or Rejected (reason and “Apply again”).	M	Pending applications cannot be edited or duplicated.	R1
FR-SEL-06	“Apply again” after rejection shall reopen the form pre-filled with the previous submission and display the rejection reason.	S	Resubmission creates a new Pending application.	R1
FR-SEL-07	The /sell page shall show a status card for the customer’s own application with the relevant next action.	S	Card hidden if the customer has never applied.	R1
FR-SEL-08	Seller onboarding shall include identity and business verification (KYC/KYB), sanctions screening and payout account set-up after approval.	M	Verification outcome stored with evidence.	R2

8.16 Marketplace Administration: Seller Application Review (ADM)

A review queue for marketplace administrators to approve or reject seller applications. Decisions are reflected immediately on the applicant’s pages.

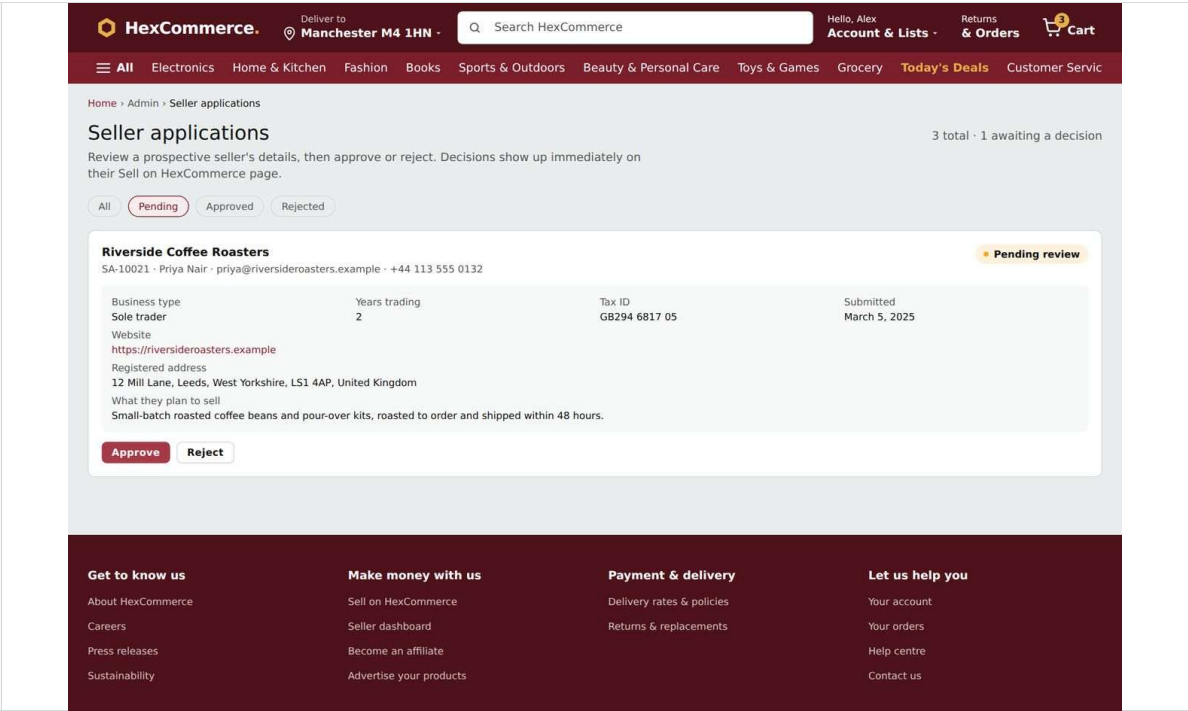


Figure 21: Admin seller-application queue

Table 27: Marketplace Administration: Seller Application Review requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-ADM-01	The /admin/sellers page shall list applications newest first with filter chips All / Pending (default) / Approved / Rejected and a summary “N total · M awaiting a decision”.	M	Empty filter shows “No applications here”.	R1
FR-ADM-02	Each application card shall show business name, reference, contact, e-mail, phone, status, business type, years trading, tax ID, submitted date, website (opens in a new tab), registered address and description.	M	External website link uses rel="noreferrer".	R1
FR-ADM-03	Pending applications shall offer “Approve” (single action) and “Reject”, which requires a confirmation step with an optional reason that is shown to the applicant.	M	Cancel discards the draft reason.	R1
FR-ADM-04	Decided applications shall show the decision, decision date and any note, and offer “Move back to pending”, which clears the decision and note.	S	—	R1
FR-ADM-05	Access to administration functions shall be restricted to authenticated users holding an administrator role, and every decision shall be recorded in an immutable audit log (who, what, when, before/after).	M	Unauthorised access returns 403; audit entries retained ≥ 7 years.	R2

8.17 Seller Dashboard (SDB)

The seller’s home: headline performance indicators, quick links, recent orders and items needing attention.

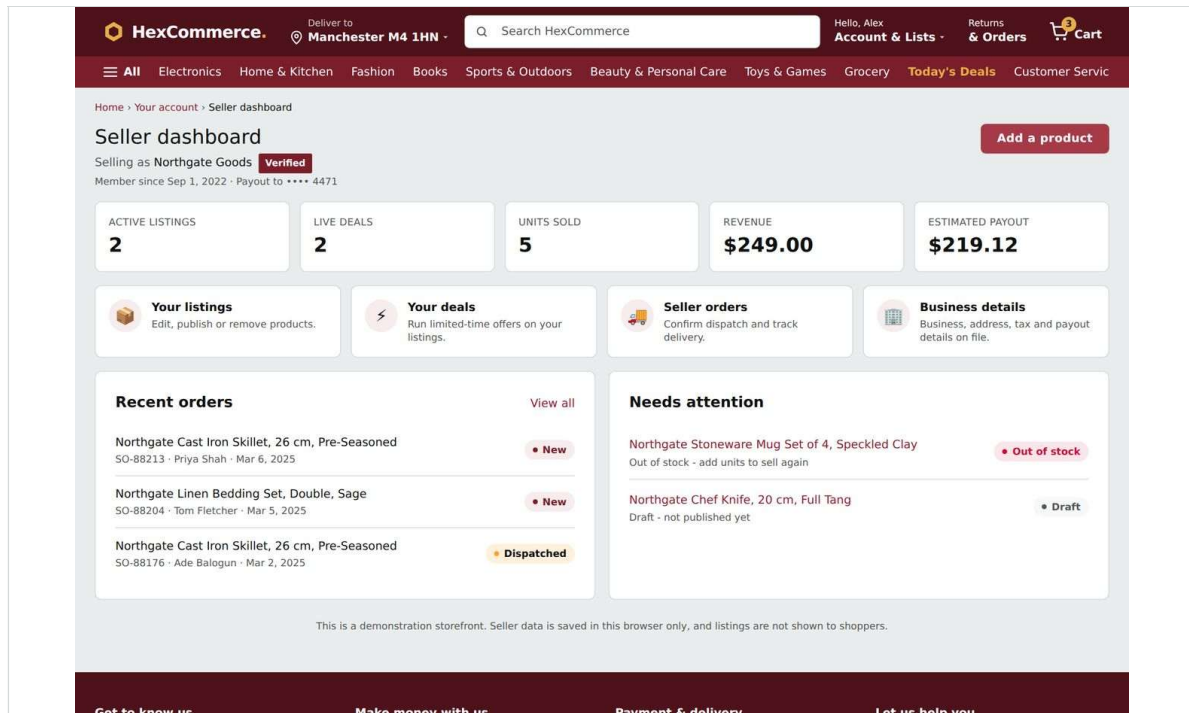


Figure 22: Seller dashboard: KPI tiles, quick links, recent orders and “Needs attention”

Table 28: Seller Dashboard requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SDB-01	The dashboard shall show the business name, verification status badge, member-since date, masked payout account and an “Add a product” action.	M	—	R1
FR-SDB-02	KPI tiles shall show Active listings, Live deals, Units sold, Revenue and Estimated payout, calculated per BR-SLR-05.	M	Cancelled orders excluded from units, revenue and payout.	R1
FR-SDB-03	Quick links shall lead to Your listings, Your deals, Seller orders and Business details.	S	—	R1
FR-SDB-04	A “Recent orders” card shall show the three most recent seller orders (title, reference, buyer, date, status).	S	Removed listings show “Listing removed”.	R1
FR-SDB-05	A “Needs attention” card shall list Draft and Out-of-stock listings with guidance and a link to edit; when none, show “Every listing is live and in stock.”	S	—	R1
FR-SDB-06	The dashboard should provide sales, traffic and conversion reporting over selectable periods with export to CSV.	S	Figures reconcile with settlement reports.	R2

8.18 Seller Listing Management (SLS)

Sellers create, edit, publish, unpublish and remove product listings using a four-step wizard with a live storefront preview.

Figure 23: Add a product: four-step wizard with live card and detail-page preview

Table 29: Seller Listing Management requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SLS-01	The /sell/listings page shall show the listing count, status filter chips (All, Active, Draft, Out of stock, Unpublished), a title search box and one row per listing (cover image, title, status pill, reference, department, condition, last updated, price with discount, stock).	M	Filter and search combine (AND).	R1
FR-SLS-02	Row actions shall be: Edit; Unpublish (Active or Out of stock); Publish (Draft or Unpublished → Active, or Out of stock when stock = 0); Remove with an inline “Remove this listing? Yes, remove / Keep” confirmation.	M	Removing a listing also removes all its deals (BR-SLR-04).	R1
FR-SLS-03	The listing wizard shall comprise four steps — Product details, Pricing & stock, Description, Review & publish — with a stepper showing completed (✓), current and pending steps, and Back/Continue controls.	M	Continue is blocked until the current step validates (BR-SLR-01).	R1
FR-SLS-04	Step 1 shall capture title, brand, department, sub-department (options depend on department; reset when department changes), condition (New, Renewed, Used – like new, Used – good), optional emoji icon and product images (FR-SIM).	M	—	R1
FR-SLS-05	Step 2 shall capture selling price, optional “was” price (must exceed selling price) and units in stock.	M	Hint: a listing with no stock publishes as “Out of stock”.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-SLS-06	Step 3 shall capture a short summary (max 160 characters with live counter), a full description and a dynamic list of key features (add/remove).	M	Blank features are discarded on save.	R1
FR-SLS-07	Step 4 shall summarise the listing and offer “Publish now” (“Publish (out of stock)” when stock = 0) and “Save as draft”. Submission re-validates steps 1–3 and returns to the first failing step.	M	Status assigned per BR-SLR-02.	R1
FR-SLS-08	A live preview of the storefront product card and the detail-page gallery shall update as the seller types.	S	—	R1
FR-SLS-09	After saving, a result screen shall confirm “Listing published / updated” or “Saved as a draft”, with “Go to your listings” and (create mode) “Add another product”.	S	—	R1
FR-SLS-10	Edit mode (/sell/listings/{id}/edit) shall pre-fill the wizard; an unknown id shall show “Listing not found” with a link back.	M	—	R1
FR-SLS-11	Published seller listings shall appear in the customer-facing catalogue, search and product pages, and be purchasable.	M	New listing searchable within 5 minutes of publishing.	R2
FR-SLS-12	Listings should pass automated content policy checks (prohibited items, image quality) before going live.	S	Flagged listings held for review.	R2

8.19 Seller Product Image Upload (SIM)

Sellers upload product photographs that become the cover and gallery images for their listings.

Table 30: Seller Product Image Upload requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SIM-01	The image field shall accept up to 6 images via a file picker (multiple selection) or drag-and-drop.	M	Drop zone highlights while dragging.	R1
FR-SIM-02	Only image files of 8 MB or less shall be accepted; accepted images shall be downscaled so the longest edge is at most 900 px and re-encoded as JPEG (quality 0.82) on a white background.	M	Skipped files are reported: “N added · M skipped (over 8 MB, not an image, or over the 6-image limit).”	R1
FR-SIM-03	The first image shall be the cover; sellers shall be able to “Make cover” any other image and remove any image.	M	Cover shown on cards and the listings table.	R1
FR-SIM-04	When no images are uploaded, deterministic generated artwork (gradient plus product/department glyph) shall be used.	S	Same product always renders the same artwork.	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-SIM-05	Images shall be stored in a managed media service with CDN delivery, responsive renditions and malware scanning.	M	Images served in modern formats (WebP/AVIF) where supported.	R2

8.20 Seller Deal Management (SDL)

Sellers run time-limited percentage discounts on their active listings. Live deals are surfaced to customers on the Today's Deals page.

Figure 24: Create a deal with live deal-card preview

Table 31: Seller Deal Management requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SDL-01	The /sell/deals page shall show “N deals · M live now”, filter chips (All, Scheduled, Live, Ended) and one row per deal (image, listing title, status, reference, date range, deal price vs normal price, “x of y claimed” progress).	M	Sorted by creation date, newest first.	R1
FR-SDL-02	Row actions shall be: Edit; Start now (Scheduled); End deal (Live or Scheduled); Remove with inline confirmation.	M	Ended deals are removed from the customer Deals page immediately.	R1
FR-SDL-03	The deal editor shall capture the listing (Active listings only; in edit mode also the deal’s own listing), discount %, units in the deal, start date and end date, validated per BR-SLR-03.	M	Hint shows “New price <x> (was <y>)” once the discount is valid.	R1
FR-SDL-04	A live preview card shall show the listing image, discount badge, deal price, end date and deal quantity.	S	—	R1

ID	Requirement	Priority	Acceptance criteria	Release
FR-SDL-05	On save the result screen shall read “Deal is live”, “Deal scheduled” or “Deal updated” per BR-SLR-03.	S	—	R1
FR-SDL-06	If the seller has no Active listings, the editor shall explain that an active listing is required and link to “Add a product” and “Your listings”.	S	—	R1
FR-SDL-07	Scheduled deals shall start and live deals shall end automatically at their start/end date-time in the marketplace time zone.	M	Status changes within 1 minute of the boundary.	R2

8.21 Seller Order Fulfilment (SOR)

Sellers see orders placed against their listings and progress them through dispatch and delivery.

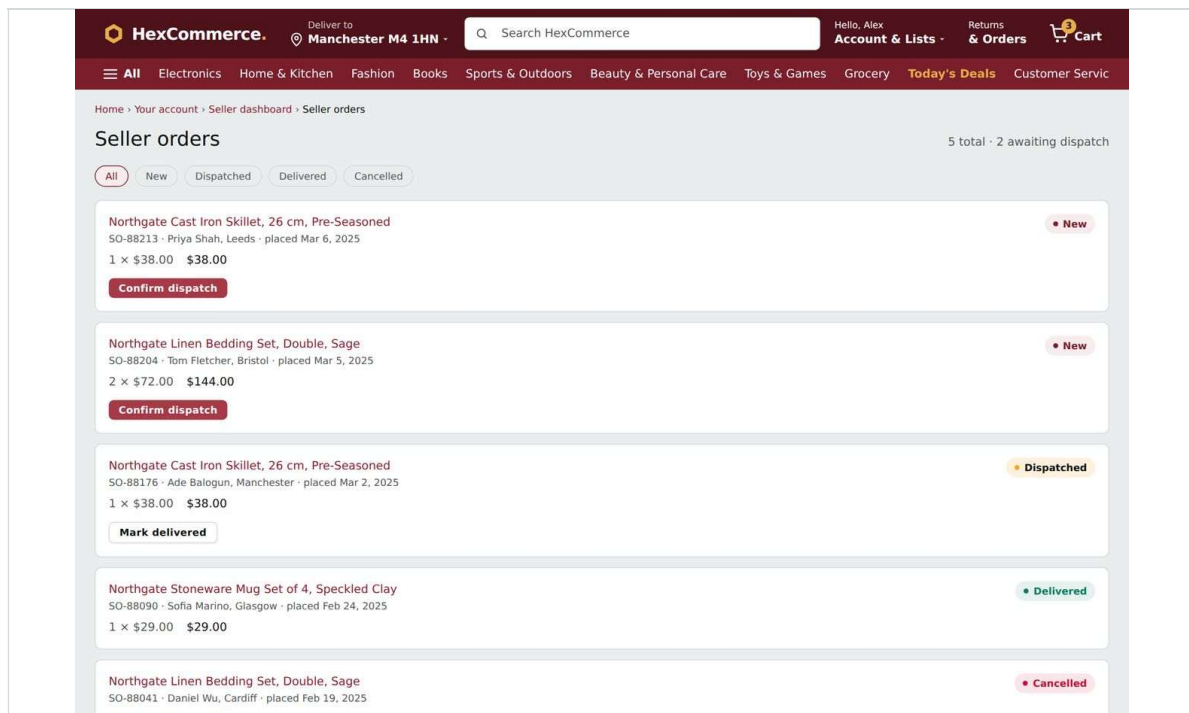


Figure 25: Seller orders with dispatch and delivery actions

Table 32: Seller Order Fulfilment requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SOR-01	The /sell/orders page shall show “N total · M awaiting dispatch”, filter chips (All, New, Dispatched, Delivered, Cancelled) and one row per order (listing title or “Listing removed”, reference, buyer, city, placed date, status, quantity × unit price, line total).	M	Sorted newest first.	R1
FR-SOR-02	“Confirm dispatch” shall move a New order to Dispatched; “Mark delivered” shall move a Dispatched order to Delivered.	M	No actions shown for Delivered or Cancelled orders.	R1
FR-SOR-03	Dispatch confirmation shall capture carrier and tracking number and notify the buyer.	M	Tracking visible on the customer order detail.	R2

ID	Requirement	Priority	Acceptance criteria	Release
FR-SOR-04	Sellers shall be able to cancel an order before dispatch with a reason, triggering a customer refund.	S	Refund initiated automatically.	R2

8.22 Seller Business Details (SBD)

A record of the business information held for the onboarded seller, used on invoices, payouts and the buyer-facing seller page.

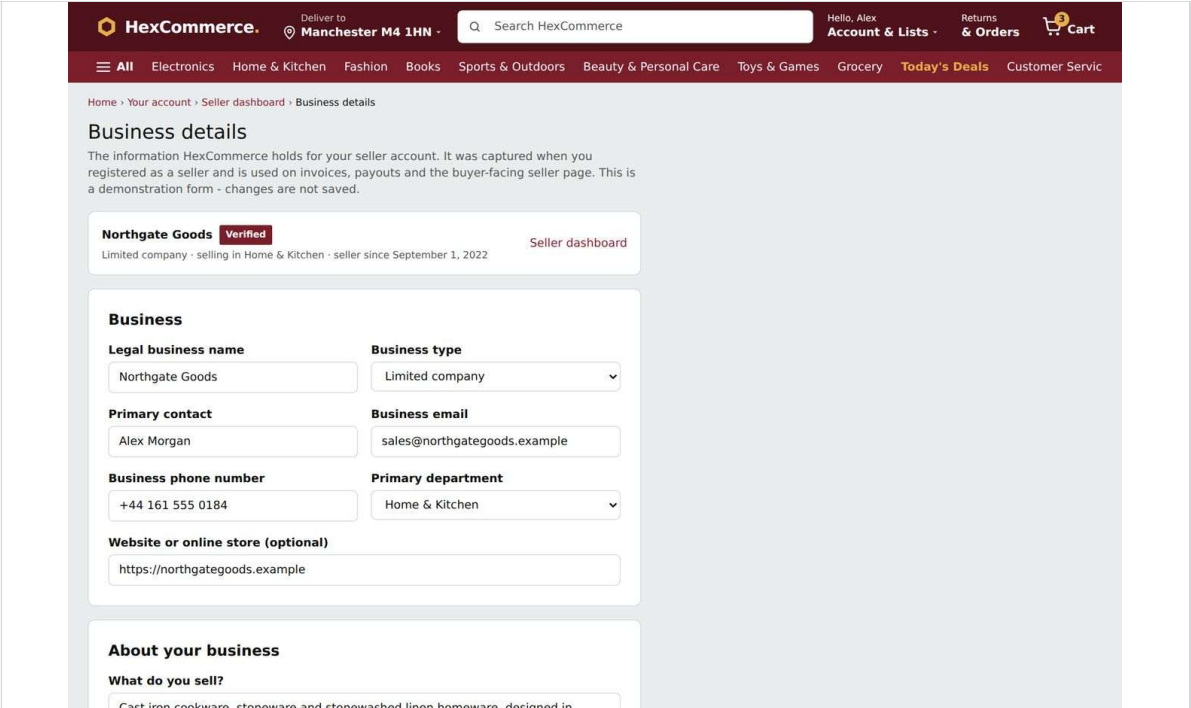


Figure 26: Seller business details

Table 33: Seller Business Details requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SBD-01	The /sell/register page shall show an identity strip (business name, verification badge, business type, primary department, seller-since date) and sections Business, About your business, Registered address, Tax details and Payout, pre-filled with the profile.	S	Payout section notes it is set up after approval.	R1
FR-SBD-02	Changes to business details shall be saved; changes to legal name, tax number or payout account shall trigger re-verification and notify the seller.	M	R1: “Save changes” does not persist (presentational).	R2

8.23 Configuration and White-labelling (CFG)

The storefront can be re-branded and localised purely by configuration, with no code changes.

Table 34: Configuration and White-labelling requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CFG-01	Site name and tagline shall be configurable and used in the logo, page titles, metadata, footer and legal copy.	M	Changing NEXT_PUBLIC_SITE_NAME updates all occurrences after rebuild.	R1
FR-CFG-02	The colour theme shall be selectable from six presets (maroon – default, midnight, forest, slate, amber, plum) with optional per-token overrides for primary, primary-dark, primary-light, accent, header, sub-header and footer, accepting hex (#800000, #800) or “r,g,b”.	M	No component hard-codes a brand colour; blank override inherits the preset.	R1
FR-CFG-03	Currency (ISO 4217) and locale (BCP 47) shall be configurable and drive all price, number and date formatting.	M	Setting GBP/en-GB renders £ and day-month-year dates throughout.	R1
FR-CFG-04	Free-delivery threshold and sales-tax rate shall be configurable.	M	Defaults: 35 and 0.0875.	R1
FR-CFG-05	The standard delivery fee, price-band boundaries/labels and referral-fee rate should also be configurable rather than fixed in code.	S	Price-band labels follow the configured currency.	R2

8.24 Multi-seller Orders, Shipping Rules and Zones (MSO) – R2

A single basket may contain items from several sellers and from HexCommerce itself. The order must be split into shipments per fulfilling party, each with its own delivery options, charges, tracking and cancellation.

HexCommerce Account & Lists · Cart

Checkout (3 items from 2 sellers)

Shipment 1 of 2 — Sold and dispatched by Northgate Goods

Item 1: \$38.00

Item 2: \$72.00

Delivery: Standard (3-5 days) — FREE ▼ Seller shipping rules apply

Shipment 2 of 2 — Sold by HexCommerce

Item 1: \$24.99

Delivery: Next day — \$4.99 ▼

Payment

Card number (hosted by payment provider) MM/YY CVC

3-D Secure challenge shown by the provider when required.

Voucher or promotion code

SPRING10 Apply

✓ SPRING10 applied: 10% off eligible items (– \$11.00)

Order summary

Items (3)	\$134.99
Voucher SPRING10	–\$11.00
Delivery (2 shipments)	\$4.99
Tax	\$10.85
Order total	\$139.83

Place your order

Figure 27: Wireframe (R2): checkout with two shipments, per-shipment delivery, hosted payment fields and voucher**Table 35:** Multi-seller Orders, Shipping Rules and Zones: wireframe annotations

#	Annotation
1	Items grouped into one shipment per fulfilling seller, with “Sold and dispatched by” attribution.
2	Per-shipment delivery option list derived from the seller’s shipping rules and the destination zone.
3	Card fields hosted by the payment provider (PCI DSS SAQ A); 3-D Secure where required.

#	Annotation
4	Voucher entry with inline eligibility result (FR-PRM).
5	Summary showing voucher discount, combined delivery for all shipments, tax and total.

Table 36: Multi-seller Orders, Shipping Rules and Zones requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-MSO-01	The basket and checkout shall group items into shipments by fulfilling party and show “Sold by” and “Dispatched by” for each shipment.	M	A basket with items from two sellers shows two shipment groups.	R2
FR-MSO-02	Each seller shall define shipping rules per destination zone: available services (e.g. standard, next day), price (flat, per item, weight band or free over a threshold), handling time and carrier.	M	Seller can create, edit and deactivate rules; changes apply to new baskets within 5 minutes.	R2
FR-MSO-03	Shipping zones shall be defined by the marketplace (country, region and postcode patterns) and used to determine eligibility and price for each shipment.	M	A postcode outside every zone shows “Does not deliver to <postcode>” and blocks checkout for that item only.	R2
FR-MSO-04	The customer shall select a delivery option per shipment; the delivery date promise shall reflect seller handling time and carrier transit time.	M	Arrival dates differ per shipment where applicable.	R2
FR-MSO-05	On placement, one customer order shall be created with one seller order per shipment; each seller sees only their own lines and the buyer’s dispatch details.	M	Seller B cannot see Seller A’s items or prices.	R2
FR-MSO-06	Shipments shall be tracked, cancelled and refunded independently; the customer order status shall summarise its shipments (e.g. “Partially dispatched”).	M	Cancelling shipment 1 refunds only its items and delivery charge.	R2
FR-MSO-07	The free-delivery threshold shall be evaluated per fulfilling party unless the marketplace funds free delivery across sellers.	S	Rule is configurable; basket messaging states which items count towards the threshold.	R2

8.25 Vouchers and Promotion Codes (PRM) – R2

Promotion codes let marketing and sellers reward customers beyond time-limited deals. Codes must be controlled, auditable and resistant to abuse.

Table 37: Vouchers and Promotion Codes requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-PRM-01	Authorised users shall create vouchers with: code (or auto-generated batch), discount type (percentage, fixed amount, free delivery), value, currency, start/end date-time, minimum spend, eligible departments/sellers/products, exclusions, total and per-customer redemption limits, and funding party (marketplace or seller).	M	Voucher cannot be saved without an end date and a redemption limit.	R2

ID	Requirement	Priority	Acceptance criteria	Release
FR-PRM-02	Customers shall enter one voucher code at checkout (or in the basket); the system shall validate it and display either the discount applied or a specific reason for rejection.	M	Messages: expired, not yet valid, minimum spend not met, no eligible items, already used, invalid code.	R2
FR-PRM-03	Discounts shall apply only to eligible lines, be pro-rated across those lines for refund purposes, and be shown as a separate line in the order summary, confirmation and invoice.	M	Partial return refunds the pro-rated amount paid for the returned line.	R2
FR-PRM-04	Stacking rules shall be configurable: by default a voucher may combine with deals but not with another voucher.	S	Second code replaces the first with a confirmation prompt.	R2
FR-PRM-05	Redemption shall be recorded atomically at order placement; limits shall hold under concurrent checkouts.	M	Load test: a 100-use code is never redeemed more than 100 times.	R2
FR-PRM-06	Voucher performance reporting shall show redemptions, discount cost by funding party, and incremental order value.	S	Report exportable to CSV.	R2

8.26 Returns and Refunds (RET) – R2

A structured returns and refunds process that meets statutory rights, keeps customers informed and allocates cost correctly between marketplace and seller.

Figure 28: Wireframe (R2): returns request with reason, photos, resolution and drop-off method

Table 38: Returns and Refunds: wireframe annotations

#	Annotation
1	Eligible items with quantity selector; items outside the return window explain why.
2	Mandatory reason (support topics) and photo upload for damage/defect claims.

#	Annotation
3	Resolution choice: refund to original payment method or replacement.
4	Return method: printer-free QR drop-off or paid collection.
5	Refund estimate with return-delivery charge and processing time.
6	Return status timeline visible afterwards on the order.

Table 39: Returns and Refunds requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-RET-01	Customers shall be able to request a return for delivered items within the applicable return window (the longer of the policy window — currently 30 days — and any statutory cancellation period).	M	Window end date shown per item; expired items direct the customer to Customer Service.	R2
FR-RET-02	A return request shall capture items, quantities, reason, optional photos (mandatory for damage claims) and preferred resolution (refund or replacement).	M	Up to 5 photos, JPEG/PNG ≤ 10 MB each.	R2
FR-RET-03	The system shall issue a return label or QR code for the chosen method; return delivery shall be free when the reason is the seller's fault (damaged, wrong, not as described).	M	Charge applied only for change-of-mind returns where policy allows.	R2
FR-RET-04	Return status shall progress Requested → Label issued → In transit → Received → Inspected → Refunded/Replaced/Rejected, visible to customer, seller and agents.	M	Each transition triggers a notification (see Notification catalogue).	R2
FR-RET-05	Refunds shall be issued to the original payment method within 14 days of the return being received (or evidence of return), including the original standard delivery charge where legally required.	M	Refund amount = item price paid (after pro-rated voucher) + applicable delivery.	R2
FR-RET-06	Sellers shall be able to accept, inspect and dispute returns within an SLA; disputes are escalated to the marketplace for a final decision.	S	Seller has 2 business days to inspect after receipt; otherwise auto-accepted.	R2
FR-RET-07	Refund cost shall be allocated to the responsible party (seller or marketplace) and reflected in settlement (FR-PAY-03).	M	Settlement line of type "Refund" with referral-fee reversal.	R2

8.27 Seller Payouts and Settlement (PAY) – R2

Sellers are paid their sales less fees, refunds and reserves on a regular schedule, with transparent statements that reconcile to orders.

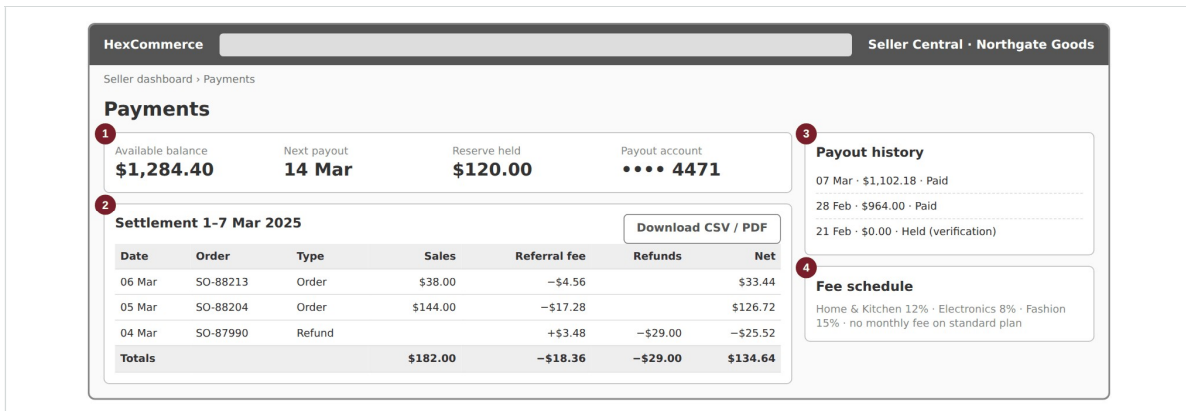


Figure 29: Wireframe (R2): seller payments page with balance, settlement statement, payout history and fee schedule

Table 40: Seller Payouts and Settlement: wireframe annotations

#	Annotation
1	Headline balance, next payout date, reserve held and masked payout account.
2	Settlement statement per period with order-level sales, referral fee, refunds and net; downloadable.
3	Payout history including holds and reasons.
4	Category referral-fee schedule applicable to the seller.

Table 41: Seller Payouts and Settlement requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-PAY-01	Payout accounts shall be captured and verified after application approval (bank account ownership check) before any payout is made.	M	Unverified account → payouts held with reason shown.	R2
FR-PAY-02	Referral fees shall be calculated per order line using the category fee schedule in force at the time of sale.	M	Fee schedule versioned; historic orders keep their original rate.	R2
FR-PAY-03	Settlement shall aggregate, per period, order sales, delivery charges collected, referral fees, refunds (with fee reversals), voucher costs funded by the seller, and adjustments.	M	Statement totals reconcile to the sum of lines to the cent.	R2
FR-PAY-04	Funds for an order shall become available a configurable number of days after delivery (default 7) and be paid on a configurable schedule (default weekly).	M	Balance breakdown shows pending vs available.	R2
FR-PAY-05	The marketplace shall be able to hold a rolling reserve or freeze payouts for risk reasons, with the reason communicated to the seller.	M	Hold and release actions audited.	R2
FR-PAY-06	Sellers shall be able to download statements (CSV, PDF) and tax documents (e.g. fee invoices, annual summaries).	S	Fee invoice issued per period.	R2

8.28 Fraud and Risk Management (FRD) – R2

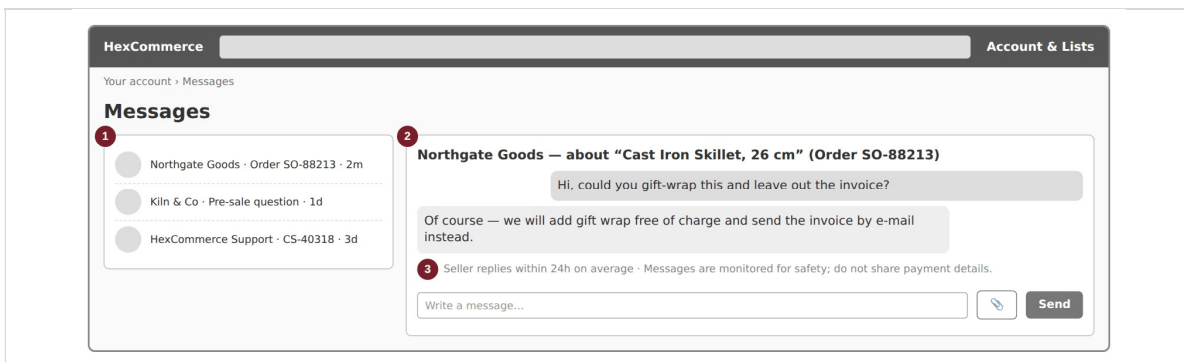
Controls to prevent payment fraud, account takeover, promotion abuse and fraudulent sellers while keeping friction low for legitimate users.

Table 42: *Fraud and Risk Management requirements*

ID	Requirement	Priority	Acceptance criteria	Release
FR-FRD-01	Every order shall be risk-scored before payment capture using payment, device, account-age, address and velocity signals; high-risk orders shall be held for manual review.	M	Held orders are not released to sellers until approved.	R2
FR-FRD-02	Velocity limits shall apply to sign-in attempts, account creation, voucher attempts, payment attempts and review submission.	M	Limits configurable; breaches logged and trigger step-up (CAPTCHA/MFA).	R2
FR-FRD-03	New sign-ins from unrecognised devices, and changes to e-mail, password, addresses or payout accounts, shall trigger notifications and, for sensitive changes, re-authentication.	M	Payout-account change holds the next payout for 72 hours.	R2
FR-FRD-04	Seller risk monitoring shall flag anomalies (e.g. sudden price drops, high cancellation or not-received rates, counterfeit reports) for Trust & Safety review.	S	Flags appear in the moderation queue (FR-MOD).	R2
FR-FRD-05	Chargebacks shall be ingested from the PSP, linked to the order and, where the seller is responsible, debited through settlement.	S	Chargeback case visible to agents and the seller.	R2

8.29 Buyer–Seller Messaging (MSG) – R2

A monitored, in-platform channel for pre-sale questions and order-related communication between customers and sellers, without exposing personal contact details.

**Figure 30:** *Wireframe (R2): customer message centre with seller conversation***Table 43:** *Buyer–Seller Messaging: wireframe annotations*

#	Annotation
1	Conversation list across sellers and support, with context (order or product).
2	Thread linked to an order or product, with attachments.
3	Seller response-time indicator and safety notice.

Table 44: Buyer–Seller Messaging requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-MSG-01	Customers shall be able to message a seller from a product page (pre-sale) or an order (post-sale); each conversation is linked to that product or order.	M	“Contact seller” available on product and order pages.	R2
FR-MSG-02	Personal e-mail addresses and phone numbers shall not be exposed; messages are relayed through the platform.	M	Neither party sees the other’s contact details.	R2
FR-MSG-03	Messages shall be screened for prohibited content (payment details, off-platform payment requests, abusive language) and blocked or flagged.	M	A message containing a card number is blocked with an explanation.	R2
FR-MSG-04	Sellers shall respond within 24 hours on business days; response-time metrics shall be shown to the seller and used in seller performance.	S	Average response time displayed on the seller page.	R2
FR-MSG-05	Either party shall be able to escalate a conversation to Customer Service, which creates a support issue with the thread attached.	S	Escalation visible in the agent console.	R2

8.30 Notifications (E-mail and In-app) (NTF) – R2

Transactional notifications keep customers, sellers and staff informed at each lifecycle event. The trigger catalogue in Table “Notification trigger catalogue” is normative.

Table 45: Notifications (E-mail and In-app) requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-NTF-01	The system shall send the transactional notifications listed in the notification trigger catalogue, through the channels specified, within the stated timing.	M	Each trigger verified end-to-end in system test.	R2
FR-NTF-02	Templates shall be branded with the configured site name and theme, localised to the recipient’s locale and currency, and editable without code change.	M	Template preview available to administrators.	R2
FR-NTF-03	Users shall manage marketing preferences separately from transactional notifications; transactional messages cannot be switched off.	M	Unsubscribe link in every marketing e-mail; honoured within 48 hours.	R2
FR-NTF-04	An in-app notification centre (bell icon) shall list recent notifications with unread counts for customers and sellers.	S	Mirrors the unread indicator pattern used in Customer Service.	R2
FR-NTF-05	Delivery status (sent, delivered, bounced, opened) shall be logged and failed transactional e-mails retried and surfaced to agents.	S	Bounce on order confirmation flags the customer record.	R2

Table 46: Notification trigger catalogue

ID	Event	Recipient	Channel	Timing
NT-01	Account created / e-mail verification	Customer	E-mail	Immediately
NT-02	Password reset requested; password or e-mail changed; new-device sign-in	Customer / Seller	E-mail	Immediately
NT-03	Order confirmation (items, shipments, totals, delivery promise)	Customer	E-mail + in-app	≤ 5 minutes after placement
NT-04	New order received (per seller order)	Seller	E-mail + in-app	≤ 5 minutes
NT-05	Shipment dispatched (carrier, tracking link)	Customer	E-mail + in-app	On dispatch confirmation
NT-06	Out for delivery / delivered / delivery exception	Customer	In-app (+ e-mail for exceptions)	On carrier event
NT-07	Order or shipment cancelled; refund issued	Customer	E-mail + in-app	Immediately
NT-08	Return requested / label issued / received / refunded / rejected	Customer, Seller	E-mail + in-app	On each status change
NT-09	Support issue acknowledged; agent reply; status change	Customer	E-mail + in-app	Immediately
NT-10	New buyer message; unanswered message reminder (20 h)	Seller	E-mail + in-app	Immediately / 20 h
NT-11	Seller application submitted; approved; rejected (with reason)	Applicant	E-mail + in-app	Immediately
NT-12	Listing rejected or needs changes; listing out of stock	Seller	E-mail + in-app	Immediately
NT-13	Deal started / ending in 24 h / ended	Seller	In-app	On event
NT-14	Payout sent; payout held (with reason); statement available	Seller	E-mail + in-app	On event
NT-15	Review published or rejected	Review author	E-mail	On moderation decision
NT-16	Price drop / back in stock on a wish-listed item (opt-in marketing)	Customer	E-mail	Daily digest
NT-17	SLA breach imminent; order held for fraud review	Staff	In-app (console)	Real time

8.31 Customer Service Agent Console (CSA) – R2

The internal tool agents use to work the issues customers raise in the help desk (FR-CS), with SLAs, macros and the ability to act on orders.

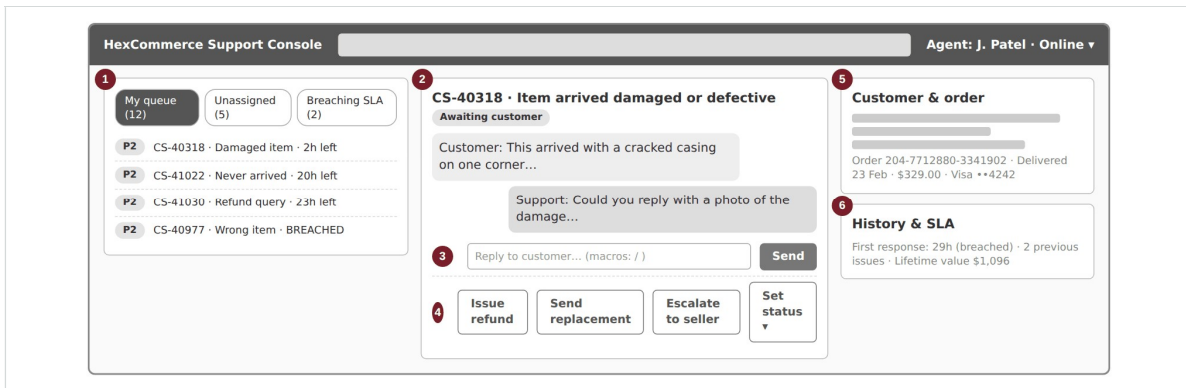


Figure 31: Wireframe (R2): agent console with SLA-ordered queue, thread, actions and customer context

Table 47: Customer Service Agent Console: wireframe annotations

#	Annotation
1	Queues (mine, unassigned, breaching SLA) ordered by time to SLA breach, with priority.
2	Issue thread identical to the customer view, with status control.
3	Reply composer with macros/templates.
4	Order actions: refund, replacement, escalate to seller, set status.
5	Customer and order context: order, payment method (masked), delivery.
6	SLA timers, previous issues and customer value.

Table 48: Customer Service Agent Console requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CSA-01	Agents shall work issues from queues (assigned to me, unassigned, breaching SLA) sorted by time remaining to SLA breach; issues can be assigned, reassigned and prioritised (P1–P4).	M	An unassigned issue older than 1 hour is highlighted.	R2
FR-CSA-02	Agents shall reply to customers, add internal notes (not visible to customers) and set status (In review, Awaiting your reply, Resolved) per BR-CS-01.	M	Customer sees replies and status, never internal notes.	R2
FR-CSA-03	Agents shall be able to issue full or partial refunds, trigger replacements and create returns from the console, within role-based monetary limits.	M	Refunds above the agent limit require supervisor approval.	R2
FR-CSA-04	Macros/templates with placeholders (customer name, order number) shall be available and centrally maintained.	S	Macro usage reported.	R2
FR-CSA-05	SLA policies (first response 24 h; resolution targets per topic) shall be configurable and breaches reported.	M	Dashboard shows % within SLA by day and agent.	R2
FR-CSA-06	Agents shall see the customer's orders, previous issues and messages in one view, with sensitive data masked.	M	Full card numbers never displayed.	R2

8.32 Catalogue, Review and Content Moderation (MOD) – R2

Trust & Safety tooling to review listings, images and reviews against policy, act on customer reports and keep an audit trail.

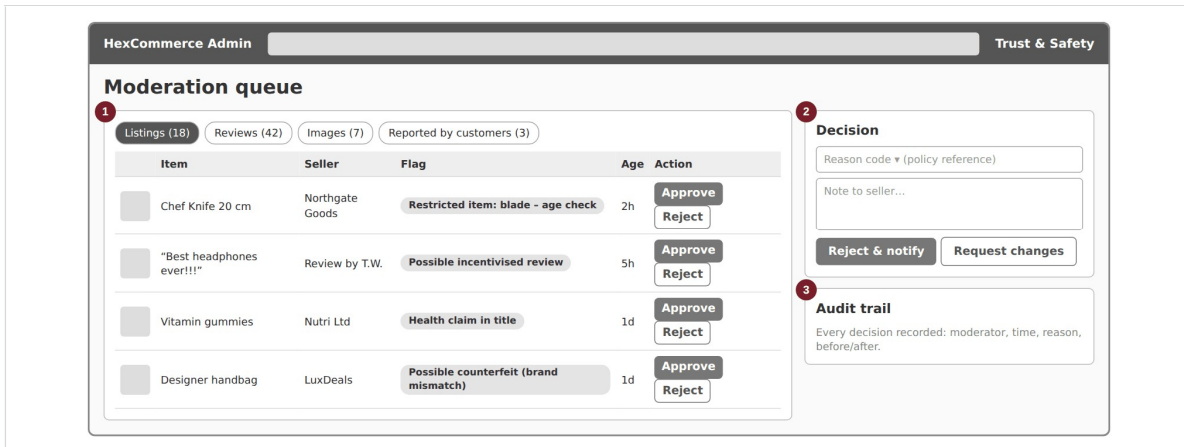


Figure 32: Wireframe (R2): moderation queue for listings, reviews, images and customer reports

Table 49: Catalogue, Review and Content Moderation: wireframe annotations

#	Annotation
1	Tabbed queues with automated flag reason, seller and age; bulk approve/reject.
2	Decision panel with policy reason code and note to seller.
3	Audit trail for every decision.

Table 50: Catalogue, Review and Content Moderation requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-MOD-01	New and edited listings, images and reviews shall be screened automatically (prohibited/restricted items, counterfeit indicators, profanity, personal data, incentivised-review signals) and flagged items routed to a moderation queue.	M	Flagged listings are not visible to customers until approved.	R2
FR-MOD-02	Moderators shall approve, reject or request changes, selecting a policy reason code; sellers and review authors are notified with the reason.	M	Reason code mandatory on reject.	R2
FR-MOD-03	Customers shall be able to report a listing or review ("Report this"), creating a moderation case.	M	Report form offers reasons; duplicate reports aggregate.	R2
FR-MOD-04	Every moderation decision shall be recorded in an audit log and sellers shall be able to appeal once.	M	Appeal routed to a different moderator.	R2
FR-MOD-05	Restricted categories (e.g. knives, age-restricted goods) shall require additional seller approval and customer age verification at checkout where legally required.	S	Chef knife listing blocked until category approval.	R2

8.33 User, Role and Order Administration (UAD) – R2

Back-office administration of customer and seller accounts, staff roles and orders, complementing the seller-application queue (FR-ADM).

Table 51: User, Role and Order Administration requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-UAD-01	Administrators shall search customers and sellers by name, e-mail, order number or seller id and view account details, status and history.	M	Search returns within 2 seconds.	R2
FR-UAD-02	Administrators shall be able to suspend, reinstate and close accounts, with a mandatory reason, and trigger password resets.	M	Suspended seller listings are hidden immediately.	R2
FR-UAD-03	Staff roles shall be managed with least privilege (e.g. Administrator, Trust & Safety, CS Agent, CS Supervisor, Merchandiser, Finance) and changes audited.	M	A CS Agent cannot access the seller-application queue.	R2
FR-UAD-04	Administrators shall search and view any order and its shipments, payments, refunds, returns, messages and issues, and perform permitted interventions (cancel, refund, re-send confirmation).	M	All interventions audited.	R2
FR-UAD-05	Data subject requests (access, rectification, erasure) shall be fulfilled through an administrative workflow.	M	Erasure anonymises personal data while retaining financial records required by law.	R2

8.34 Merchandising Content Management (CMS) – R2

Merchandisers manage the home-page hero slides, promotional tiles, editorial banners and curated rails without code changes or redeployment.

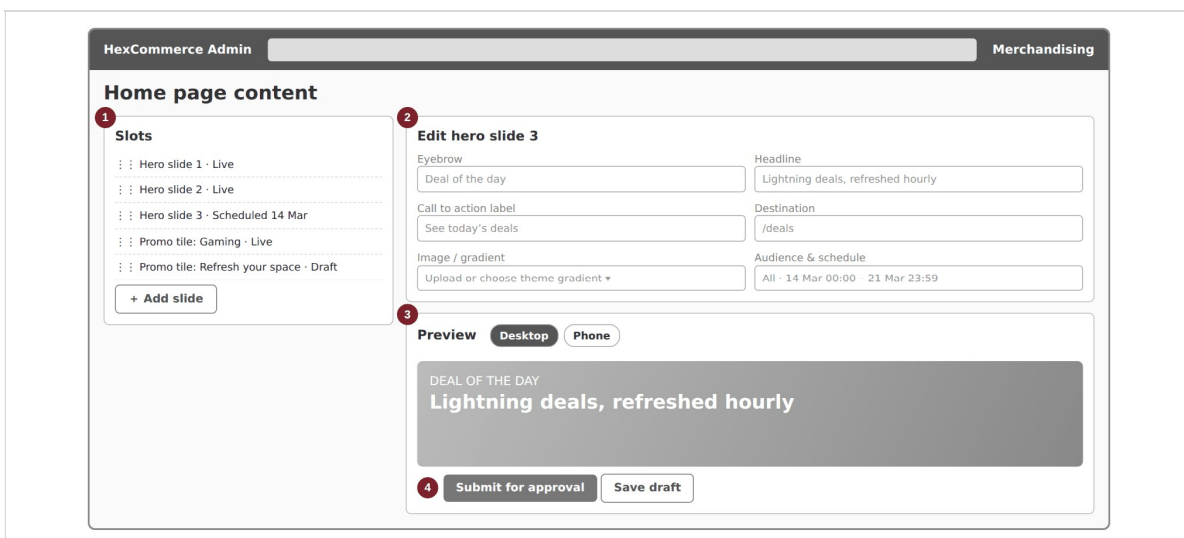


Figure 33: Wireframe (R2): home-page content editor with slots, slide editor, preview and approval

Table 52: Merchandising Content Management: wireframe annotations

#	Annotation
1	Ordered content slots with status (Live, Scheduled, Draft); drag to reorder.
2	Slide fields matching the current HeroSlide model, plus audience and schedule.
3	Desktop and phone preview using the live theme.
4	Two-step publishing: submit for approval, then publish.

Table 53: Merchandising Content Management requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-CMS-01	Merchandisers shall create, edit, reorder, schedule and retire hero slides, promotional tiles, editorial banners and curated product rails.	M	Changes visible on the storefront within 5 minutes of publishing.	R2
FR-CMS-02	Content items shall have start/end date-times and optional audience targeting (e.g. new vs returning customers).	S	Expired content disappears automatically.	R2
FR-CMS-03	A preview shall render content in desktop and phone layouts using the active theme before publishing.	M	Preview matches the published result.	R2
FR-CMS-04	Publishing shall require approval by a second user with the Merchandiser Lead role; all versions retained with rollback.	S	Rollback restores the previous version in one action.	R2
FR-CMS-05	Content shall enforce accessibility rules: alt text for images, minimum colour contrast for text on gradients, meaningful link text.	M	Publishing blocked when alt text is missing.	R2

8.35 Search Engine Optimisation and Discoverability (SEO) – R2

Requirements that make catalogue pages discoverable and correctly represented in search engines and social previews.

Table 54: Search Engine Optimisation and Discoverability requirements

ID	Requirement	Priority	Acceptance criteria	Release
FR-SEO-01	Product, department and search pages shall have unique titles, meta descriptions and canonical URLs; filtered/sorted variants shall canonicalise to the unfiltered page unless explicitly indexable.	M	No duplicate-title warnings in site audit.	R1/R2
FR-SEO-02	An XML sitemap (products, departments, informational pages) and robots.txt shall be generated and kept current; account, checkout, seller and admin routes shall be excluded from indexing.	M	Sitemap updates within 24 hours of catalogue changes.	R2
FR-SEO-03	Product pages shall include schema.org structured data (Product, Offer, AggregateRating, BreadcrumbList).	M	Passes rich-results validation.	R2

ID	Requirement	Priority	Acceptance criteria	Release
FR-SEO-04	Open Graph and social card metadata (title, description, image) shall be provided for product and department pages.	S	Shared links render a preview card.	R2
FR-SEO-05	Removed products shall return 404/410 or redirect (301) to the closest department; URL slugs shall remain stable once published.	S	No redirect chains longer than one hop.	R2

9. Business Rules

Business rules are the policies and calculations the solution enforces. They are referenced by functional requirements and must be implemented consistently in every channel and service.

Table 55: Business rules catalogue

ID	Category	Rule
BR-PRC-01	Pricing	Items subtotal = Σ (unit price $\times$ quantity) over basket lines.
BR-PRC-02	Pricing	Savings = Σ ((list price – price) $\times$ quantity) for lines whose product has a list price.
BR-PRC-03	Delivery	Delivery is FREE when the basket is empty or the subtotal $\geq$ free-delivery threshold (default 35.00); otherwise a flat fee of 5.99 applies. Remaining amount to qualify = threshold – subtotal.
BR-PRC-04	Tax	Estimated tax = subtotal $\times$ tax rate (default 8.75%). Tax is calculated on the item subtotal only, not on delivery.
BR-PRC-05	Pricing	Order total = subtotal + delivery + tax. Discount percentage = $\text{round}((\text{list} - \text{price}) / \text{list} \times 100)$; shown only when list price > price.
BR-CRT-01	Basket	Maximum 10 units of any single product per basket; every add path (add, set quantity, move from wish list, move from saved) is capped at 10.
BR-CRT-02	Basket	Setting a line quantity to 0 removes the line. “Save for later” moves the whole line; moving back adds 1 unit.
BR-INV-01	Inventory	Stock = 0 $\rightarrow$ “Currently unavailable” (purchase controls disabled); 1–10 $\rightarrow$ “Only N left in stock - order soon”; > 10 $\rightarrow$ “In stock”. Buy-box quantity options = 1 to $\min(10, \text{stock})$.
BR-DLV-01	Delivery	Express (default) arrives next day; Standard arrives in four days; both free in R1. Product-card and basket estimates use next-day.
BR-ORD-01	Orders	Order reference format 204-NNNNNNN-NNNNNNN. Tracking stages in order: Processing, Shipped, Out for delivery, Delivered; Cancelled is terminal and suppresses the tracker.
BR-CAT-01	Merchandising	Popularity score = rating $\times \log_{10}(\text{review count} + 10)$. “Best sellers” = top 12 by score. “Featured” sort = badged products first, then by popularity score.
BR-CAT-02	Sorting	Price ascending/descending by price; Avg. Customer Review by rating then review count; Most Reviewed by review count; Newest Arrivals by catalogue insertion order (latest first).
BR-CAT-03	Filtering	Filters combine with AND. Price bands: Under \$25; \$25–\$50; \$50–\$100; \$100–\$200; \$200–\$500; \$500 & above (inclusive lower and upper bounds). Rating filter = average $\geq$ N. In-stock = stock > 0. On sale = list price > price.
BR-CAT-04	Pagination	12 results per page; page number clamped to [1, page count]; window shows first, last, current ± 1 with ellipses when page count > 7.
BR-SRC-01	Search	Query split on whitespace, case-insensitive. Each term must appear in title, brand, summary, tags or department. Score per term: title +10, brand +6, tags +4, department +3, sub-department +3, summary +2. Ties broken by rating. Suggestions: ≥ 2 characters, ≤ 8 results.
BR-REV-01	Reviews	Rating 1–5 required; headline 3–80 characters; body 10–2,000 characters (after trimming).
BR-REV-02	Reviews	“Verified purchase” applies when the product appears in any order of the customer.

ID	Category	Rule
BR-REV-03	Reviews	Displayed average = (catalogue rating × catalogue count + Σ own ratings) / (catalogue count + own count). Histogram buckets weighted by $\exp(-1.6 \times \text{stars} - \text{average})$, normalised to 100%.
BR-CS-01	Support	New issue → Open. A customer reply to an issue in “Awaiting your reply” or “Resolved” moves it to “In review”; other statuses are unchanged. “Awaiting your reply” is labelled “Your reply needed”. Cancelled orders are not eligible for issues.
BR-SEL-01	Onboarding	Required: business name, contact, e-mail (pattern x@y.z), phone, years trading (≥ 0), description, address line 1, city, state/county, postcode, country, tax number. Optional: website, address line 2.
BR-SEL-02	Onboarding	Application lifecycle: Pending → Approved Rejected (with optional note); an admin can move a decided application back to Pending, clearing the decision date and note.
BR-SLR-01	Listings	Title and brand required; price > 0; optional “was” price must be > price; stock is an integer ≥ 0 ; summary (≤ 160 chars) and description required.
BR-SLR-02	Listings	Status on save: “Save as draft” → Draft. Publish with stock ≤ 0 → Out of stock. Publish otherwise → Active (an Unpublished, Draft or Out-of-stock listing becomes Active). Publish from table: stock > 0 → Active else Out of stock.
BR-SLR-03	Deals	Discount integer 5%–70%; units ≥ 1 and $\leq$ listing stock; end date > start date; only Active listings eligible. On create/edit: start date after today → Scheduled, otherwise Live; an Ended deal stays Ended when edited. Deal price = $\text{round}(\text{price} \times (1 - \% / 100), 2)$.
BR-SLR-04	Listings	Deleting a listing deletes all deals on that listing.
BR-SLR-05	Seller KPIs	Billable orders = orders not Cancelled. Units sold = Σ quantity; Revenue = Σ unit price × quantity; Estimated payout = revenue × (1 – referral fee 12%). Active listings and Live deals are counts by status.
BR-SLR-06	Fulfilment	Seller order lifecycle: New → Dispatched → Delivered; Cancelled terminal.
BR-IMG-01	Media	Max 6 images per listing; each original ≤ 8 MB and of type image/*; stored at ≤ 900 px longest edge, JPEG q = 0.82; first image is the cover.
BR-MSO-01	Orders (R2)	One seller order (shipment) is created per fulfilling party per customer order; delivery charges and free-delivery thresholds are evaluated per shipment unless the marketplace funds delivery.
BR-PRM-01	Vouchers (R2)	Voucher discount applies to eligible lines only, never exceeds their value, excludes delivery unless it is a free-delivery voucher, and is pro-rated across eligible lines by value for refunds.
BR-RET-01	Returns (R2)	Return window = the longer of the policy window (30 days) and any statutory cancellation period, counted from the delivery date; refund issued within 14 days of receipt of the returned item.
BR-PAY-01	Settlement (R2)	Net payout = sales + delivery charges collected – referral fees – refunds + referral-fee reversals on refunds – seller-funded voucher cost ± adjustments – chargebacks – reserve movement.
BR-GEN-01	Identifiers	Reference formats: support CS-NNNNN, seller application SA-NNNNN, listing SL-NNNNN, deal SD-NNNNN, seller order SO-NNNNN. R2: identifiers shall be generated server-side and guaranteed unique.

Worked example – basket totals (default configuration)

Basket: 1 × \$24.99 (list \$29.99) and 2 × \$12.50. Subtotal = \$49.99. Savings = \$5.00. Subtotal $\geq$ \$35 → Delivery FREE. Estimated tax = $\$49.99 \times 8.75\% = \4.37 . Order total = \$54.36. Had the subtotal been

\$30.00, delivery would be \$5.99 and the message would read “Add \$5.00 of eligible items to qualify for FREE delivery.”

Worked example – seller payout

Seller orders: 1 × \$38 (New), 2 × \$72 (New), 1 × \$38 (Dispatched), 1 × \$29 (Delivered), 1 × \$72 (Cancelled). Billable units = 5; Revenue = \$249.00; Estimated payout = $\$249.00 \times (1 - 12\%) = \219.12 .

10. Data Requirements

10.1 Logical Data Model

The following entities are required. Attributes marked “?” are optional. All timestamps use ISO 8601 (UTC); monetary amounts are decimal values in the configured ISO 4217 currency.

Table 56: Entity catalogue

Entity	Description	Key attributes
Product	Purchasable catalogue item	id, slug, title, brand, category, subcategory?, price, listPrice?, rating (0–5), reviewCount, stock, badge?, summary, description, features[], specs{}, colors[]?, sizes[]?, tags[], glyph?, image?, images[]?
Category	Department with optional sub-departments	id, slug, name, blurb, glyph, subcategories[{slug, name}]
Review	Customer review of a product	id, productId, author, rating (1–5), title, body, date (ISO 8601), verified, helpfulCount
CartLine	Basket or order line	productId, quantity
Order	Customer order	id, placedOn, deliveredOn?, status (Processing Shipped Out for delivery Delivered Cancelled), total, shipTo, items[CartLine]
Address	Saved delivery address	id, label, name, line1, line2?, city, state, postalCode, country, isDefault
PaymentMethod	Saved payment instrument (masked)	id, brand, last4, expiry, isDefault
User	Signed-in customer	name, email, memberSince
SupportIssue	Customer-service case	id, orderId, productId, topic, status (Open In review Awaiting your reply Resolved), createdAt, updatedAt, read, messages[]
SupportMessage	Message in a case thread	id, from (customer support), author, body, sentAt
Deal (marketplace)	Lightning deal on a catalogue product	productId, percentOff (1–99), claimed (0–1), endsIn (label)
HeroSlide	Home carousel slide	id, eyebrow, title, body, ctaLabel, ctaHref, gradient[2], glyph
SellerApplication	Prospective seller application	id, businessName, businessType, contactName, email, phone, website?, description, yearsTrading, line1, line2?, city, state, postcode, country, taxId, status (Pending Approved Rejected), submittedAt, decidedAt?, note?
SellerProfile	Onboarded seller business record	businessName, businessType, contactName, email, phone, website?, description, registeredSince, category, address fields, taxId, payoutAccount (masked), status (Verified Pending verification)
SellerListing	Product listed by a seller	id, title, brand, category, subcategory?, condition, price, listPrice?, stock, status (Active Draft Out of stock Unpublished), summary, description, features[], glyph?, images[]?, createdAt, updatedAt
SellerOrder	Order against a seller listing	id, listingId, buyer, city, quantity, unitPrice, placedOn, status (New Dispatched Delivered Cancelled)

Entity	Description	Key attributes
SellerDeal	Seller-run time-limited discount	id, listingId, percentOff (5–70), dealStock, claimed, status (Scheduled Live Ended), startsOn, endsOn, createdAt, updatedAt

The diagram shows the target logical model for the core entities (primary key PK, foreign key FK; crow's foot = many). Order lines carry the seller so that one customer order can span several sellers.

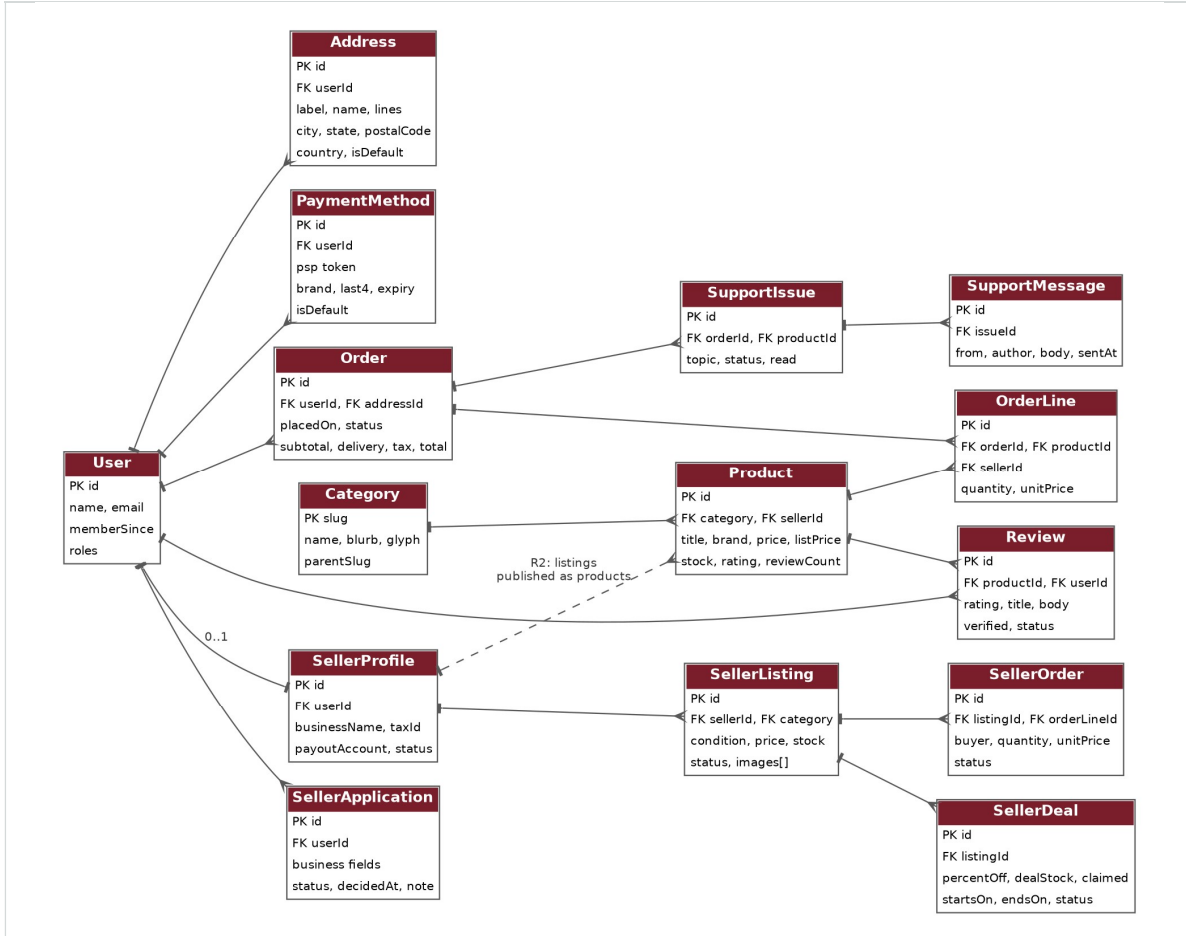


Figure 34: Logical data model – core entities

10.2 Additional Entities for Production (R2)

Table 57: R2 entity catalogue

Entity	Description	Key attributes
Shipment	Portion of an order fulfilled by one party	id, orderId, sellerId, deliveryOption, charge, carrier, trackingNumber, status, promisedDate
ShippingRule / Zone	Seller delivery offer for a destination zone	id, sellerId, zoneId, service, pricingType, price, freeOver?, handlingDays; Zone: id, countries, regions, postcodePatterns
Voucher / Redemption	Promotion code and its usage	id, code, type, value, validFrom, validTo, minSpend, eligibility, limits, fundedBy; Redemption: voucherId, orderId, userId, amount
Return	Return request against shipment lines	id, orderId, lines[], reason, photos[], resolution, method, status, refundAmount, responsibleParty

Entity	Description	Key attributes
Refund	Money returned to the customer	id, orderId, returnId?, amount, reason, pspReference, status, issuedBy
SettlementLine / Payout	Seller ledger entry and payout batch	Line: id, sellerId, type (sale, fee, refund, adjustment, chargeback), amount, orderLineId, period; Payout: id, sellerId, amount, status, holdReason?
MessageThread / Message	Buyer–seller conversation	Thread: id, customerId, sellerId, productId?, orderId?, status; Message: id, threadId, author, body, attachments[], screeningResult
Notification	Record of a notification sent	id, templateId, recipientId, channel, payload, status, sentAt
ModerationCase	Item under Trust & Safety review	id, subjectType, subjectId, flagReason, source (automated, report), status, decision, reasonCode, moderatorId
ContentSlot	Merchandising content item	id, slotType, fields, audience, startsAt, endsAt, status, version, approvedBy
AuditEvent	Immutable record of privileged actions	id, actorId, role, action, subjectType, subjectId, before, after, at

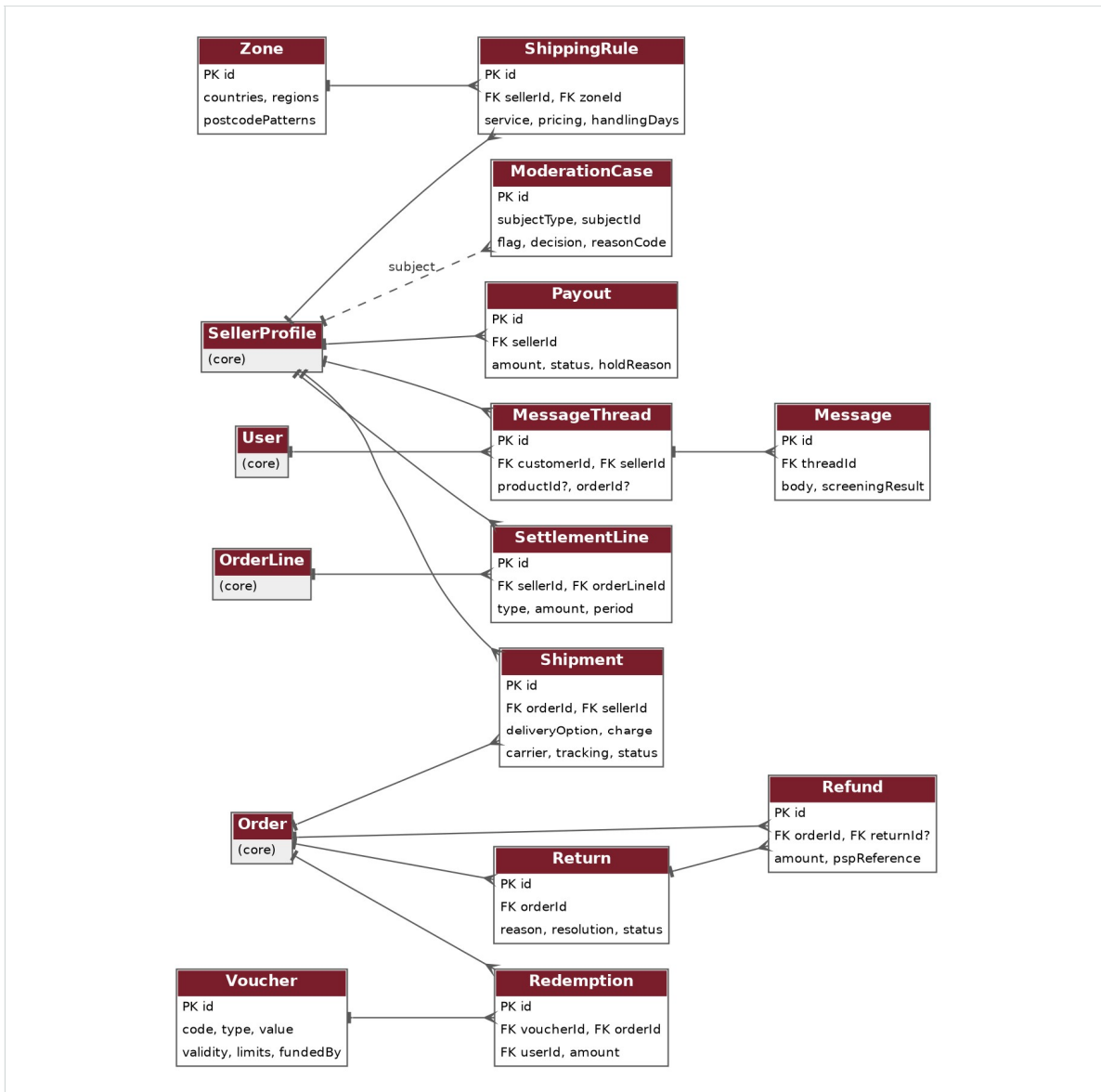


Figure 35: Logical data model – R2 entities and their links to core entities (grey)

10.3 Entity Relationships

- A **Category** contains many **Products**; a Product belongs to exactly one Category and at most one sub-department.
- A **Product** has many **Reviews**; a Review belongs to one Product and one author.
- A **User** has many **Addresses**, **PaymentMethods**, **Orders**, **SupportIssues** and one basket, wish list and saved-for-later list.
- An **Order** contains one or more lines, each referencing a Product and quantity; an Order ships to one Address.
- A **SupportIssue** references one Order and one Product and contains an ordered list of **SupportMessages**.
- A **SellerProfile** owns many **SellerListings**; a SellerListing has many **SellerOrders** and **SellerDeals**; deleting a listing deletes its deals.
- A **SellerApplication** belongs to one applicant User and, when approved, results in a SellerProfile (R2).

10.4 Status Lifecycles

Table 58: Entity lifecycles

Entity	Primary flow	Alternative transitions
Customer order	Processing → Shipped → Out for delivery → Delivered	Cancelled (terminal, no payment taken)
Support issue	Open → In review ⇄ Awaiting your reply → Resolved	Customer reply on Resolved or Awaiting → In review
Seller application	Pending → Approved Rejected	Admin “Move back to pending” → Pending
Seller listing	Draft → Active ⇄ Unpublished; Active ⇄ Out of stock (by stock)	Remove (deletes listing and its deals)
Seller deal	Scheduled → Live → Ended	Start now (Scheduled → Live); End deal (→ Ended); Remove
Seller order	New → Dispatched → Delivered	Cancelled (terminal)

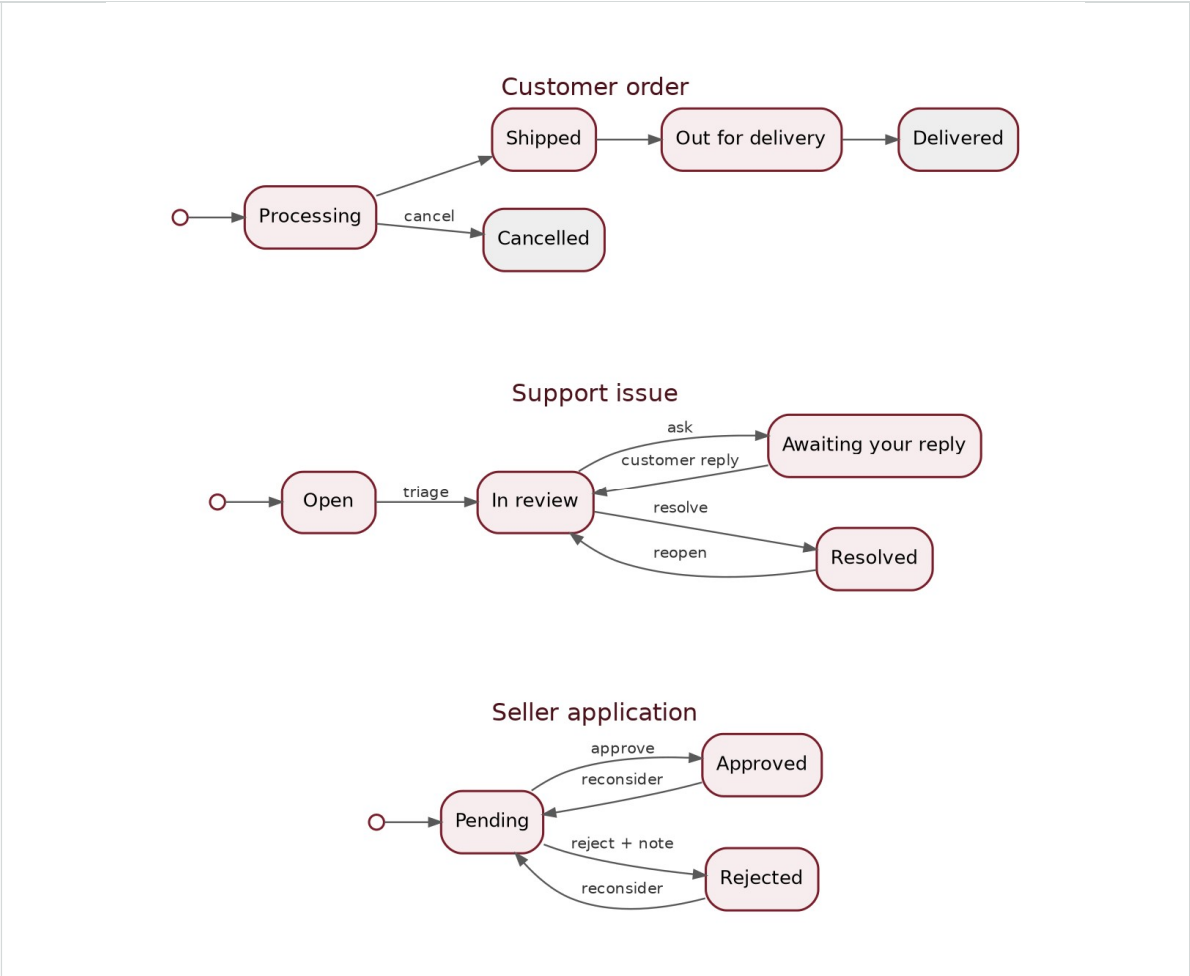


Figure 36: State diagrams: customer order, support issue, seller application

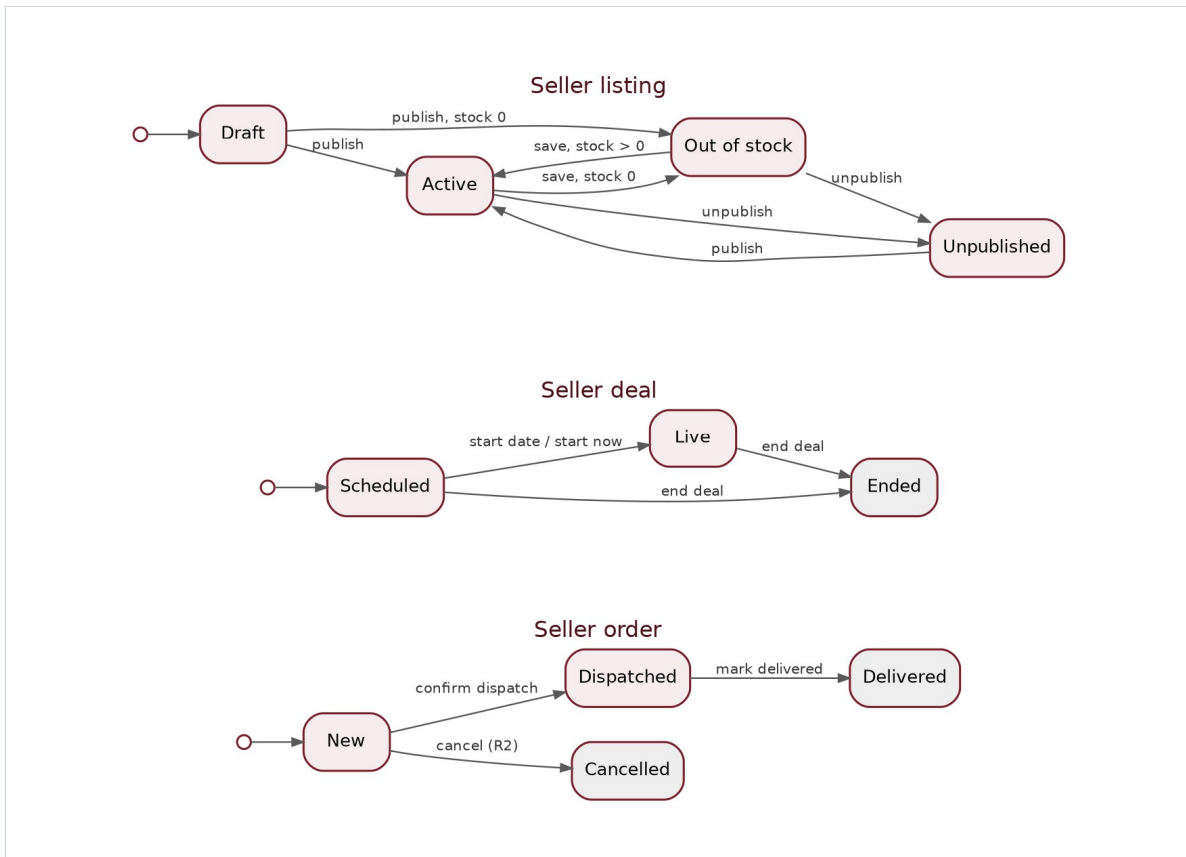


Figure 37: State diagrams: seller listing, seller deal, seller order

10.5 Reference Data

Reference set	Values
Departments (8)	Electronics (Audio, Computers, Cameras, Wearables); Home & Kitchen (Cookware, Small appliances, Décor, Bedding); Fashion (Men's, Women's, Footwear, Accessories); Books (Fiction, Non-fiction, Children's); Sports & Outdoors (Fitness, Camping, Cycling); Beauty & Personal Care (Skincare, Haircare, Fragrance); Toys & Games (Building sets, Board games, Outdoor play); Grocery (Coffee & tea, Snacks, Pantry).
Listing conditions	New; Renewed; Used – like new; Used – good.
Business types	Sole trader; Partnership; Limited company; Non-profit.
Support topics	See FR-CS-02 (nine values).
Sort orders	Featured; Price: Low to High; Price: High to Low; Avg. Customer Review; Most Reviewed; Newest Arrivals.
Theme presets	maroon (default); midnight; forest; slate; amber; plum.

10.6 Data Persistence and Retention

In R1 all mutable state is stored in the browser's local storage under the keys below. Each store reads its snapshot after first render (to keep server and client output identical), falls back to seeded data when storage is empty, unavailable or corrupted, and writes back only after loading.

Table 59: R1 browser storage keys

Storage key	Content	Owner
hexcommerce.store.v1	Basket lines, wish list, saved-for-later	Customer
hexcommerce.deliverTo.v1	Selected delivery address id	Customer
hexcommerce.reviews.v1	Customer-written reviews keyed by product	Customer
hexcommerce.support.v1	Customer-service issues and threads	Customer
hexcommerce.sellerApplications.v1	Seller application queue and the customer's own application id	Customer / Admin
hexcommerce.seller.v1	Seller listings (incl. image data URLs), seller orders, seller deals	Seller

R2 requirement: all of the above shall be persisted server-side, associated with the authenticated account, and subject to a documented retention schedule. Indicative periods, to be confirmed by Legal/DPO: orders and invoices 7 years for tax purposes; support threads 3 years; abandoned baskets 90 days; rejected seller applications 2 years. Legacy browser data shall be merged into the account at first sign-in.

11. Non-Functional Requirements

Non-functional requirements are classified using the ISO/IEC 25010 product quality model. They apply to all modules unless stated otherwise.

Table 60: Non-functional requirements

ID	Quality characteristic	Requirement	Priority
NFR-PERF-01	Performance efficiency	Pages shall meet Core Web Vitals “good” thresholds at the 75th percentile on mobile: LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1.	M
NFR-PERF-02	Performance efficiency	Catalogue, listing, product, deals, orders and account pages shall be server-rendered; client-side JavaScript shall be limited to interactive islands (e.g. buy box, search, menus, rails).	S
NFR-PERF-03	Performance efficiency	Type-ahead suggestions shall appear within 100 ms of a keystroke (R1: computed locally; R2: search API p95 ≤ 150 ms).	M
NFR-PERF-04	Performance efficiency	The platform shall support at least 5× average peak traffic during promotional events without degradation beyond NFR-PERF-01.	M
NFR-USA-01	Interaction capability	The UI shall follow established marketplace conventions (header search, buy box, basket, stepper checkout) so first-time users complete a purchase unaided.	M
NFR-USA-02	Interaction capability	Every list or collection shall have a purposeful empty state with an onward action; every mutating action shall give immediate inline feedback.	M
NFR-USA-03	Interaction capability	Form validation messages shall be specific, in plain language, and shown adjacent to the form.	M
NFR-ACC-01	Accessibility	The solution shall conform to WCAG 2.2 Level AA (ISO/IEC 40500 for WCAG 2.0 baseline) and EN 301 549.	M
NFR-ACC-02	Accessibility	All functionality shall be operable by keyboard with visible focus indicators; menus and dialogs close on Escape; modal dialogs trap scroll.	M
NFR-ACC-03	Accessibility	Custom widgets shall expose correct ARIA roles/states: combobox/listbox (search), radiogroup (star input), progressbar (deals, delivery), dialog (menus), aria-current (pagination, tabs), aria-pressed (wish list).	M
NFR-ACC-04	Accessibility	Auto-advancing content shall pause on hover/focus and stop when the user prefers reduced motion.	M
NFR-ACC-05	Accessibility	Prices shall have a plain-text equivalent for assistive technology; decorative glyphs shall be hidden from assistive technology.	S
NFR-COM-01	Compatibility	The UI shall be responsive from 320 px to wide desktop with no horizontal page scrolling; layout adapts at sm/md/lg/xl breakpoints.	M
NFR-COM-02	Compatibility	Supported browsers: latest two major versions of Chrome, Edge, Firefox and Safari (desktop and mobile).	M
NFR-REL-01	Reliability	The UI shall degrade gracefully when browser storage is unavailable or corrupted, falling back to seeded/in-session state without errors.	M
NFR-REL-02	Reliability	Server-rendered and client-hydrated markup shall match (deterministic dates and artwork) to avoid hydration errors.	M

ID	Quality characteristic	Requirement	Priority
NFR-REL-03	Reliability	Production availability $\geq 99.9\%$ monthly for customer-facing journeys; RPO ≤ 15 min, RTO ≤ 1 h.	M
NFR-SEC-01	Security	All traffic shall use TLS 1.2+ with HSTS; a Content Security Policy shall restrict script and image sources.	M
NFR-SEC-02	Security	The application shall meet OWASP ASVS v4.0 Level 2; all input shall be validated server-side regardless of client validation.	M
NFR-SEC-03	Security	Passwords: minimum 12 characters (NIST SP 800-63B), breached-password screening, salted adaptive hashing; MFA available.	M
NFR-SEC-04	Security	Role-based access control shall separate Customer, Seller and Administrator capabilities; /admin routes require the Administrator role.	M
NFR-SEC-05	Security	The platform shall never store full card numbers or CVV; card capture shall use the PSP's hosted fields (PCI DSS v4.0 SAQ A scope).	M
NFR-PRV-01	Privacy	Personal data processing shall comply with UK GDPR / EU GDPR and applicable local law (lawful basis, data minimisation, retention schedule, DSAR within one month).	M
NFR-PRV-02	Privacy	Non-essential cookies/storage shall require prior consent (PECR / ePrivacy); a cookie notice and preference centre shall be provided.	M
NFR-MNT-01	Maintainability	Domain types shall be declared once and shared by data, query and UI layers; colour shall be expressed only through design tokens.	S
NFR-MNT-02	Maintainability	Code shall pass linting and type-checking in CI; unit test coverage $\geq 80\%$ for business rules (pricing, filtering, status transitions).	S
NFR-PRT-01	Flexibility	The storefront shall build and run without external fonts or remote images; real product imagery can be enabled per record.	C
NFR-I18N-01	Flexibility	All user-facing strings shall be externalised to support additional languages; formatting shall use the configured locale and currency.	S
NFR-OBS-01	Operability	Production shall provide structured logging, metrics, tracing and alerting for all critical journeys (search, add to basket, checkout).	M

11.1 Accessibility Features Observed in the Current UI

The current UI already implements the following accessibility techniques, which shall be preserved and extended:

- Skip link to main content; semantic landmarks (header, nav, main, aside, footer).
- Visible focus rings; keyboard-operable menus, drawers and type-ahead; Escape to close.
- ARIA combobox/listbox for search, radiogroup for star rating input, menuitemradio for address picker, progressbar for deal and delivery progress, aria-current for pagination and tabs, aria-pressed for wish-list toggles, aria-expanded for disclosure controls.
- Screen-reader-only plain-text prices alongside visual superscript prices; decorative glyphs hidden with aria-hidden.
- Carousel pauses on hover/focus and honours prefers-reduced-motion.
- Descriptive accessible names (e.g. "Add <product> to cart", "Shopping cart, 3 items").

11.2 Capacity and Sizing Assumptions

Planning assumptions – not a forecast
No business volumes were available for this analysis. The figures below are order-of-magnitude assumptions for a mid-sized marketplace, used only to make performance and cost requirements testable. They must be replaced with Finance and Marketing forecasts (OI-07) before gate G1.

Table 61: Sizing assumptions

Dimension	Launch	Year 1 peak	Drives
Active catalogue items (SKUs)	50,000	500,000	Search index size, image storage, sitemap size
Active sellers	500	5,000	Onboarding queue staffing, settlement batch size
Registered customers	200,000	2,000,000	Identity provider tier, personal-data volume
Peak concurrent sessions	5,000	25,000	Web tier and CDN capacity (NFR-PERF-04)
Peak orders per hour	2,000	10,000	Checkout, PSP and order-management throughput
Peak search queries per second	200	1,000	Search cluster sizing; type-ahead p95 ≤ 150 ms
Support issues per day	1,500	8,000	Agent headcount, SLA planning
Images stored	300,000	3,000,000	Media storage and CDN egress
Notifications per day	60,000	400,000	E-mail provider plan, deliverability

NFR-PERF-04 (5× average peak) and the performance tests in Section 17 shall be run against the Year 1 peak figures once validated.

12. Regulatory and Compliance Requirements

Indicative – not legal advice

The obligations below are listed to make compliance work plannable. Applicability depends on the launch market(s) (OI-01) and must be confirmed by Legal and the Data Protection Officer.

Table 62: Compliance obligations for production

ID	Obligation	Requirement
CMP-01	Payment security (PCI DSS v4.0)	Use PSP-hosted payment fields; no storage/transmission of PAN or CVV through HexCommerce systems; display only brand, last four digits and expiry; support Strong Customer Authentication (3-D Secure 2) where required (PSD2).
CMP-02	Data protection (UK GDPR / EU GDPR)	Privacy notice; lawful basis register; data minimisation; DSAR, rectification and erasure workflows; DPIA for seller KYC data; breach notification within 72 hours.
CMP-03	Cookies and storage (PECR / ePrivacy)	Consent before non-essential cookies or local storage; granular preference centre; record of consent.
CMP-04	Consumer protection (e.g. UK Consumer Contracts Regulations 2013, Consumer Rights Act 2015, EU CRD)	Clear total price including taxes and delivery before order; 14-day cancellation right for distance sales; clear returns policy; order confirmation on a durable medium.
CMP-05	Pricing transparency (e.g. UK DMCC Act 2024 / EU Omnibus Directive)	“Was” prices and discount claims must reference a genuine prior price (e.g. lowest price in prior 30 days for EU); deal urgency claims must be accurate.
CMP-06	Reviews integrity	Prohibit fake or incentivised reviews; disclose verification method (“Verified purchase”); moderation policy published.
CMP-07	Online marketplace obligations (e.g. EU Digital Services Act)	Trader traceability (KYB of sellers); display seller identity to buyers; notice-and-action for illegal content/products.
CMP-08	Accessibility (Equality Act 2010, European Accessibility Act 2025, ADA)	WCAG 2.2 AA conformance; accessibility statement published.
CMP-09	Tax	Correct VAT/sales tax calculation per jurisdiction; compliant invoices; marketplace-facilitator tax obligations where applicable.

13. Reporting and Analytics Requirements

13.1 Seller KPIs (in product)

Table 63: Seller dashboard KPI definitions

KPI	Definition	Source rule
Active listings	Count of listings with status Active.	BR-SLR-05
Live deals	Count of deals with status Live.	BR-SLR-05
Units sold	Σ quantity over seller orders not Cancelled.	BR-SLR-05
Revenue	Σ unit price $\times$ quantity over seller orders not Cancelled.	BR-SLR-05
Estimated payout	Revenue $\times$ (1 – referral fee).	BR-SLR-05
Awaiting dispatch	Count of seller orders with status New.	FR-SOR-01
Deal claimed %	$\text{round}(\text{claimed} \div \text{deal units} \times 100)$.	FR-SDL-01

13.2 Business Analytics (R2)

The business shall capture analytics events to measure the objectives in Section 2.3. Minimum event set:

Event	Key properties	Supports
search_performed	query, result_count, category scope, suggestion_used	BO-01, zero-result rate
filter_applied / sort_changed	facet, value, result_count	BO-01
product_viewed	product_id, source (rail, search, deal)	BO-01, BO-02
add_to_cart / buy_now	product_id, quantity, price, source	BO-01, BO-02
checkout_step	step (1–4), delivery_speed	BO-01 funnel
order_placed	order_id, value, items, delivery, tax	BO-01, BO-02
review_submitted	product_id, rating, verified	BO-05
support_issue_raised / replied	topic, order_age, time_to_first_response	BO-04
seller_application_submitted / decided	decision, time_to_decision	BO-03
listing_published / deal_started	listing_id, status, discount	BO-03, BO-02

14. External Interface Requirements (R2)

Integration points correspond to the dashed systems in the context diagram (Section 5.1).

Table 64: Integration points

ID	System	Direction	Purpose / data
INT-01	Identity provider	Bi-directional	Authentication, sessions, roles (Customer, Seller, Administrator), MFA, password reset.
INT-02	Payment service provider	Outbound / webhook	Tokenised card capture, authorisation, capture, refunds, 3-D Secure; payment status webhooks.
INT-03	Catalogue and search	Outbound	Products, seller listings, facets, relevance, suggestions.
INT-04	Order management / inventory	Bi-directional	Order creation, stock reservation, status events (Processing → Delivered), cancellations.
INT-05	Carrier / tracking	Inbound	Dispatch tracking numbers and delivery events.
INT-06	Tax engine	Outbound	Jurisdictional tax calculation for basket, checkout and invoices.
INT-07	Notification service	Outbound	Order confirmation, dispatch, support replies, seller decisions.
INT-08	Customer-service platform	Bi-directional	Issue creation, agent replies, status changes, attachments.
INT-09	KYC/KYB and sanctions screening	Outbound	Seller identity and business verification outcomes.
INT-10	Seller payouts / settlement	Outbound	Revenue, referral fees, payouts to verified accounts.
INT-11	Media service / CDN	Outbound	Image upload, renditions and delivery.
INT-12	Analytics	Outbound	Events defined in Section 13.2, subject to consent.

15. Gap Analysis and Recommendations

The UI review identified the following gaps between current behaviour and the business need. Each gap is rated by business impact and linked to the requirement that closes it.

Table 65: Gap register

ID	Observation	Impact	Recommendation
GAP-01	No authentication: sign-in, registration and sign-out are presentational; a single persona is always signed in; /admin/sellers is unguarded.	High	Implement identity service, sessions and RBAC (FR-AUT-03/04, NFR-SEC-04) before any public release.
GAP-02	Checkout takes no payment and creates no order; the generated reference never appears in order history.	High	Integrate PSP and order management (FR-CHK-08).
GAP-03	Default currency is USD with en-US formatting, while all sample addresses, the seller profile and tax number are UK-based; price-band and filter-chip labels hard-code “\$”.	High	Confirm the launch market; set GBP/en-GB if UK; make price-band labels currency-aware (FR-CFG-05).
GAP-04	Tax messaging is contradictory: the product page states “Inclusive of all taxes” while basket and checkout add “Estimated tax” on top of the subtotal.	High	Adopt one model per market (VAT-inclusive for UK/EU; tax-exclusive for US) and align copy and calculations.
GAP-05	Seller listings are never shown in the customer catalogue; seller deal cards on /deals are display-only.	High	Unify seller listings with the catalogue (FR-SLS-11, FR-DEA-05).
GAP-06	Product variant chips (colour/size) are not selectable; one price and stock level per product.	Medium	Introduce SKU/variant model (FR-PDP-05).
GAP-07	Order detail always shows Delivery “FREE” and recomputes tax from current prices, which may not reconcile with the stored order total.	Medium	Persist price, delivery and tax at order time and display stored values.
GAP-08	Addresses and payment methods cannot be added or edited; the corresponding tiles and “Add or edit an address” link back to the account hub.	Medium	Deliver FR-ACC-04 and FR-CHK-09.
GAP-09	Several links are placeholders: “Get invoice”, “Add a protection plan”, footer items such as Careers, Press releases, Become an affiliate, Advertise your products.	Medium	Provide destination content or hide links until available.
GAP-10	Review “Helpful” button has no effect; reviews are not moderated or shared across customers.	Medium	Deliver FR-REV-07 (R2) and FR-REV-08.
GAP-11	Deal countdowns are static labels; claimed percentages are not tied to stock; scheduled seller deals do not start automatically.	Medium	Deliver FR-DEA-04 and FR-SDL-07.
GAP-12	Business-details “Save changes” does not persist.	Low	Deliver FR-SBD-02.
GAP-13	Customer service asks for photos but has no attachment capability.	Medium	Deliver FR-CS-09.
GAP-14	“Recently viewed” is a fixed list; “Newest Arrivals” sort uses catalogue order because products have no date attribute.	Low	Track browsing events; add a published date to products.

ID	Observation	Impact	Recommendation
GAP-15	All customer/seller data lives in one browser's storage: no cross-device continuity, and large image data URLs may exhaust storage quota.	High	Move state server-side (FR-CRT-07 R2, FR-SIM-05).
GAP-16	Identifiers are generated client-side with random numbers and may collide.	Medium	Generate identifiers server-side (BR-GEN-01).
GAP-17	Sample data inconsistency: the most recent order has a delivered-on date but status "Out for delivery".	Low	Correct seed data; derive delivered date from status events.
GAP-18	At 1280 px the department strip clips the final item ("Customer Servic...") without a visible scroll affordance.	Low	Add scroll fade/arrow or move Customer Service to the "All" menu at that width.
GAP-19	Standard delivery fee (5.99) and referral fee (12%) are hard-coded; the referral fee on /sell is described as 8%–15% by category but a flat 12% is used for payout estimates.	Medium	Make fees configurable per category (FR-CFG-05).
GAP-20	Search has no typo tolerance, synonyms or analytics on zero-result queries.	Medium	Deliver FR-SRC-08 with search analytics.

15.1 Change Control

Once approved, this BRD is baselined. Proposed changes shall be submitted as change requests stating the requirement IDs affected, rationale, impact on scope, cost, schedule and risk. Changes are assessed by the Product Owner and approved by the Executive Sponsor for Must-priority items; approved changes increment the minor version (e.g. 1.1) and are recorded in the revision history.

16. Risks, Issues and Decisions Required

16.1 Risks

Table 66: Risk register

ID	Risk	Likelihood	Impact	Mitigation
RSK-01	Stakeholders mistake the high-fidelity prototype for a production-ready system.	Medium	High	Clear R1/R2 labelling in this BRD; demonstration disclaimers retained in UI until R2.
RSK-02	Regulatory non-compliance at launch (payments, privacy, consumer rights, accessibility).	Medium	High	Compliance workstream with legal sign-off; PCI SAQ A scope; WCAG audit before go-live.
RSK-03	Market ambiguity (US vs UK) delays tax, currency and returns design.	High	Medium	Decision required from Sponsor (see Open Issues, OI-01).
RSK-04	Seller catalogue quality (images, descriptions) degrades customer experience.	Medium	Medium	Content policy checks (FR-SLS-12) and seller guidance.
RSK-05	Peak-event traffic overwhelms search and checkout.	Low	High	Load testing to NFR-PERF-04; autoscaling; CDN caching of server-rendered pages.
RSK-06	Fraudulent sellers pass onboarding.	Medium	High	KYC/KYB and sanctions screening (FR-SEL-08); manual review of high-risk categories.
RSK-07	R1 acceptance criteria codify prototype behaviour (e.g. hard-coded fees, tax model) that the business does not actually want.	Medium	Medium	Validation workshops review every R1 criterion; known cases logged as gaps (Section 15).
RSK-08	Objectives, KPI targets, priorities and sizing are analyst proposals, not validated business inputs.	High	High	Phase 0 validation with named owners (Section 1.4.5); no build commitment before gate G1.
RSK-09	Multi-seller orders increase operational complexity (split shipments, partial refunds, settlement).	Medium	Medium	Pilot with a small seller cohort; settlement reconciliation tests (FR-PAY-03).

16.2 Open Issues and Decisions Required

Table 67: Open issues

ID	Issue / decision	Owner
OI-01	Confirm launch market(s), currency and tax model (VAT-inclusive vs sales-tax-exclusive).	Executive Sponsor / Finance
OI-02	Confirm fulfilment model: seller-fulfilled, marketplace-fulfilled, or both, and associated fees.	Seller Operations
OI-03	Confirm return policy (30-day refund or replacement shown in the buy box) against statutory rights in each market.	Legal
OI-04	Select identity provider, payment service provider, search engine and e-mail service.	Engineering Lead / Procurement

ID	Issue / decision	Owner
OI-05	Define referral-fee schedule per category (current UI: 8%–15%; payout estimate: flat 12%).	Finance
OI-06	Define review moderation policy and service levels.	Trust & Safety
OI-07	Provide demand and catalogue forecasts to replace the sizing assumptions in Section 11.2.	Finance / Marketing
OI-08	Decide who funds voucher discounts and free delivery across sellers.	Finance / Merchandising
OI-09	Decide payout schedule, availability delay and reserve policy.	Finance / Trust & Safety
OI-10	Select fraud-screening provider and define manual-review staffing.	Security / Finance
OI-11	Schedule validation workshops per Section 1.4.5 and name attendees.	Product Owner

17. Transition Requirements

Transition requirements (BABOK) describe the capabilities needed to move from the current state to the production solution; they are temporary and end with hypercare exit.

17.1 Acceptance Approach

1. **Requirements validation** – workshops confirm every item marked Proposed in Section 1.4.5 before build commitment (gate G1).
2. **System testing** – every acceptance criterion in Section 8 is converted into at least one test case (Appendix D); business rules in Section 9 are covered by automated unit tests.
3. **Accessibility audit** – independent WCAG 2.2 AA audit with assistive-technology testing (screen reader, keyboard, zoom 400%).
4. **Performance testing** – load and soak tests at 5× forecast peak against NFR-PERF-01/04.
5. **Security testing** – penetration test against OWASP ASVS L2; PCI SAQ A attestation.
6. **User acceptance testing** – representatives of each user class execute the processes in Section 6 using production-like data; sign-off by the Product Owner.
7. **Operational readiness** – support runbooks, monitoring dashboards, seller and CS training, and data-migration of any pilot data.

17.2 Cutover Plan

Table 68: Cutover timeline

When	Activity	Owner
T-4 weeks	Freeze scope; complete UAT; sign off training materials; rehearsal 1 of cutover in staging.	Product Owner
T-2 weeks	Onboard and train pilot sellers; load and verify their listings, shipping rules and payout accounts; rehearsal 2.	Seller Ops
T-1 week	Final security and accessibility re-test of fixes; configure production (theme, currency, tax, fees); verify PSP live credentials in test mode.	Eng. Lead
T-1 day	Go/no-go meeting against readiness checklist; communications prepared; support rota confirmed.	Sponsor
T0	Deploy; remove demonstration notices and sample data; enable payments; smoke test critical journeys with real low-value orders and refunds.	Eng. Lead
T0 + 1 day	Open to customers in stages (e.g. 10% → 50% → 100% of traffic) with KPI and error-rate gates at each step.	Product Owner
T0 + 4 weeks	Hypercare exit review; handover to business-as-usual support.	Product Owner

Rollback: until real orders exist, rollback is a redeploy of the previous release. After the first real order, rollback is replaced by forward-fix; payments can be disabled independently via configuration to stop new orders while fulfilment of placed orders continues.

17.3 Data Migration

Table 69: Data migration approach

Data	Approach	Control
Browser-stored basket, wish list, saved-for-later, reviews and support issues (R1 keys)	On first sign-in after launch, merge browser data into the account (de-duplicated); then clear the local copies.	Only for pilot/beta users; sample seed data is never migrated.
Sample catalogue, orders, addresses, cards, sellers and applications	Not migrated. Removed from production build; replaced by real catalogue and seller data.	Verify no sample data or demo notices remain (checklist item).
Pilot sellers' business details and listings	Imported via bulk upload template (CSV + images) and validated against BR-SLR-01 and moderation rules.	Seller sign-off on imported data before go-live.
Configuration (theme, currency, tax, fees, delivery)	Promoted from staging through change control; values recorded in the release note.	Finance sign-off on fees and tax.

17.4 Training and Enablement

Table 70: Training plan

Audience	Content	Format
Customer service agents	Agent console, SLAs, refunds and returns, escalation to sellers, data-protection handling	Instructor-led (1 day) + sandbox practice
CS supervisors	Queue management, SLA reporting, refund approvals, macros	Instructor-led (½ day)
Marketplace administrators / Trust & Safety	Seller-application review, KYC outcomes, moderation policy and reason codes, audit	Instructor-led (1 day)
Merchandisers	CMS slots, scheduling, preview, approval, accessibility rules for content	Workshop (½ day)
Finance	Settlement, payouts, holds, fee schedule changes, refund allocation, reports	Workshop (½ day)
Sellers	Onboarding, listings wizard, images, shipping rules, deals, orders, returns, payouts	Self-serve help centre + recorded webinars

17.5 Hypercare

- Duration four weeks from general availability, extendable by the Sponsor.
- Daily 30-minute triage of incidents and defects with Product, Engineering, CS and Seller Ops; P1 fixes deployed same day.
- Monitored KPIs: checkout completion, payment failure rate, order error rate, search zero-result rate, support volume and SLA, seller onboarding throughput.
- Exit criteria: no open P1/P2 defects; KPIs within agreed tolerance for 7 consecutive days; support team operating within SLA; runbooks accepted by BAU owners.

18. Requirements Traceability Matrix

The matrix traces each business objective to the functional modules and principal requirements that realise it, and to the screens where it is verified. Traceability from every requirement to its test case is in Appendix D.

Table 71: Traceability: business objectives to requirements and screens

Objective	Modules	Principal requirements	Screens (site map #)
BO-01 Conversion	GLB, HOM, CAT, SRC, PDP, CRT, CHK, MSO, SEO, CMS	FR-GLB-01...05, FR-CAT-02...12, FR-SRC-01...07, FR-PDP-01...08, FR-CRT-01...05, FR-CHK-01...08, FR-MSO-01...06, FR-SEO-01...05, FR-CMS-01...03	1–7
BO-02 Order value	HOM, PDP, CRT, WSH, DEA, PRM	FR-HOM-04...08, FR-PDP-09, FR-CRT-03, FR-CRT-05, FR-WSH-01...03, FR-DEA-01...03, FR-PRM-01...06	1, 4–6, 8
BO-03 Seller supply	SEL, ADM, SDB, SLS, SIM, SDL, SOR, SBD, PAY, MSO	FR-SEL-01...08, FR-ADM-01...05, FR-SDB-01...05, FR-SLS-01...11, FR-SIM-01...05, FR-SDL-01...07, FR-SOR-01...03, FR-PAY-01...06, FR-MSO-02	16–27
BO-04 Self-service support	CS, ORD, RET, CSA, MSG, NTF	FR-CS-01...09, FR-ORD-02...05, FR-RET-01...07, FR-CSA-01...06, FR-MSG-01...05, FR-NTF-01...05	9, 10, 12
BO-05 Trust	REV, PDP, CHK, ORD, MOD, FRD, UAD	FR-REV-01...08, FR-PDP-04, FR-PDP-07, FR-CHK-07, FR-ORD-03, FR-MOD-01...05, FR-FRD-01...05, FR-UAD-03	4, 7, 10
BO-06 White-label	CFG, GLB, NTF	FR-CFG-01...05, FR-GLB-07, FR-GLB-11, FR-GLB-12, FR-NTF-02	All
BO-07 Inclusive and usable	All	NFR-ACC-01...05, NFR-COM-01/02, NFR-PERF-01, FR-GLB-08/09	All

Appendix A. UI States Gallery

Screens in states other than their default: validation errors, empty states, confirmations, menus and decision outcomes. Captured from the current UI (default theme, 1280 px unless noted). The seller-onboarding states were produced by submitting a test application and rejecting it in the admin queue.

Table 72: *UI states index*

#	State	Requirement
1	Account & Lists flyout (hover)	FR-GLB-03
2	Deliver-to address picker	FR-GLB-02
3	“All” slide-over menu	FR-GLB-06
4	Search type-ahead suggestions for “head”	FR-SRC-02/03
5	Search with no results (empty state)	FR-CAT-10
6	Phone: filter drawer open (390 px)	FR-CAT-09
7	Review form validation error (“Select a star rating.”)	FR-REV-03
8	Empty basket with a saved-for-later item	FR-CRT-06
9	Order confirmation after “Place your order”	FR-CHK-07
10	Customer service form validation error	FR-CS-01
11	Customer service issue expanded with thread and reply box	FR-CS-04/05
12	Seller application submitted confirmation	FR-SEL-04
13	Seller application pending review	FR-SEL-05
14	Admin rejecting an application with a reason	FR-ADM-03
15	Applicant view after rejection, with reason and “Apply again”	FR-SEL-05/06
16	Sell page status card showing the rejected application	FR-SEL-07
17	Listing wizard validation error (“Give your product a title.”)	FR-SLS-03
18	Edit listing with unknown id (“Listing not found”)	FR-SLS-10
19	Deal editor validation error (“Enter a discount percentage.”)	FR-SDL-03
20	Page not found	FR-GLB-10

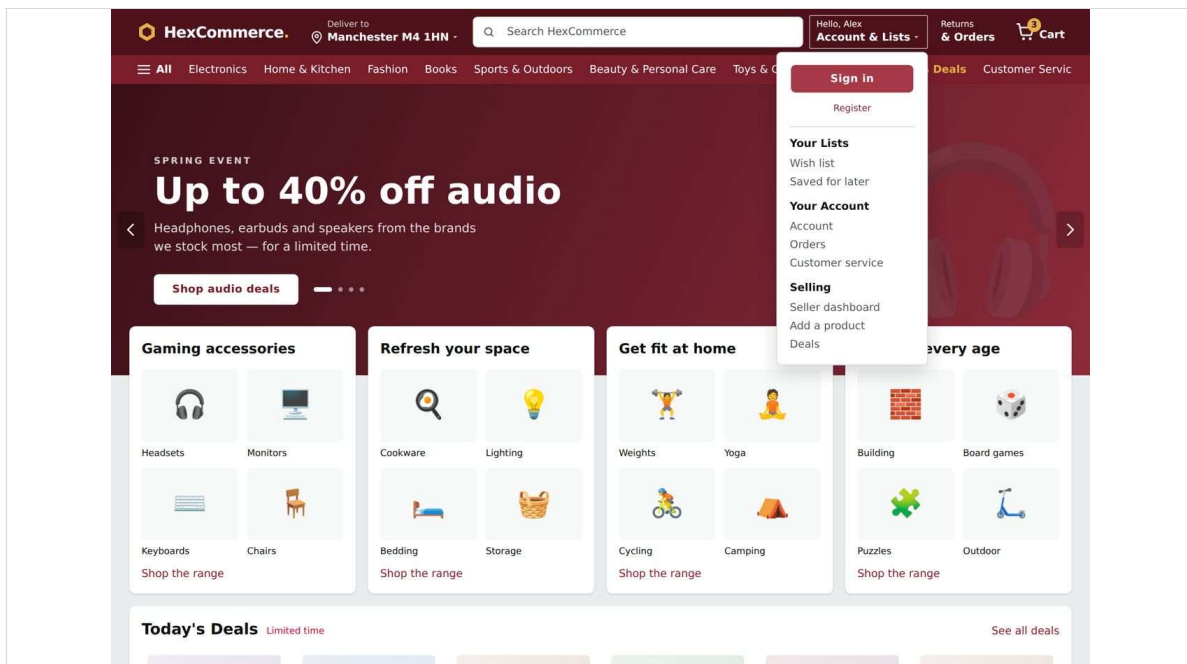


Figure 38: Account & Lists flyout (hover) (FR-GLB-03)

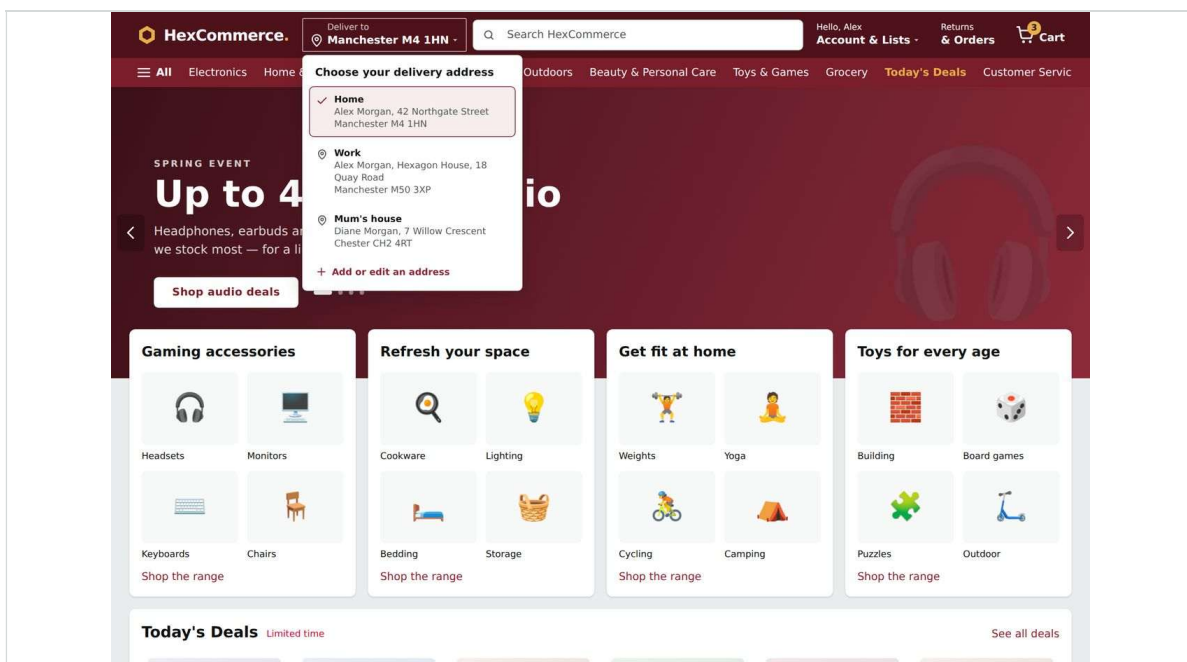


Figure 39: Deliver-to address picker (FR-GLB-02)

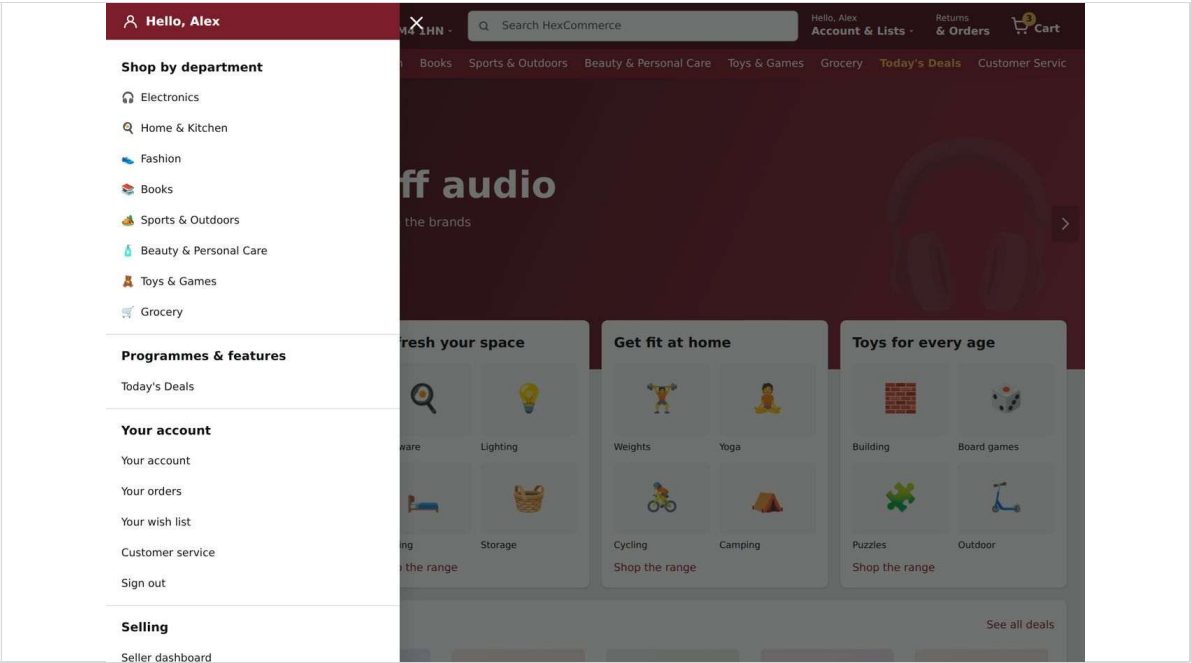


Figure 40: “All” slide-over menu (FR-GLB-06)

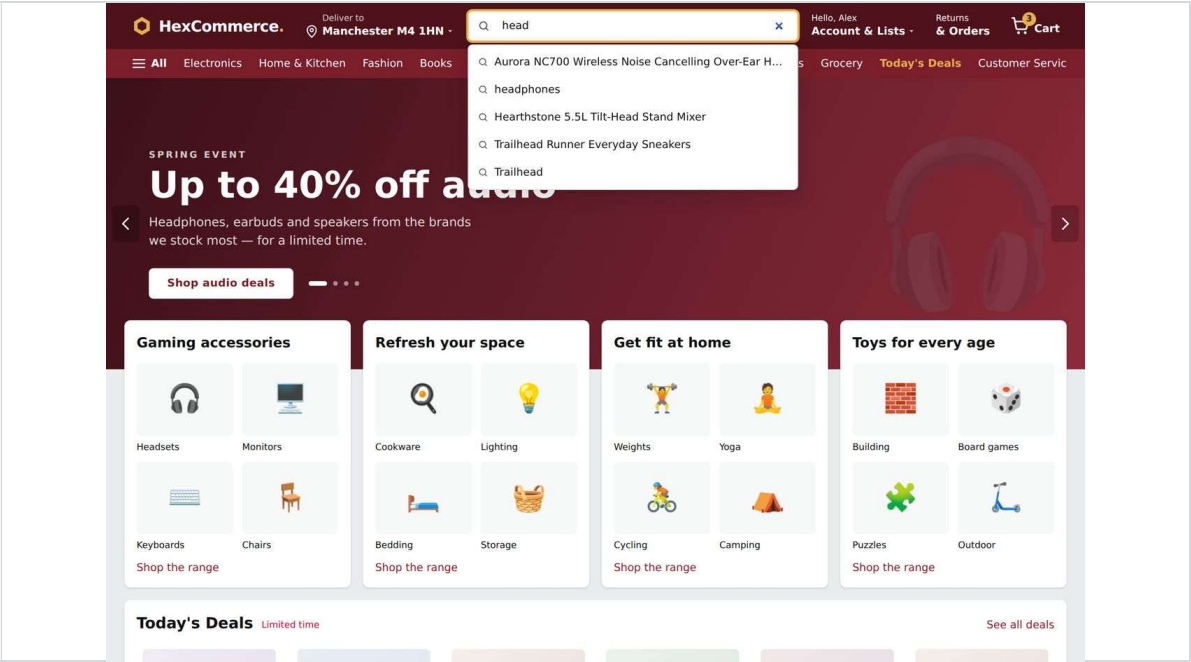


Figure 41: Search type-ahead suggestions for “head” (FR-SRC-02/03)

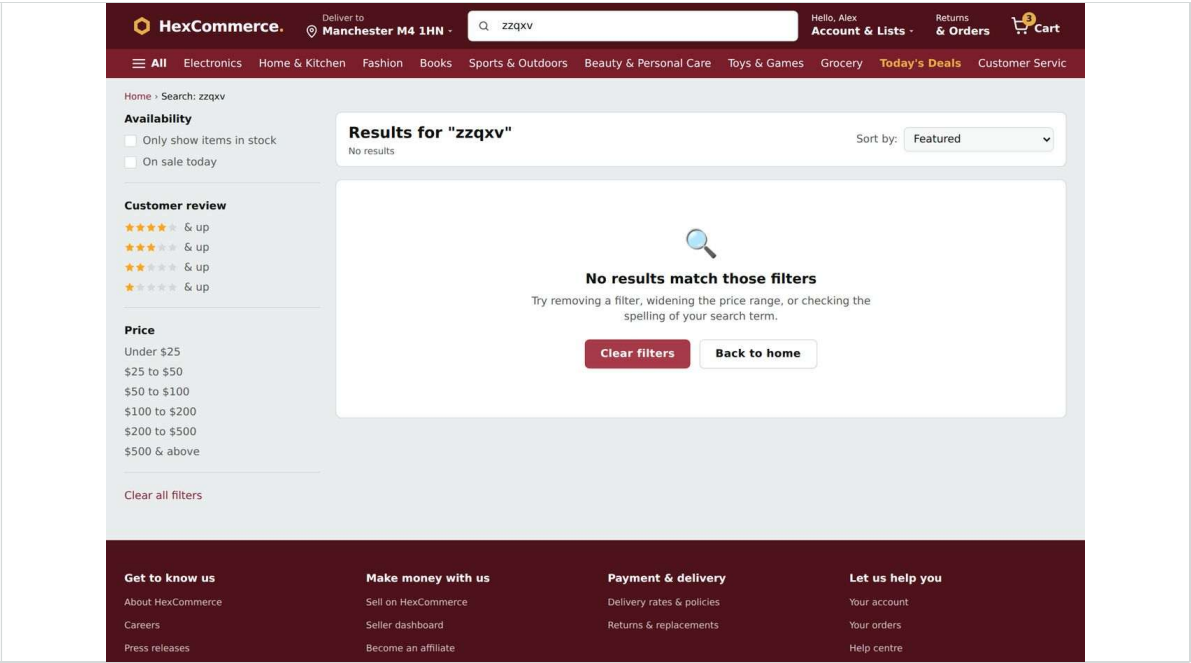


Figure 42: Search with no results (empty state) (FR-CAT-10)

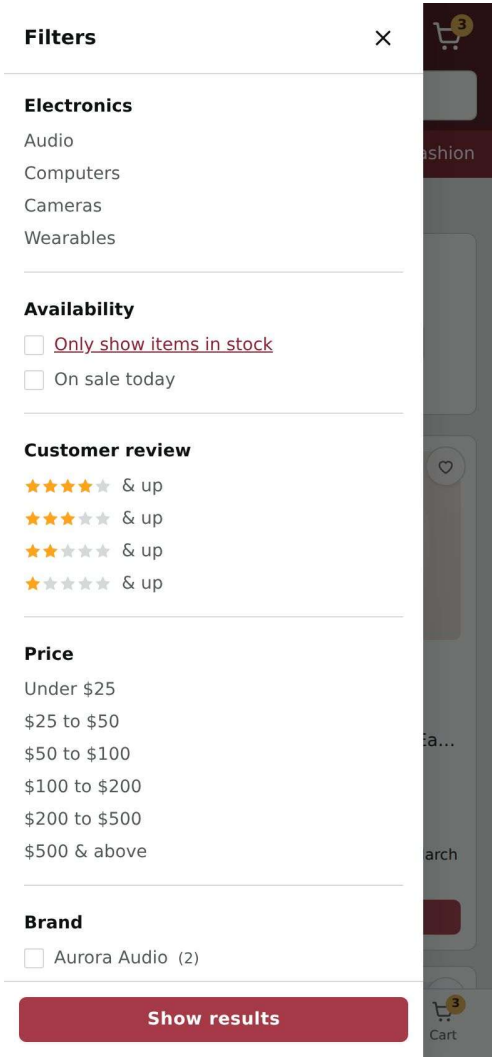


Figure 43: Phone: filter drawer open (390 px) (FR-CAT-09)

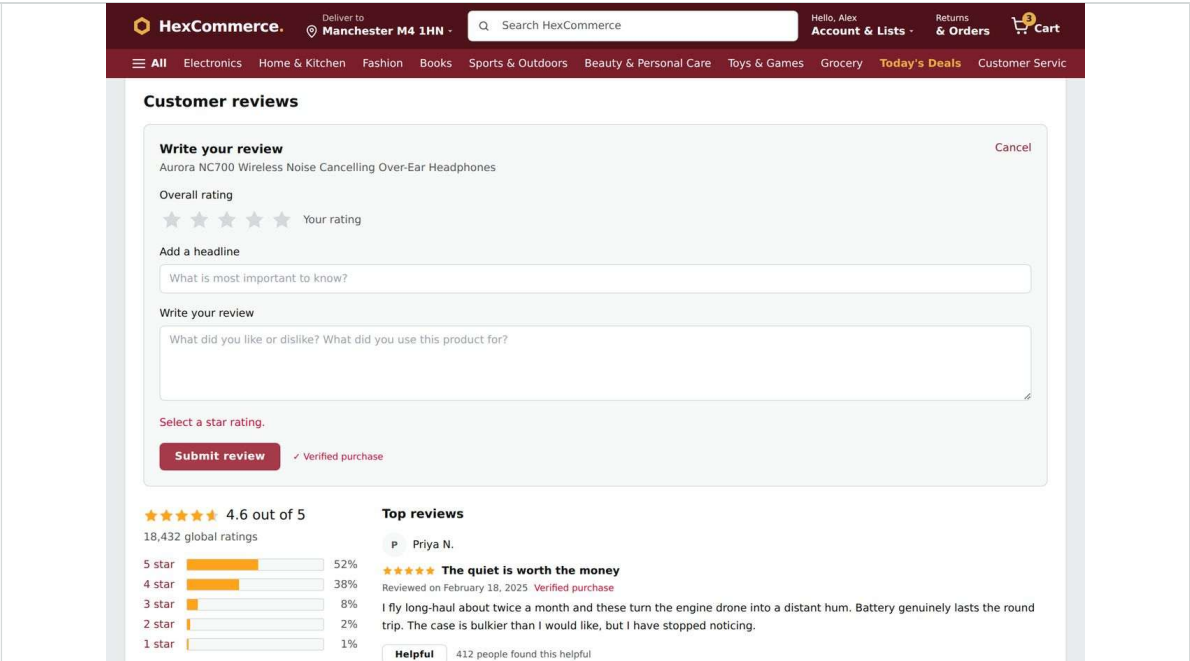


Figure 44: Review form validation error (“Select a star rating.”) (FR-REV-03)

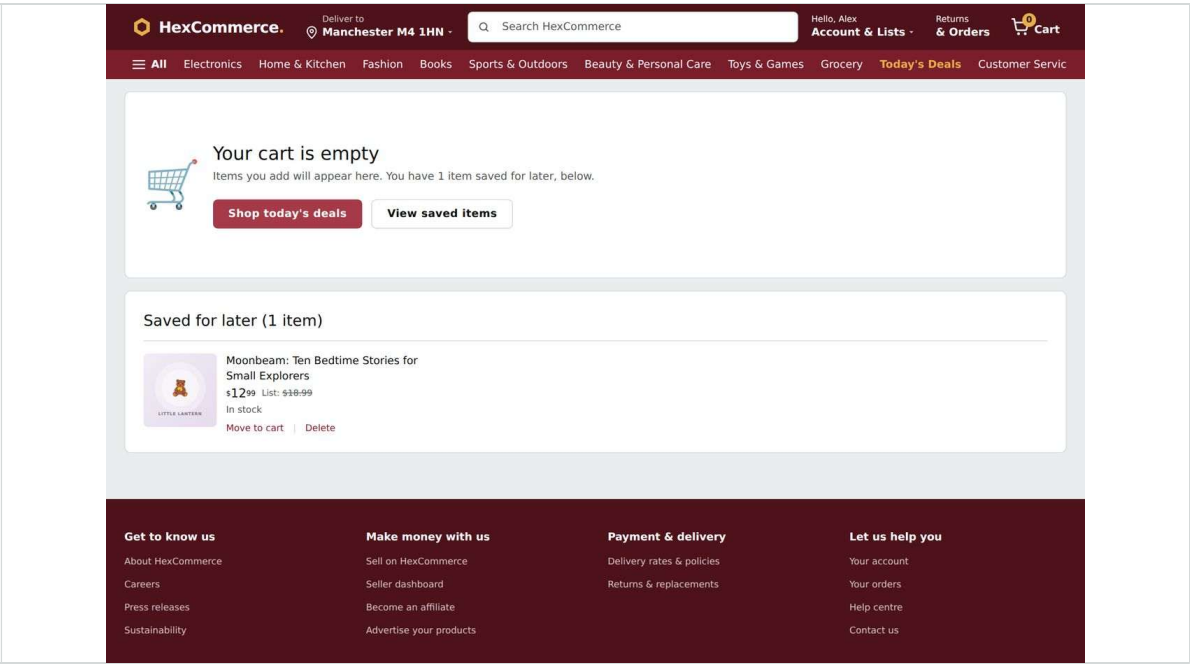


Figure 45: Empty basket with a saved-for-later item (FR-CRT-06)

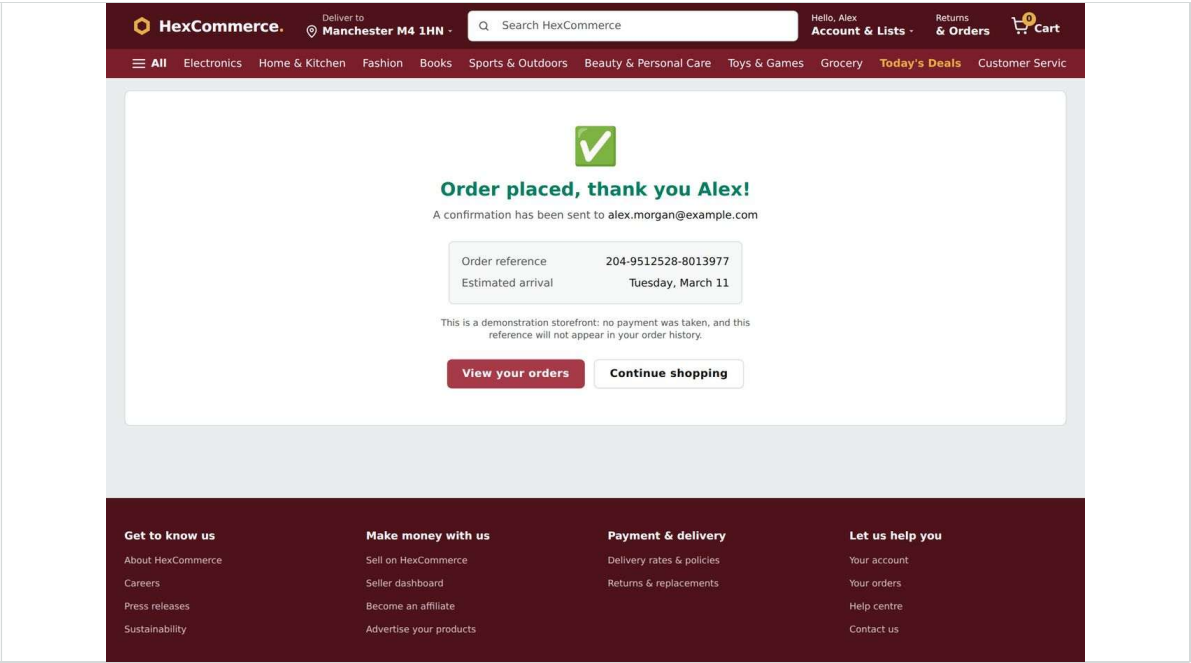


Figure 46: Order confirmation after “Place your order” (FR-CHK-07)

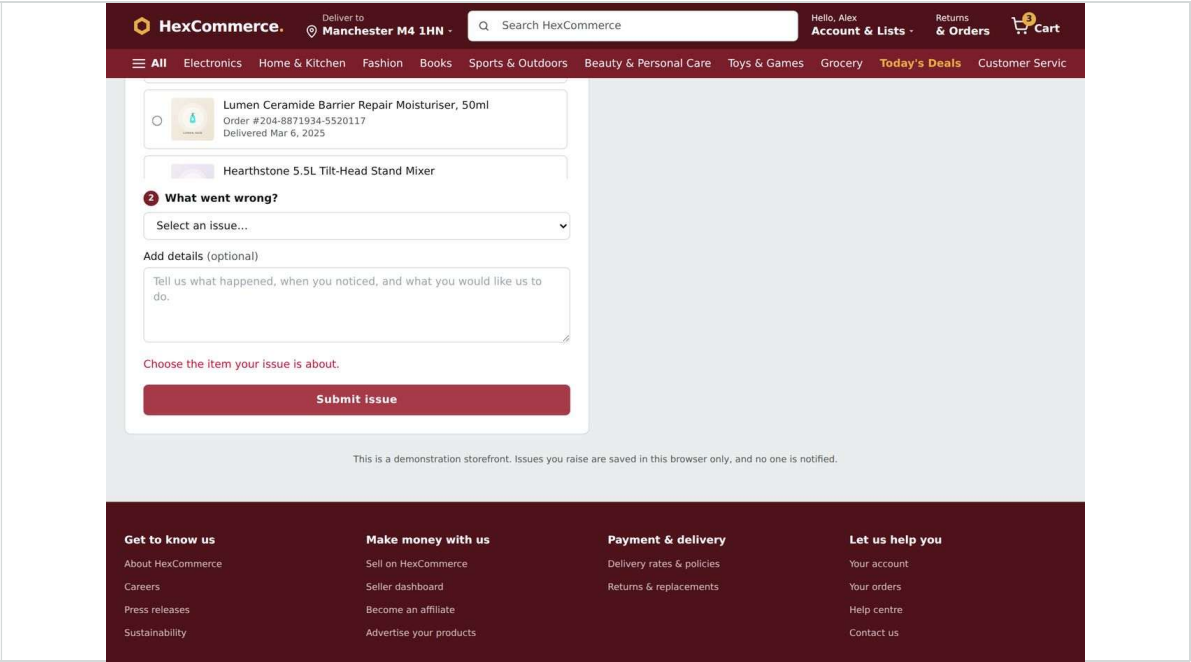


Figure 47: Customer service form validation error (FR-CS-01)

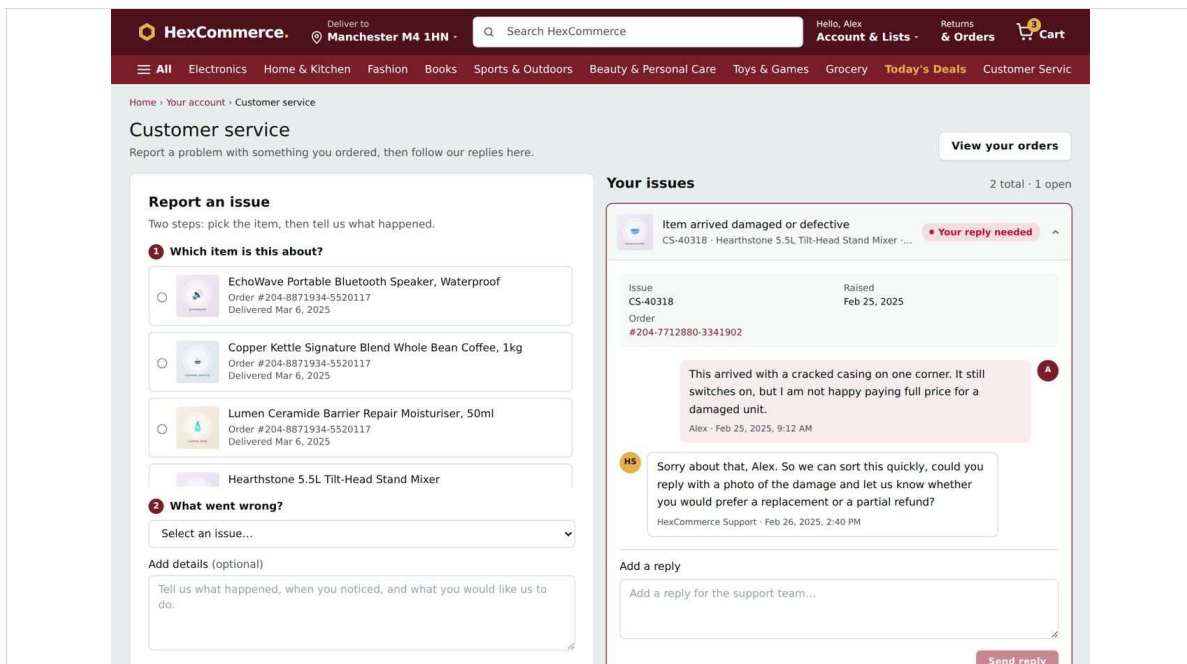


Figure 48: Customer service issue expanded with thread and reply box (FR-CS-04/05)

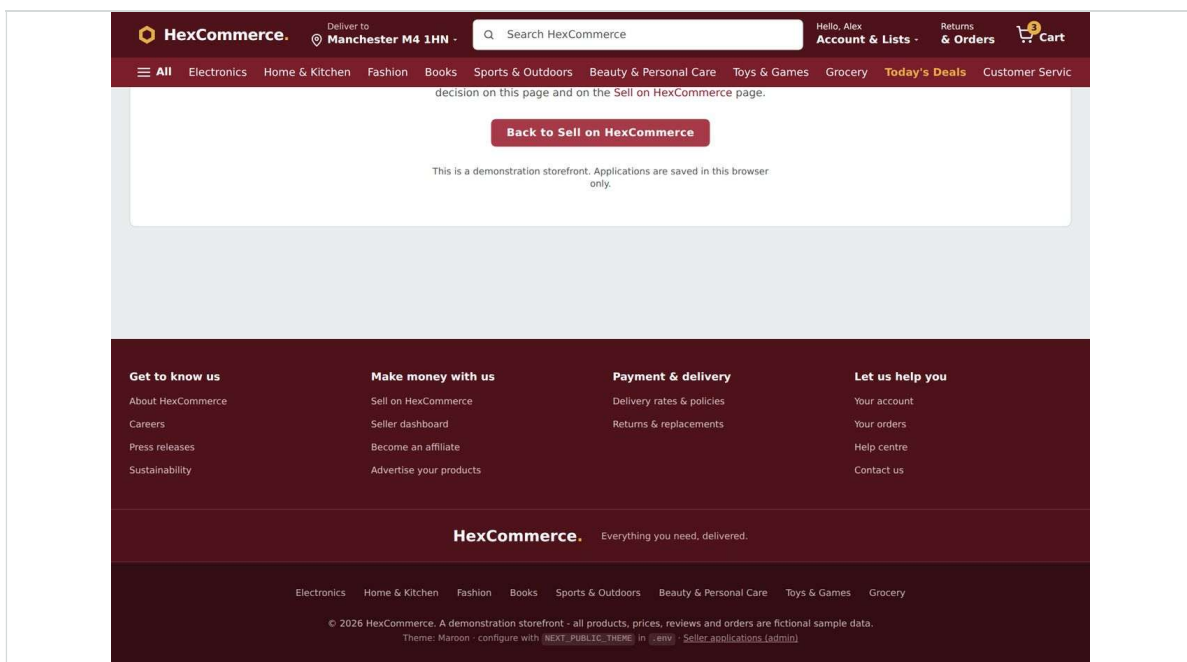


Figure 49: Seller application submitted confirmation (FR-SEL-04)

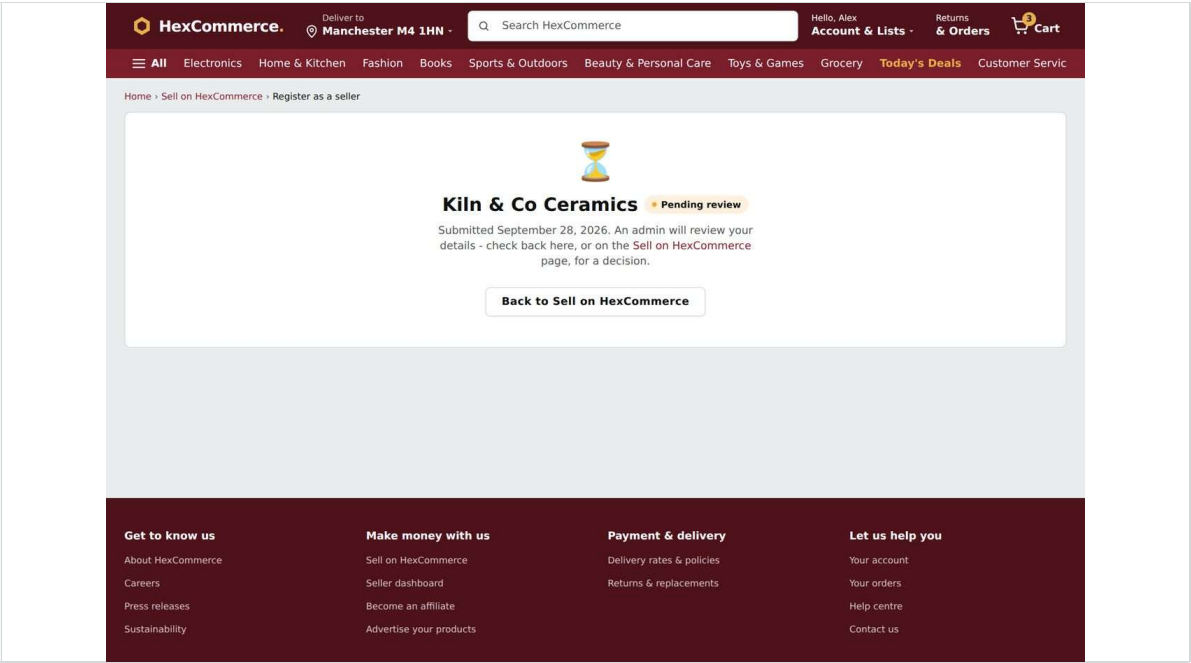


Figure 50: Seller application pending review (FR-SEL-05)

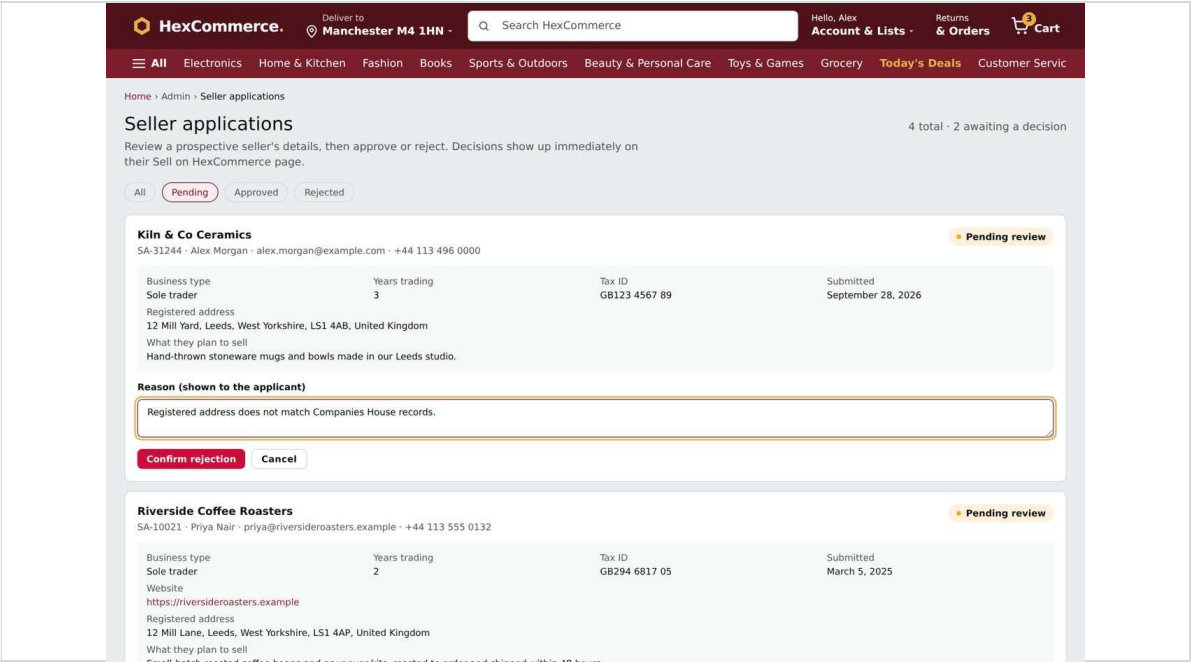


Figure 51: Admin rejecting an application with a reason (FR-ADM-03)

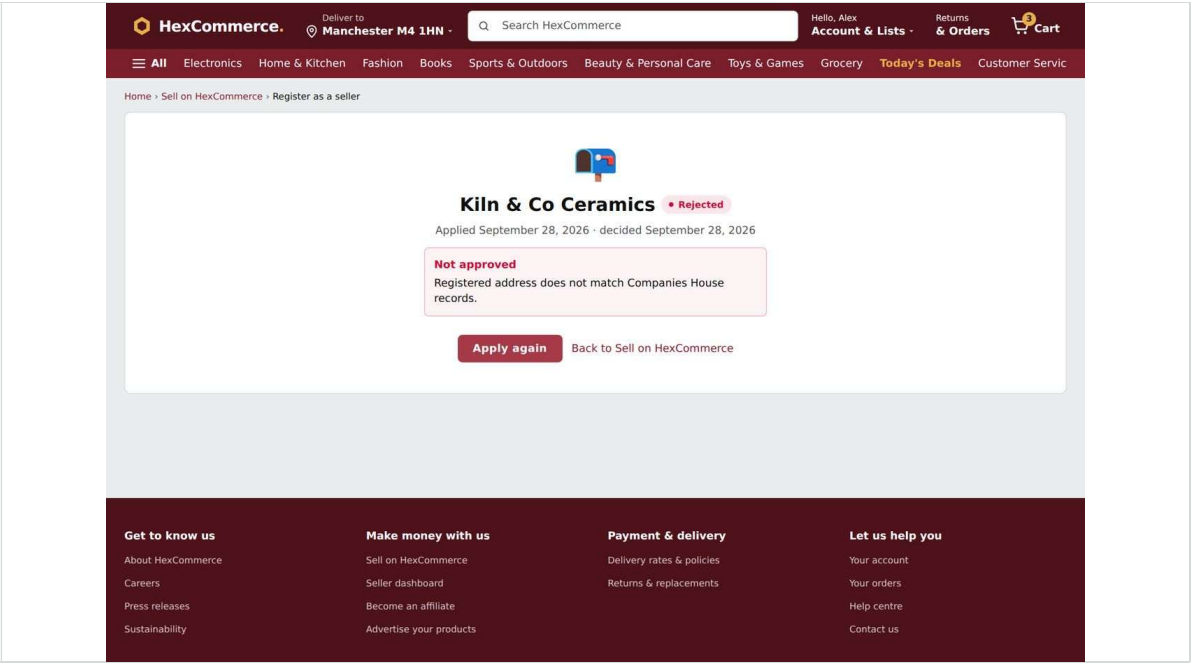


Figure 52: Applicant view after rejection, with reason and “Apply again” (FR-SEL-05/06)

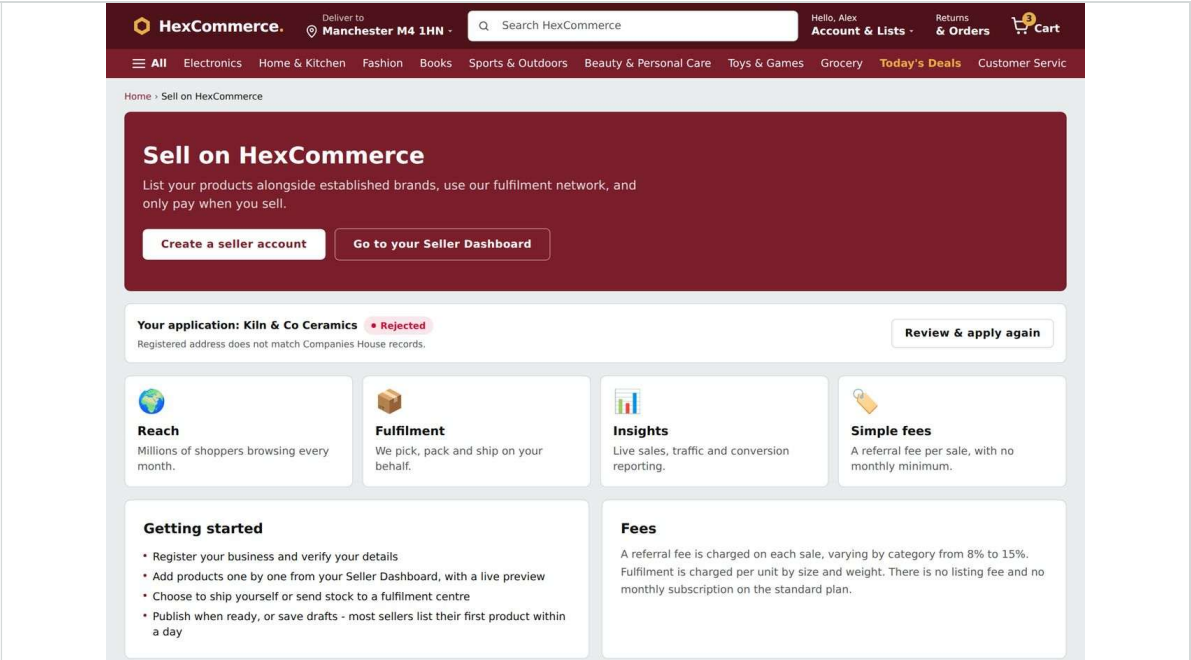


Figure 53: Sell page status card showing the rejected application (FR-SEL-07)

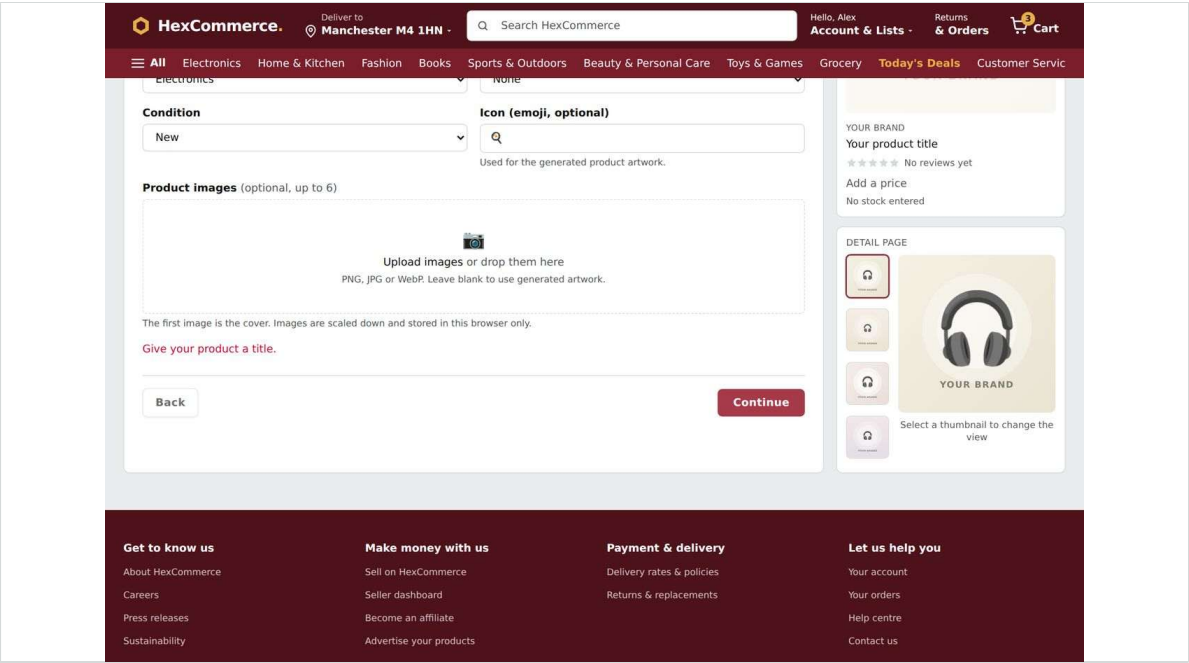


Figure 54: Listing wizard validation error (“Give your product a title.”) (FR-SLS-03)

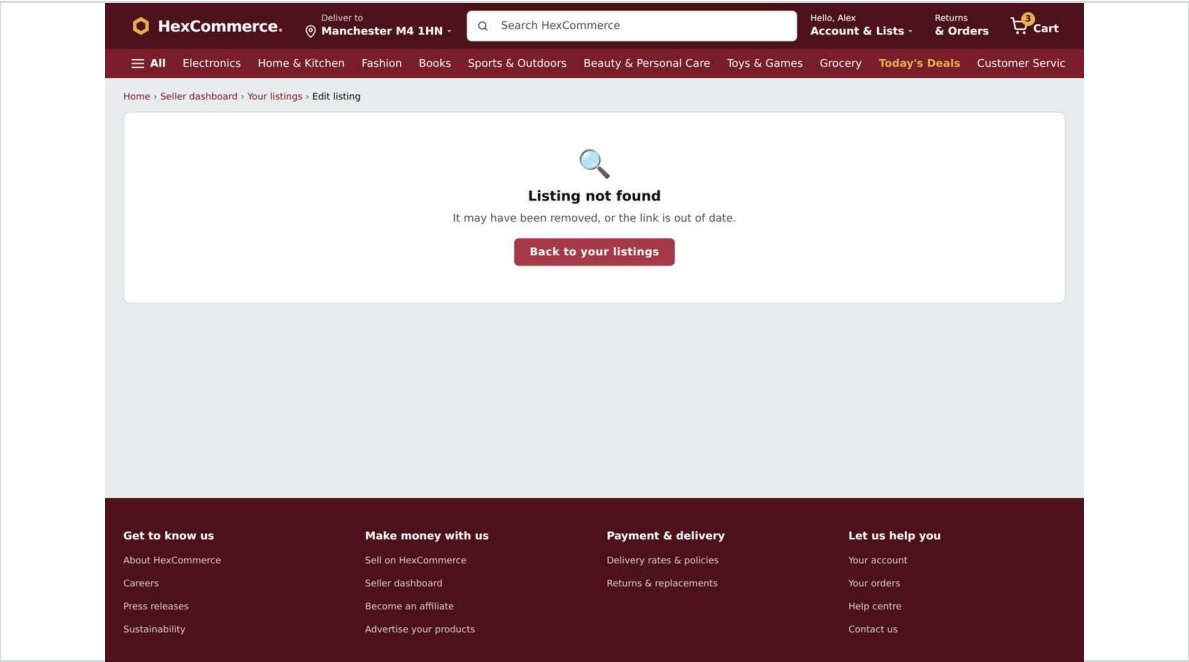


Figure 55: Edit listing with unknown id (“Listing not found”) (FR-SLS-10)

The screenshot shows the 'Create a deal' page on the HexCommerce seller dashboard. The page has a dark red header with the HexCommerce logo, delivery location (Manchester M4 1HN), a search bar, and user links (Hello, Alex, Account & Lists, Returns & Orders, Cart). Below the header is a navigation bar with categories: All, Electronics, Home & Kitchen, Fashion, Books, Sports & Outdoors, Beauty & Personal Care, Toys & Games, Grocery, Today's Deals, and Customer Service. The main content area is titled 'Create a deal' and includes a breadcrumb: Home > Seller dashboard > Your deals > Create a deal. Below this is a sub-header: 'Run a limited-time discount on one of your active listings.' The 'Deal details' section contains a 'Listing' dropdown (Northgate Cast Iron Skillet, 26 cm, Pre-Seasoned — \$38.00), a 'Discount' input field (20, with a note '5-70% off the listing price.'), 'Units in this deal' (0, with a note 'You have 64 in stock.'), 'Starts on' (03/10/2025), and 'Ends on' (03/17/2025). A red validation error message 'Enter a discount percentage.' is displayed below the discount input. A 'Cancel' link and a 'Start deal' button are at the bottom of the form. To the right is a 'PREVIEW' section showing a magnifying glass icon over a skillet, the product name 'Northgate Cast Iron Skillet, 26 cm, Pre-Seasoned', and a summary of the deal: 'Set a discount', 'Ends Mar 17, 2025', and 'Set the deal quantity'. The footer contains four columns of links: 'Get to know us' (About HexCommerce, Careers, Press releases), 'Make money with us' (Sell on HexCommerce, Seller dashboard, Become an affiliate), 'Payment & delivery' (Delivery rates & policies, Returns & replacements), and 'Let us help you' (Your account, Your orders, Help centre).

Figure 56: Deal editor validation error ("Enter a discount percentage.") (FR-SDL-03)

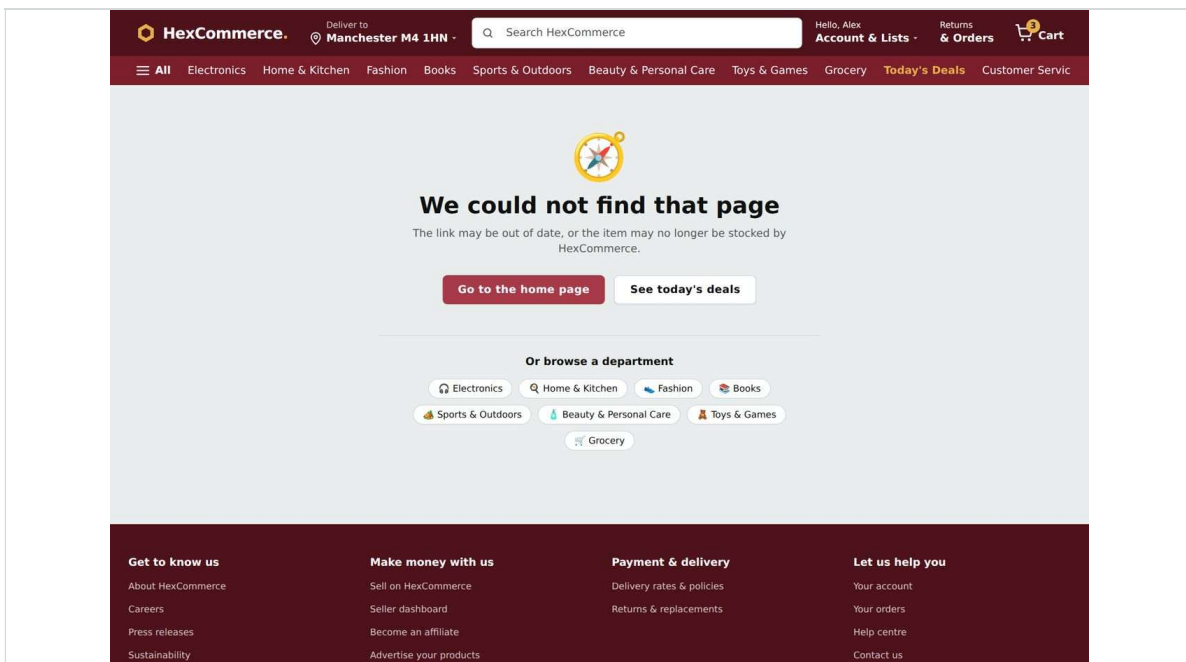


Figure 57: Page not found (FR-GLB-10)

Appendix B. Screen Gallery

Additional reference screenshots of the current UI (default “maroon” theme, 1280 px viewport).

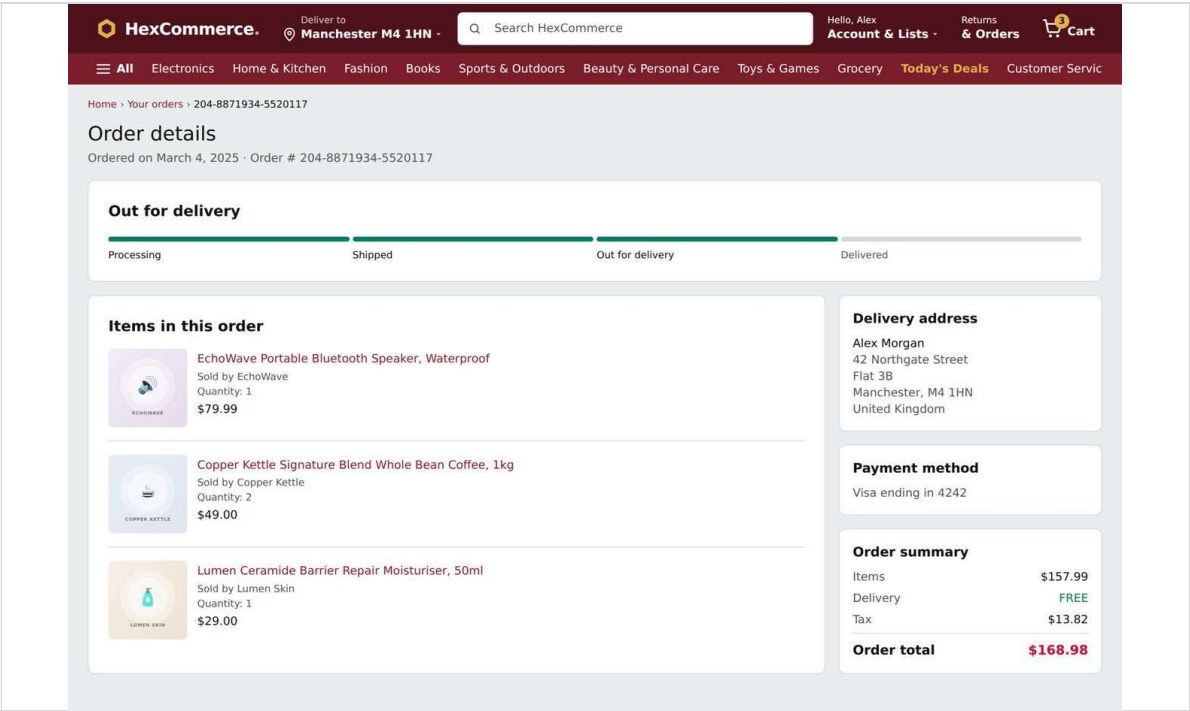


Figure 58: Order details with delivery tracker

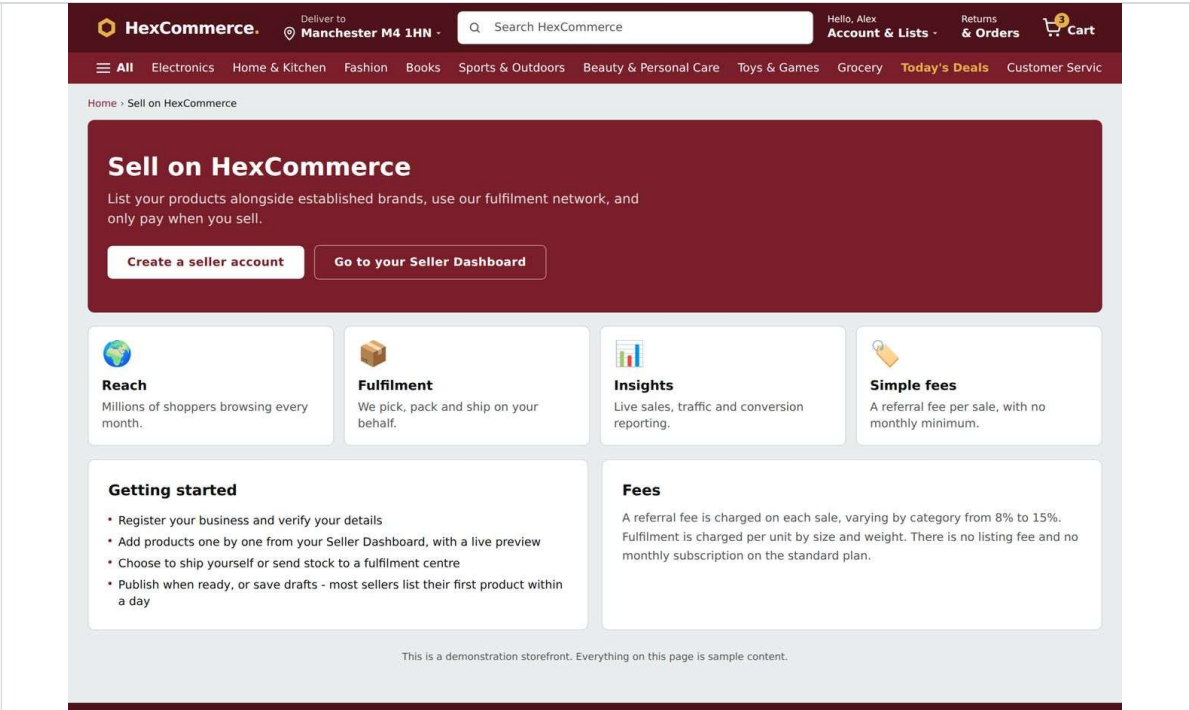


Figure 59: Sell on HexCommerce recruitment page

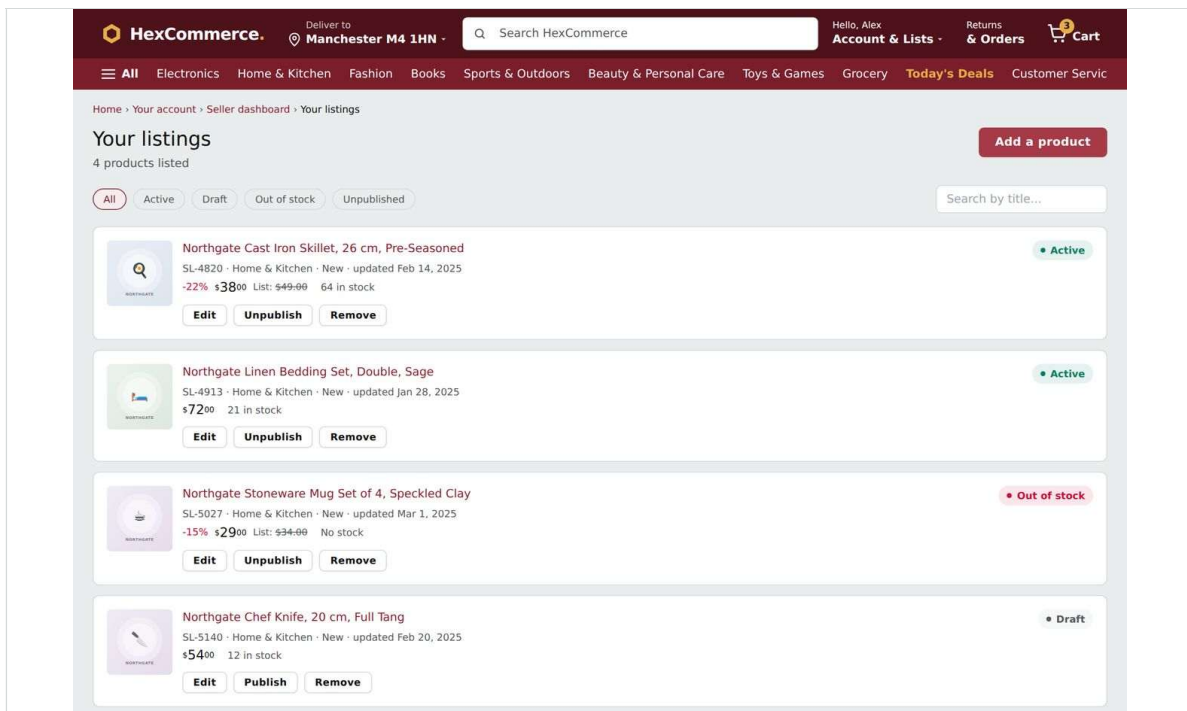


Figure 60: Seller listings management

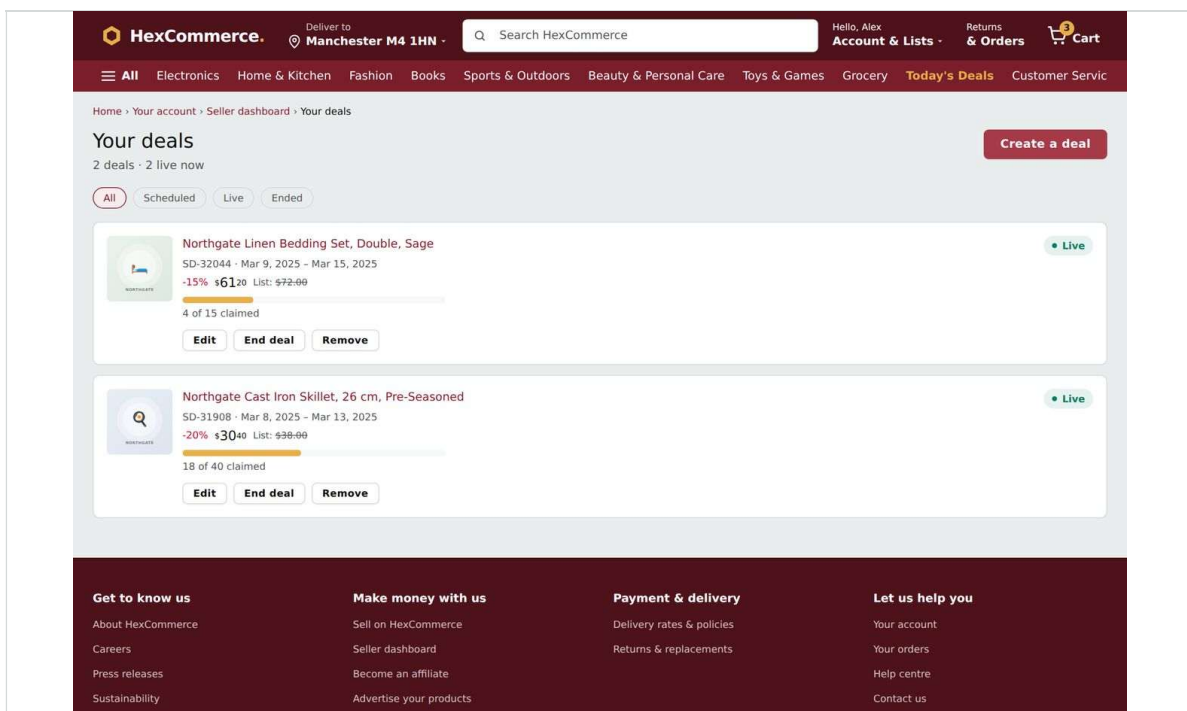


Figure 61: Seller deals management

Appendix C. Configuration Parameters

Table 73: Environment configuration

Parameter	Default	Description
NEXT_PUBLIC_SITE_NAME	HexCommerce	Store name in logo, titles, metadata, footer.
NEXT_PUBLIC_SITE_TAGLINE	Everything you need, delivered.	Marketing line in metadata and footer.
NEXT_PUBLIC_THEME	maroon	Theme preset: maroon, midnight, forest, slate, amber, plum.
NEXT_PUBLIC_THEME_PRIMARY / _PRIMARY_DARK / _PRIMARY_LIGHT	(preset)	Brand colour overrides (hex or r,g,b).
NEXT_PUBLIC_THEME_ACCENT	(preset)	Call-to-action colour override.
NEXT_PUBLIC_THEME_HEADER / _SUBHEADER / _FOOTER	(preset)	Chrome colour overrides.
NEXT_PUBLIC_CURRENCY	USD	ISO 4217 currency code.
NEXT_PUBLIC_LOCALE	en-US	BCP 47 locale for number and date formatting.
NEXT_PUBLIC_FREE_SHIPPING_THRESHOLD	35	Subtotal at or above which delivery is free.
NEXT_PUBLIC_TAX_RATE	0.0875	Sales-tax rate applied to the subtotal.
(code constant) shippingFee	5.99	Flat delivery fee below threshold (recommend configurable – FR-CFG-05).
(code constant) SELLER_REFERRAL_FEE	0.12	Referral fee used for payout estimate (recommend per-category – OI-05).

Appendix D. Requirements-to-Test Traceability

Every requirement is assigned a verification method (ISO/IEC/IEEE 29148: Test, Inspection, Analysis, Demonstration), a test level and a test-case identifier. Test cases have **not yet been designed**; this appendix is the register QA completes during Phase 0–5 and the basis for acceptance reporting.

Table 74: Requirements-to-test register (296 requirements)

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-GLB-01	R1	Test	System (E2E) + UAT	TC-GLB-01	To be designed
FR-GLB-02	R1	Test	System (E2E) + UAT	TC-GLB-02	To be designed
FR-GLB-03	R1	Test	System (E2E) + UAT	TC-GLB-03	To be designed
FR-GLB-04	R1	Test	System (E2E) + UAT	TC-GLB-04	To be designed
FR-GLB-05	R1	Test	System (E2E) + UAT	TC-GLB-05	To be designed
FR-GLB-06	R1	Test	System (E2E) + UAT	TC-GLB-06	To be designed
FR-GLB-07	R1	Test	System (E2E) + UAT	TC-GLB-07	To be designed
FR-GLB-08	R1	Test	System (E2E) + UAT	TC-GLB-08	To be designed
FR-GLB-09	R1	Test	System (E2E) + UAT	TC-GLB-09	To be designed
FR-GLB-10	R1	Test	System (E2E) + UAT	TC-GLB-10	To be designed
FR-GLB-11	R1	Test	System (E2E) + UAT	TC-GLB-11	To be designed
FR-GLB-12	R1	Test	System (E2E) + UAT	TC-GLB-12	To be designed
FR-HOM-01	R1	Test	System (E2E) + UAT	TC-HOM-01	To be designed
FR-HOM-02	R1	Test	System (E2E) + UAT	TC-HOM-02	To be designed
FR-HOM-03	R1	Test	System (E2E) + UAT	TC-HOM-03	To be designed
FR-HOM-04	R1	Test	System (E2E) + UAT	TC-HOM-04	To be designed
FR-HOM-05	R1	Test	System (E2E) + UAT	TC-HOM-05	To be designed
FR-HOM-06	R1	Test	System (E2E) + UAT	TC-HOM-06	To be designed
FR-HOM-07	R1	Test	System (E2E) + UAT	TC-HOM-07	To be designed
FR-HOM-08	R1	Test	System (E2E) + UAT	TC-HOM-08	To be designed
FR-HOM-09	R1	Test	System (E2E) + UAT	TC-HOM-09	To be designed
FR-HOM-10	R1	Test	System (E2E) + UAT	TC-HOM-10	To be designed
FR-HOM-11	R2	Test	System + UAT	TC-HOM-11	To be designed
FR-CAT-01	R1	Test	System (E2E) + UAT	TC-CAT-01	To be designed
FR-CAT-02	R1	Test	System (E2E) + UAT	TC-CAT-02	To be designed
FR-CAT-03	R1	Test	System (E2E) + UAT	TC-CAT-03	To be designed
FR-CAT-04	R1	Test	System (E2E) + UAT	TC-CAT-04	To be designed
FR-CAT-05	R1	Test	System (E2E) + UAT	TC-CAT-05	To be designed
FR-CAT-06	R1	Test	System (E2E) + UAT	TC-CAT-06	To be designed
FR-CAT-07	R1	Test	System (E2E) + UAT	TC-CAT-07	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-CAT-08	R1	Test	System (E2E) + UAT	TC-CAT-08	To be designed
FR-CAT-09	R1	Test	System (E2E) + UAT	TC-CAT-09	To be designed
FR-CAT-10	R1	Test	System (E2E) + UAT	TC-CAT-10	To be designed
FR-CAT-11	R1	Test	System (E2E) + UAT	TC-CAT-11	To be designed
FR-CAT-12	R1	Test	System (E2E) + UAT	TC-CAT-12	To be designed
FR-SRC-01	R1	Test	System (E2E) + UAT	TC-SRC-01	To be designed
FR-SRC-02	R1	Test	System (E2E) + UAT	TC-SRC-02	To be designed
FR-SRC-03	R1	Test	System (E2E) + UAT	TC-SRC-03	To be designed
FR-SRC-04	R1	Test	System (E2E) + UAT	TC-SRC-04	To be designed
FR-SRC-05	R1	Test	System (E2E) + UAT	TC-SRC-05	To be designed
FR-SRC-06	R1	Test	System (E2E) + UAT	TC-SRC-06	To be designed
FR-SRC-07	R1	Test	System (E2E) + UAT	TC-SRC-07	To be designed
FR-SRC-08	R2	Test	System + UAT	TC-SRC-08	To be designed
FR-PDP-01	R1	Test	System (E2E) + UAT	TC-PDP-01	To be designed
FR-PDP-02	R1	Test	System (E2E) + UAT	TC-PDP-02	To be designed
FR-PDP-03	R1	Test	System (E2E) + UAT	TC-PDP-03	To be designed
FR-PDP-04	R1	Test	System (E2E) + UAT	TC-PDP-04	To be designed
FR-PDP-05	R1/R2	Test	System (E2E) + UAT	TC-PDP-05	To be designed
FR-PDP-06	R1	Test	System (E2E) + UAT	TC-PDP-06	To be designed
FR-PDP-07	R1	Test	System (E2E) + UAT	TC-PDP-07	To be designed
FR-PDP-08	R1	Test	System (E2E) + UAT	TC-PDP-08	To be designed
FR-PDP-09	R1	Test	System (E2E) + UAT	TC-PDP-09	To be designed
FR-PDP-10	R1/R2	Test	System (E2E) + UAT	TC-PDP-10	To be designed
FR-REV-01	R1	Test	System (E2E) + UAT	TC-REV-01	To be designed
FR-REV-02	R1	Test	System (E2E) + UAT	TC-REV-02	To be designed
FR-REV-03	R1	Test	System (E2E) + UAT	TC-REV-03	To be designed
FR-REV-04	R1	Test	System (E2E) + UAT	TC-REV-04	To be designed
FR-REV-05	R1	Test	System (E2E) + UAT	TC-REV-05	To be designed
FR-REV-06	R1	Test	System (E2E) + UAT	TC-REV-06	To be designed
FR-REV-07	R1/R2	Test	System (E2E) + UAT	TC-REV-07	To be designed
FR-REV-08	R2	Test	System + UAT	TC-REV-08	To be designed
FR-WSH-01	R1	Test	System (E2E) + UAT	TC-WSH-01	To be designed
FR-WSH-02	R1	Test	System (E2E) + UAT	TC-WSH-02	To be designed
FR-WSH-03	R1	Test	System (E2E) + UAT	TC-WSH-03	To be designed
FR-WSH-04	R1	Test	System (E2E) + UAT	TC-WSH-04	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-CRT-01	R1	Test	System (E2E) + UAT	TC-CRT-01	To be designed
FR-CRT-02	R1	Test	System (E2E) + UAT	TC-CRT-02	To be designed
FR-CRT-03	R1	Test	System (E2E) + UAT	TC-CRT-03	To be designed
FR-CRT-04	R1	Test	System (E2E) + UAT	TC-CRT-04	To be designed
FR-CRT-05	R1	Test	System (E2E) + UAT	TC-CRT-05	To be designed
FR-CRT-06	R1	Test	System (E2E) + UAT	TC-CRT-06	To be designed
FR-CRT-07	R1/R2	Test	System (E2E) + UAT	TC-CRT-07	To be designed
FR-CRT-08	R1	Test	System (E2E) + UAT	TC-CRT-08	To be designed
FR-CHK-01	R1	Test	System (E2E) + UAT	TC-CHK-01	To be designed
FR-CHK-02	R1	Test	System (E2E) + UAT	TC-CHK-02	To be designed
FR-CHK-03	R1	Test	System (E2E) + UAT	TC-CHK-03	To be designed
FR-CHK-04	R1	Test	System (E2E) + UAT	TC-CHK-04	To be designed
FR-CHK-05	R1	Test	System (E2E) + UAT	TC-CHK-05	To be designed
FR-CHK-06	R1	Test	System (E2E) + UAT	TC-CHK-06	To be designed
FR-CHK-07	R1	Test	System (E2E) + UAT	TC-CHK-07	To be designed
FR-CHK-08	R2	Test	System + UAT	TC-CHK-08	To be designed
FR-CHK-09	R2	Test	System + UAT	TC-CHK-09	To be designed
FR-CHK-10	R2	Test	System + UAT	TC-CHK-10	To be designed
FR-ORD-01	R1	Test	System (E2E) + UAT	TC-ORD-01	To be designed
FR-ORD-02	R1	Test	System (E2E) + UAT	TC-ORD-02	To be designed
FR-ORD-03	R1	Test	System (E2E) + UAT	TC-ORD-03	To be designed
FR-ORD-04	R2	Test	System + UAT	TC-ORD-04	To be designed
FR-ORD-05	R2	Test	System + UAT	TC-ORD-05	To be designed
FR-ORD-06	R2	Test	System + UAT	TC-ORD-06	To be designed
FR-ACC-01	R1	Test	System (E2E) + UAT	TC-ACC-01	To be designed
FR-ACC-02	R1	Test	System (E2E) + UAT	TC-ACC-02	To be designed
FR-ACC-03	R1	Test	System (E2E) + UAT	TC-ACC-03	To be designed
FR-ACC-04	R2	Test	System + UAT	TC-ACC-04	To be designed
FR-ACC-05	R2	Test	System + UAT	TC-ACC-05	To be designed
FR-CS-01	R1	Test	System (E2E) + UAT	TC-CS-01	To be designed
FR-CS-02	R1	Test	System (E2E) + UAT	TC-CS-02	To be designed
FR-CS-03	R1	Test	System (E2E) + UAT	TC-CS-03	To be designed
FR-CS-04	R1	Test	System (E2E) + UAT	TC-CS-04	To be designed
FR-CS-05	R1	Test	System (E2E) + UAT	TC-CS-05	To be designed
FR-CS-06	R1	Test	System (E2E) + UAT	TC-CS-06	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-CS-07	R1	Test	System (E2E) + UAT	TC-CS-07	To be designed
FR-CS-08	R2	Test	System + UAT	TC-CS-08	To be designed
FR-CS-09	R2	Test	System + UAT	TC-CS-09	To be designed
FR-DEA-01	R1	Test	System (E2E) + UAT	TC-DEA-01	To be designed
FR-DEA-02	R1	Test	System (E2E) + UAT	TC-DEA-02	To be designed
FR-DEA-03	R1	Test	System (E2E) + UAT	TC-DEA-03	To be designed
FR-DEA-04	R2	Test	System + UAT	TC-DEA-04	To be designed
FR-DEA-05	R2	Test	System + UAT	TC-DEA-05	To be designed
FR-AUT-01	R1	Test	System (E2E) + UAT	TC-AUT-01	To be designed
FR-AUT-02	R1	Test	System (E2E) + UAT	TC-AUT-02	To be designed
FR-AUT-03	R2	Test	System + UAT	TC-AUT-03	To be designed
FR-AUT-04	R2	Test	System + UAT	TC-AUT-04	To be designed
FR-AUT-05	R2	Test	System + UAT	TC-AUT-05	To be designed
FR-SEL-01	R1	Test	System (E2E) + UAT	TC-SEL-01	To be designed
FR-SEL-02	R1	Test	System (E2E) + UAT	TC-SEL-02	To be designed
FR-SEL-03	R1	Test	System (E2E) + UAT	TC-SEL-03	To be designed
FR-SEL-04	R1	Test	System (E2E) + UAT	TC-SEL-04	To be designed
FR-SEL-05	R1	Test	System (E2E) + UAT	TC-SEL-05	To be designed
FR-SEL-06	R1	Test	System (E2E) + UAT	TC-SEL-06	To be designed
FR-SEL-07	R1	Test	System (E2E) + UAT	TC-SEL-07	To be designed
FR-SEL-08	R2	Test	System + UAT	TC-SEL-08	To be designed
FR-ADM-01	R1	Test	System (E2E) + UAT	TC-ADM-01	To be designed
FR-ADM-02	R1	Test	System (E2E) + UAT	TC-ADM-02	To be designed
FR-ADM-03	R1	Test	System (E2E) + UAT	TC-ADM-03	To be designed
FR-ADM-04	R1	Test	System (E2E) + UAT	TC-ADM-04	To be designed
FR-ADM-05	R2	Test	System + UAT	TC-ADM-05	To be designed
FR-SDB-01	R1	Test	System (E2E) + UAT	TC-SDB-01	To be designed
FR-SDB-02	R1	Test	System (E2E) + UAT	TC-SDB-02	To be designed
FR-SDB-03	R1	Test	System (E2E) + UAT	TC-SDB-03	To be designed
FR-SDB-04	R1	Test	System (E2E) + UAT	TC-SDB-04	To be designed
FR-SDB-05	R1	Test	System (E2E) + UAT	TC-SDB-05	To be designed
FR-SDB-06	R2	Test	System + UAT	TC-SDB-06	To be designed
FR-SLS-01	R1	Test	System (E2E) + UAT	TC-SLS-01	To be designed
FR-SLS-02	R1	Test	System (E2E) + UAT	TC-SLS-02	To be designed
FR-SLS-03	R1	Test	System (E2E) + UAT	TC-SLS-03	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-SLS-04	R1	Test	System (E2E) + UAT	TC-SLS-04	To be designed
FR-SLS-05	R1	Test	System (E2E) + UAT	TC-SLS-05	To be designed
FR-SLS-06	R1	Test	System (E2E) + UAT	TC-SLS-06	To be designed
FR-SLS-07	R1	Test	System (E2E) + UAT	TC-SLS-07	To be designed
FR-SLS-08	R1	Test	System (E2E) + UAT	TC-SLS-08	To be designed
FR-SLS-09	R1	Test	System (E2E) + UAT	TC-SLS-09	To be designed
FR-SLS-10	R1	Test	System (E2E) + UAT	TC-SLS-10	To be designed
FR-SLS-11	R2	Test	System + UAT	TC-SLS-11	To be designed
FR-SLS-12	R2	Test	System + UAT	TC-SLS-12	To be designed
FR-SIM-01	R1	Test	System (E2E) + UAT	TC-SIM-01	To be designed
FR-SIM-02	R1	Test	System (E2E) + UAT	TC-SIM-02	To be designed
FR-SIM-03	R1	Test	System (E2E) + UAT	TC-SIM-03	To be designed
FR-SIM-04	R1	Test	System (E2E) + UAT	TC-SIM-04	To be designed
FR-SIM-05	R2	Test	System + UAT	TC-SIM-05	To be designed
FR-SDL-01	R1	Test	System (E2E) + UAT	TC-SDL-01	To be designed
FR-SDL-02	R1	Test	System (E2E) + UAT	TC-SDL-02	To be designed
FR-SDL-03	R1	Test	System (E2E) + UAT	TC-SDL-03	To be designed
FR-SDL-04	R1	Test	System (E2E) + UAT	TC-SDL-04	To be designed
FR-SDL-05	R1	Test	System (E2E) + UAT	TC-SDL-05	To be designed
FR-SDL-06	R1	Test	System (E2E) + UAT	TC-SDL-06	To be designed
FR-SDL-07	R2	Test	System + UAT	TC-SDL-07	To be designed
FR-SOR-01	R1	Test	System (E2E) + UAT	TC-SOR-01	To be designed
FR-SOR-02	R1	Test	System (E2E) + UAT	TC-SOR-02	To be designed
FR-SOR-03	R2	Test	System + UAT	TC-SOR-03	To be designed
FR-SOR-04	R2	Test	System + UAT	TC-SOR-04	To be designed
FR-SBD-01	R1	Test	System (E2E) + UAT	TC-SBD-01	To be designed
FR-SBD-02	R2	Test	System + UAT	TC-SBD-02	To be designed
FR-CFG-01	R1	Test	System (E2E) + UAT	TC-CFG-01	To be designed
FR-CFG-02	R1	Test	System (E2E) + UAT	TC-CFG-02	To be designed
FR-CFG-03	R1	Test	System (E2E) + UAT	TC-CFG-03	To be designed
FR-CFG-04	R1	Test	System (E2E) + UAT	TC-CFG-04	To be designed
FR-CFG-05	R2	Test	System + UAT	TC-CFG-05	To be designed
FR-MSO-01	R2	Test	System + UAT	TC-MSO-01	To be designed
FR-MSO-02	R2	Test	System + UAT	TC-MSO-02	To be designed
FR-MSO-03	R2	Test	System + UAT	TC-MSO-03	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-MSO-04	R2	Test	System + UAT	TC-MSO-04	To be designed
FR-MSO-05	R2	Test	System + UAT	TC-MSO-05	To be designed
FR-MSO-06	R2	Test	System + UAT	TC-MSO-06	To be designed
FR-MSO-07	R2	Test	System + UAT	TC-MSO-07	To be designed
FR-PRM-01	R2	Test	System + UAT	TC-PRM-01	To be designed
FR-PRM-02	R2	Test	System + UAT	TC-PRM-02	To be designed
FR-PRM-03	R2	Test	System + UAT	TC-PRM-03	To be designed
FR-PRM-04	R2	Test	System + UAT	TC-PRM-04	To be designed
FR-PRM-05	R2	Test	System + UAT	TC-PRM-05	To be designed
FR-PRM-06	R2	Test	System + UAT	TC-PRM-06	To be designed
FR-RET-01	R2	Test	System + UAT	TC-RET-01	To be designed
FR-RET-02	R2	Test	System + UAT	TC-RET-02	To be designed
FR-RET-03	R2	Test	System + UAT	TC-RET-03	To be designed
FR-RET-04	R2	Test	System + UAT	TC-RET-04	To be designed
FR-RET-05	R2	Test	System + UAT	TC-RET-05	To be designed
FR-RET-06	R2	Test	System + UAT	TC-RET-06	To be designed
FR-RET-07	R2	Test	System + UAT	TC-RET-07	To be designed
FR-PAY-01	R2	Test	System + UAT	TC-PAY-01	To be designed
FR-PAY-02	R2	Test	System + UAT	TC-PAY-02	To be designed
FR-PAY-03	R2	Test	System + UAT	TC-PAY-03	To be designed
FR-PAY-04	R2	Test	System + UAT	TC-PAY-04	To be designed
FR-PAY-05	R2	Test	System + UAT	TC-PAY-05	To be designed
FR-PAY-06	R2	Test	System + UAT	TC-PAY-06	To be designed
FR-FRD-01	R2	Test	System + UAT	TC-FRD-01	To be designed
FR-FRD-02	R2	Test	System + UAT	TC-FRD-02	To be designed
FR-FRD-03	R2	Test	System + UAT	TC-FRD-03	To be designed
FR-FRD-04	R2	Test	System + UAT	TC-FRD-04	To be designed
FR-FRD-05	R2	Test	System + UAT	TC-FRD-05	To be designed
FR-MSG-01	R2	Test	System + UAT	TC-MSG-01	To be designed
FR-MSG-02	R2	Test	System + UAT	TC-MSG-02	To be designed
FR-MSG-03	R2	Test	System + UAT	TC-MSG-03	To be designed
FR-MSG-04	R2	Test	System + UAT	TC-MSG-04	To be designed
FR-MSG-05	R2	Test	System + UAT	TC-MSG-05	To be designed
FR-NTF-01	R2	Test	System + UAT	TC-NTF-01	To be designed
FR-NTF-02	R2	Test	System + UAT	TC-NTF-02	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
FR-NTF-03	R2	Test	System + UAT	TC-NTF-03	To be designed
FR-NTF-04	R2	Test	System + UAT	TC-NTF-04	To be designed
FR-NTF-05	R2	Test	System + UAT	TC-NTF-05	To be designed
FR-CSA-01	R2	Test	System + UAT	TC-CSA-01	To be designed
FR-CSA-02	R2	Test	System + UAT	TC-CSA-02	To be designed
FR-CSA-03	R2	Test	System + UAT	TC-CSA-03	To be designed
FR-CSA-04	R2	Test	System + UAT	TC-CSA-04	To be designed
FR-CSA-05	R2	Test	System + UAT	TC-CSA-05	To be designed
FR-CSA-06	R2	Test	System + UAT	TC-CSA-06	To be designed
FR-MOD-01	R2	Test	System + UAT	TC-MOD-01	To be designed
FR-MOD-02	R2	Test	System + UAT	TC-MOD-02	To be designed
FR-MOD-03	R2	Test	System + UAT	TC-MOD-03	To be designed
FR-MOD-04	R2	Test	System + UAT	TC-MOD-04	To be designed
FR-MOD-05	R2	Test	System + UAT	TC-MOD-05	To be designed
FR-UAD-01	R2	Test	System + UAT	TC-UAD-01	To be designed
FR-UAD-02	R2	Test	System + UAT	TC-UAD-02	To be designed
FR-UAD-03	R2	Test	System + UAT	TC-UAD-03	To be designed
FR-UAD-04	R2	Test	System + UAT	TC-UAD-04	To be designed
FR-UAD-05	R2	Test	System + UAT	TC-UAD-05	To be designed
FR-CMS-01	R2	Test	System + UAT	TC-CMS-01	To be designed
FR-CMS-02	R2	Test	System + UAT	TC-CMS-02	To be designed
FR-CMS-03	R2	Test	System + UAT	TC-CMS-03	To be designed
FR-CMS-04	R2	Test	System + UAT	TC-CMS-04	To be designed
FR-CMS-05	R2	Test	System + UAT	TC-CMS-05	To be designed
FR-SEO-01	R1/R2	Test	System (E2E) + UAT	TC-SEO-01	To be designed
FR-SEO-02	R2	Test	System + UAT	TC-SEO-02	To be designed
FR-SEO-03	R2	Test	System + UAT	TC-SEO-03	To be designed
FR-SEO-04	R2	Test	System + UAT	TC-SEO-04	To be designed
FR-SEO-05	R2	Test	System + UAT	TC-SEO-05	To be designed
BR-PRC-01	R1	Test	Unit + System	TC-BR-PRC-01	To be designed
BR-PRC-02	R1	Test	Unit + System	TC-BR-PRC-02	To be designed
BR-PRC-03	R1	Test	Unit + System	TC-BR-PRC-03	To be designed
BR-PRC-04	R1	Test	Unit + System	TC-BR-PRC-04	To be designed
BR-PRC-05	R1	Test	Unit + System	TC-BR-PRC-05	To be designed
BR-CRT-01	R1	Test	Unit + System	TC-BR-CRT-01	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
BR-CRT-02	R1	Test	Unit + System	TC-BR-CRT-02	To be designed
BR-INV-01	R1	Test	Unit + System	TC-BR-INV-01	To be designed
BR-DLV-01	R1	Test	Unit + System	TC-BR-DLV-01	To be designed
BR-ORD-01	R1	Test	Unit + System	TC-BR-ORD-01	To be designed
BR-CAT-01	R1	Test	Unit + System	TC-BR-CAT-01	To be designed
BR-CAT-02	R1	Test	Unit + System	TC-BR-CAT-02	To be designed
BR-CAT-03	R1	Test	Unit + System	TC-BR-CAT-03	To be designed
BR-CAT-04	R1	Test	Unit + System	TC-BR-CAT-04	To be designed
BR-SRC-01	R1	Test	Unit + System	TC-BR-SRC-01	To be designed
BR-REV-01	R1	Test	Unit + System	TC-BR-REV-01	To be designed
BR-REV-02	R1	Test	Unit + System	TC-BR-REV-02	To be designed
BR-REV-03	R1	Test	Unit + System	TC-BR-REV-03	To be designed
BR-CS-01	R1	Test	Unit + System	TC-BR-CS-01	To be designed
BR-SEL-01	R1	Test	Unit + System	TC-BR-SEL-01	To be designed
BR-SEL-02	R1	Test	Unit + System	TC-BR-SEL-02	To be designed
BR-SLR-01	R1	Test	Unit + System	TC-BR-SLR-01	To be designed
BR-SLR-02	R1	Test	Unit + System	TC-BR-SLR-02	To be designed
BR-SLR-03	R1	Test	Unit + System	TC-BR-SLR-03	To be designed
BR-SLR-04	R1	Test	Unit + System	TC-BR-SLR-04	To be designed
BR-SLR-05	R1	Test	Unit + System	TC-BR-SLR-05	To be designed
BR-SLR-06	R1	Test	Unit + System	TC-BR-SLR-06	To be designed
BR-IMG-01	R1	Test	Unit + System	TC-BR-IMG-01	To be designed
BR-MSO-01	R2	Test	Unit + System	TC-BR-MSO-01	To be designed
BR-PRM-01	R2	Test	Unit + System	TC-BR-PRM-01	To be designed
BR-RET-01	R2	Test	Unit + System	TC-BR-RET-01	To be designed
BR-PAY-01	R2	Test	Unit + System	TC-BR-PAY-01	To be designed
BR-GEN-01	R1	Test	Unit + System	TC-BR-GEN-01	To be designed
NFR-PERF-01	R2	Test	Performance	TC-NFR-PERF-01	To be designed
NFR-PERF-02	R2	Test	Performance	TC-NFR-PERF-02	To be designed
NFR-PERF-03	R2	Test	Performance	TC-NFR-PERF-03	To be designed
NFR-PERF-04	R2	Test	Performance	TC-NFR-PERF-04	To be designed
NFR-USA-01	R2	Demonstration	Usability test	TC-NFR-USA-01	To be designed
NFR-USA-02	R2	Demonstration	Usability test	TC-NFR-USA-02	To be designed
NFR-USA-03	R2	Demonstration	Usability test	TC-NFR-USA-03	To be designed
NFR-ACC-01	R2	Inspection + Test	Accessibility audit	TC-NFR-ACC-01	To be designed

Requirement	Release	Verification method	Test level	Test case ID	Status
NFR-ACC-02	R2	Inspection + Test	Accessibility audit	TC-NFR-ACC-02	To be designed
NFR-ACC-03	R2	Inspection + Test	Accessibility audit	TC-NFR-ACC-03	To be designed
NFR-ACC-04	R2	Inspection + Test	Accessibility audit	TC-NFR-ACC-04	To be designed
NFR-ACC-05	R2	Inspection + Test	Accessibility audit	TC-NFR-ACC-05	To be designed
NFR-COM-01	R2	Test	Cross-browser / device	TC-NFR-COM-01	To be designed
NFR-COM-02	R2	Test	Cross-browser / device	TC-NFR-COM-02	To be designed
NFR-REL-01	R2	Inspection	Code / design review	TC-NFR-REL-01	To be designed
NFR-REL-02	R2	Inspection	Code / design review	TC-NFR-REL-02	To be designed
NFR-REL-03	R2	Analysis + Demonstration	Operational readiness	TC-NFR-REL-03	To be designed
NFR-SEC-01	R2	Test + Inspection	Security	TC-NFR-SEC-01	To be designed
NFR-SEC-02	R2	Test + Inspection	Security	TC-NFR-SEC-02	To be designed
NFR-SEC-03	R2	Test + Inspection	Security	TC-NFR-SEC-03	To be designed
NFR-SEC-04	R2	Test + Inspection	Security	TC-NFR-SEC-04	To be designed
NFR-SEC-05	R2	Test + Inspection	Security	TC-NFR-SEC-05	To be designed
NFR-PRV-01	R2	Inspection	Compliance review	TC-NFR-PRV-01	To be designed
NFR-PRV-02	R2	Inspection	Compliance review	TC-NFR-PRV-02	To be designed
NFR-MNT-01	R2	Inspection	Code / design review	TC-NFR-MNT-01	To be designed
NFR-MNT-02	R2	Inspection	Code / design review	TC-NFR-MNT-02	To be designed
NFR-PRT-01	R2	Inspection	Code / design review	TC-NFR-PRT-01	To be designed
NFR-I18N-01	R2	Inspection	Code / design review	TC-NFR-I18N-01	To be designed
NFR-OBS-01	R2	Analysis + Demonstration	Operational readiness	TC-NFR-OBS-01	To be designed

Appendix E. Sign-off

I confirm that I have reviewed this Business Requirements Document (HC-BRD-001, version 1.1) and that it represents the agreed business requirements for the HexCommerce marketplace.

Name	Role	Decision (Approve / Approve with comments / Reject)	Signature and date
	Executive Sponsor		
	Product Owner		
	Engineering Lead		
	QA / Test Lead		

— End of document —